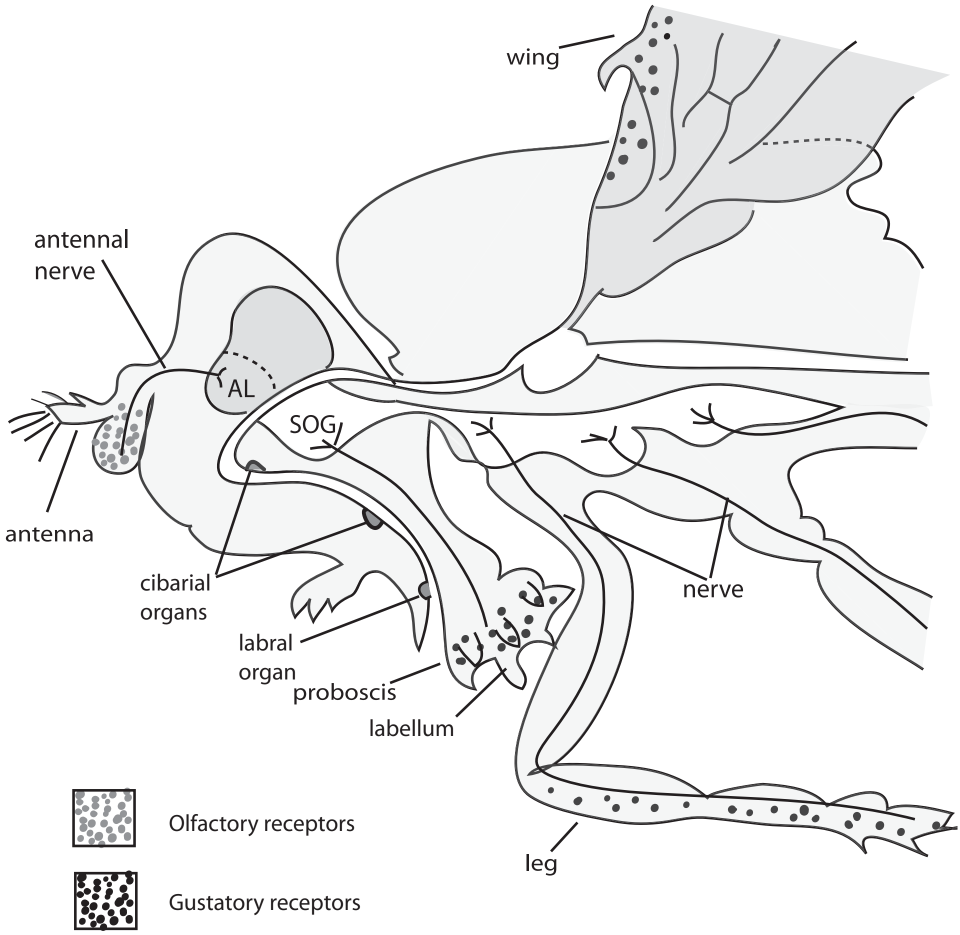

Figure 13-6. Olfactory and gustatory receptors are located on many parts of an insect's body, here showing *Drosophila.* Redrawn from (Stocker 1994), with permission. The SOG (*subesophageal ganglion*) and AL (*antennal lobe*) are parts of the insect brain that receive signals from ORs and GRs. *Labral* and *cibarial organs* are sense organs in the mouth, each with chemosensory receptors that project to the CNS.

processing in human terms, that we generally associate with our experience with consciousness.

Olfaction appears to be another example of the evolution of systems that involve shared genetic mechanisms used in independently evolved organs, as also seen in mechanoreception mechanisms in hearing, and will be seen for vision. The location and morphology of invertebrate olfactory organs is highly variable, but there are shared neural similarities. It will be interesting to see the degree to which the signaling mechanisms that induce the development of chemosensory structures are shared. Yet, although chemosensation involving the basic *7TMRs* probably existed very early in metazoan life, the particulars do not seem to have been shared since these diverse animal groups' common ancestor. For example, crustaceans don't have olfactory glomeruli or the characteristic mushroom bodies. Instead, the ORs of decapod crustaceans are found in *antennules*, appendages on the heads of these arthropods, and their axons terminate in the *antennular lobes* of the brain, suggesting that these structures evolved multiple times (Strausfeld and Hildebrand 1999).

HOW DOES IT WORK?

Unlike light and sound, odorants are not sampled from a continuous or coherent frequency spectrum. Similar molecules can "smell" different, and different compounds can smell the same. But the senses of sight and hearing do have some similarities with smell. Each photoreceptor (Chapter 14) or hair cell (Chapter 12) responds optimally to a particular part of its corresponding spectrum, but a given light or sound frequency can trigger responses in more than one photoreceptor gene or hair cell. Similarly, an odorant molecule can trigger response from various ORs.

Different individuals in a species appear to detect and respond similarly to the same odorant, at least to some extent. Receptor genes are highly organized in chromosomal clusters and have at least some coordinate expression in segmentally arranged tissue regions. The wiring among individuals of the same species is similar but not identical. Different classes of receptors send signals to similar glomeruli, from which the signals are distributed to different regions of the brain, still maintaining at least some clustering (Zou, Horowitz et al. 2001). What the brain then does is *compare* the signal from the different classes of receptors. The set of ONs sending signals can be integrated in a kind of binary algebra, resulting in an "address" or signature of a given odor, and signature can be remembered.

This description does not enlighten in regard to how a given odorant may control behavior, however. Behavior response is somewhat clearer with pheromones, which bind to preset receptors; thus, in principle, they can have developmentally prewired responses.

Vertebrate and invertebrate olfaction are different in many ways, but similarities do exist. Both groups of animals actively sample their olfactory environment: vertebrates repeatedly sniff to sample ambient odors, whereas insects may flick their olfactory appendages (e.g., Laurent 1999). Both groups also track the source of an odor in searching for mates or food from which a signal emanates. Regardless of how the information is processed cognitively, as an animal moves it compares the changing signal strength (and right-left differences). Thus, an insect may fly in and out of a signal plume, adjusting its direction as the signal changes. Olfactory pursuit requires that a signal be cleared and the system refreshed quickly enough to detect small changes in intensity or direction (e.g., Laurent 1999; MacLeod et al. 1998).

Molecules of a given odorant may have low concentration or spatial density. By chance, any given OR might not be "hit" by a ligand molecule in a given sniff or might not be hit from sniff to sniff; the aroma would seem to the animal to come and go. This could be a problem if each odorant could only be detected by a single ON or if all the ONs expressing a given receptor were tightly arranged in the olfactory epithelia. Instead, like-expressing ONs are multiple and scattered at least around a region of the olfactory epithelium. In this way, approximately the same *number* of OR-specific hits may occur from sniff to sniff, yielding roughly the same signal strength; of course, the brain then must be able to do its sums. The specificity of an odorant can be perceived because ONs expressing the same OR send their signal to similar glomeruli, where a sufficient integration of signal strength that does not depend on a single hit can occur; but again, the brain must be able to do its bookkeeping.

These comments of course relate to detecting a signal and not to its cognitive perception as an "odor" with a behavioral message. That is a separate topic, which, although important, is still very poorly understood (but we review it briefly in Chapter 16).

GUSTATION

Taste is the detection of soluble chemicals to elicit feeding and perhaps other behaviors. Gustatory receptors (GRs or, alternatively, TRs for "taste receptors") are found in the epithelium of vertebrate body parts used for the ingestion of food: lips, oral cavity, tongue, and pharynx (see Figure 13-7 for location of GRs on the mammalian tongue). In invertebrates, they are located in various appendages of the head (especially the proboscis), where they would be expected, but also in the wings, feet, and genitalia (Scott et al. 2001) (see Figure 13-6); insects can thus explore potential food sources in more varied ways than vertebrates. Insect larvae also make use of GRs. Aquatic animals, both vertebrate and invertebrate, can have chemoreceptors on their body surfaces, which are used for locating food. These chemoreceptors are the basis of a sense that is only somewhat analogous to the sense of taste, however, and are not structurally the same as "true" taste receptors. Of course, we cannot really guess what the sea "tastes" like to a fish in this respect.

Reflecting the common evolutionary origins of chemosensation, *GR* genes are *7TMR*s, but they again form a subfamily different from the *OR*s. *GR*s have introns. Mammals have 50–100 *GR* genes in at least two subfamilies, far fewer than their number of *OR* genes. The *T2R* family of genes are used for bitter taste reception and may have distant homology to the *V1R* family of rodent VRs; putative sweet or other taste receptors in the *T1R* family have homology to the *V2R*s (Mombaerts 2001). A given taste receptor cell expresses multiple, related GRs (Mombaerts 2001).

There are about 50 *GR* genes in flies, roughly corresponding to their *OR* repertoire, suggesting that soluble and volatile chemosensation are of comparable importance to insects. Nematodes (*C. elegans*) use chemoreceptors in a way that resembles taste in vertebrates, where, as noted earlier, multiple receptors are expressed in a few sensory neurons (Firestein 2001).

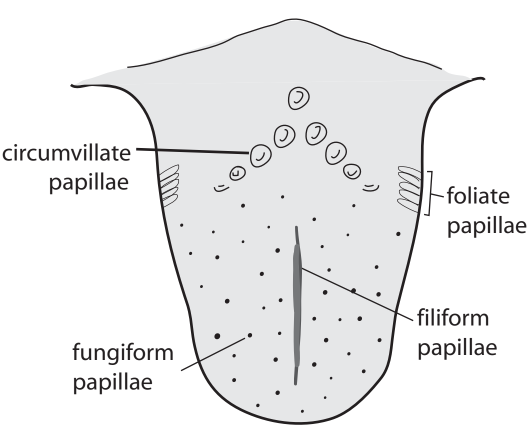

Figure 13-7. Location of gustatory receptors on a mammalian tongue; four major classes.

The taste mechanism in vertebrates is patterned on the tongue and develops separately from the olfactory and vomeronasal systems. Taste receptor cells have voltage-gated ion channels, which in vertebrates synapse into three of the primary cranial nerves (facial, glosopharyngeal, and vagus); remarkably, this is true even of distal taste buds on the body surface of fish (Finger 1997). Taste neurons are secondary receptors, which is in contrast to olfactory neurons, which are primary receptors; that is, taste receptors have no axon but synapse with sensory nerves that go to the brain. Different taste regions use different molecular reception mechanisms because of the chemical properties of the molecules being detected. Salt and acids can pass directly through ion channels, whereas sugars and bitter substances must activate a second messenger to be detected. In insects, the neurons are bipolar with a distal process that extends to the surface of the epithelium, usually ending at an opening in the cuticle or exoskeleton, which chemicals can penetrate, and a central process extending into the CNS (Finger and Simon 2000).

Gustation and olfaction are related in interesting ways, not least of course being their use of related chemoreceptor genes. The organization of taste receptors resembles that of the olfactory epithelium, but this is due to developmentally different patterning events. The classic senses of salt, sweet, sour, and bitter (and umami, the taste elicited by glutamate, found naturally in some foods and added to others as monosodium glutamate, or MSG) are detected by receptors for specific chemicals, and the long-held idea (going back to the classics) that these are located exclusively in different parts of the tongue has been shown to be incorrect. The regional differentiation of the olfactory epithelium and VNO is not highly correlated (if at all) with particular odorant properties; similar ORs are located similarly, but odorant perception is not obviously regionalized in the nose. "Taste" as we usually refer to it, really is an integral use of taste and smell receptors. However, taste receptors map to distinct regions of the vertebrate brain even though the senses lead to related and integrated percepts. A separation of smell and taste in the brain occurs also in insects, but the distinction is less clear and at least some insect *GR* genes also function in olfactory circuits.

OTHER VERTEBRATE CHEMOSENSORY MECHANISMS

Chemosensation is a very general phenomenon, and two mechanisms unrelated to immune detection and olfaction merit mention (e.g., Finger 1997). One is the human "common chemical sense," which is a property of free somatosensory nerve endings in epithelia such as on the exposed surfaces of eyes, nose, mouth, and throat. These respond to substances such as ammonia, mint, and pepper and provide sensations of burning and coolness, often leading to avoidance reactions. A second and perhaps related system involves the solitary chemoreceptor cell (SCC). Cells of this type are secondary sensory neurons, as in taste cells, and are located on external surfaces of nonamniotic aquatic craniates and used in feeding and predator avoidance. The SCC may be related evolutionarily to taste but is connected to the somatosensory system.

CHEMOSENSATION IN PLANTS AND SINGLE CELLED ORGANISMS

Although we concentrate in this chapter on olfaction and gestation in multicellular animals, other organisms detect external chemical signals in other ways that we can briefly mention (again acknowledging that our singling out of animal taste and smell as separate senses is somewhat arbitrary and artificial relative to chemosensation in the world as a whole). Plants do not have sensory systems equivalent to taste or smell but they do respond to external chemical signals including by differential growth corresponding to chemical gradients in the environment, ripening induced by ethylene, responses to herbicides, and the like.

Single-celled organisms respond to chemicals as well. As mentioned in Chapter 12, planktonic *Dictyostelium discoideum* induce aggregation among their peers by emitting a chemical signal and as aggregates they respond to environmental chemicals, particularly ammonia, in various ways. *E. coli* and *Salmonella*, and many other single celled organisms, navigate toward and away from chemical attractants or repellants, for example, toward nutrients or away from toxins.

SOME EVOLUTIONARY THOUGHTS

Chemosensory evolution presents various challenges. For example, there is a potential conflict of interest between mechanisms that preserve specificity and those that generate diversity. The immune system provides interesting contrasts and similarities.

Chemosensory receptor clusters have a history of frequent gene duplication, with the subsequent accumulation of diversity between *OR* genes. That is typical of tandemly repeated clusters of related genes, including the immunoglobulin and MHC clusters. Indeed, the pressure for olfactory diversity is shown by the fact that the *OR* family is even larger. The similarity between *OR* and immune allelic exclusion provides a kind of perceptive specificity, within diversity, sequestered in specific cells to keep things orderly and coherent. This tempts the speculation that *OR*s are also rearranged somatically during development of the olfactory epithelium (e.g., Mombaerts 1999). However, if this is going on it has not yet been demonstrated.

A partial similarity between immune and chemodetection is that both systems bind their target molecules combinatorially. Many antibody molecules may bind the same or different haptens of a circulating antigen, and a given odorant will typically be a ligand for several different ORs. Unlike the vertebrate somatic recombination-generated diversity, however, the strategy for detection of an odorant appears to involve a combinatorial process. The brain senses a signal from a particular set of simultaneous signals from multiple, presumably replicable and inherited, ORs. Once an antibody molecule has been generated, the combinatorial aspect of somatic rearrangement is over for that particular cell; combinatorial immune attacks involve multiple cells each with rearranged genomes, but there needs to be no centralized accounting of which cells are at work. By contrast, such accounting is vital for organismal response to chemosensation.

Another difference is that in mammalian immunity a response gradually becomes "focused" by selecting for the preferential amplification of cells producing the best among the diversity of antibodies that recognize an intruding molecule. Immune focusing can continually occur, so as to track mutational changes in the pathogen. To some extent, each generation of vertebrates faces a different diversity of pathogenic organisms. The olfactory environment may change from moment to moment just as the immune environment can, but changes occur much less rapidly and unpredictably across generations. Although immune recognition sharpens, olfactory perception becomes dulled with prolonged exposure; the evidence to date doesn't seem to suggest that there is olfactory focusing. A far as we know, however, our immune system does not require any form of cognitive integration—so what ensures that different individuals, who smell the same thing in combinatorally different ways, will react appropriately?

We should remember that somatic recombination is not a requisite for effective immune resistance, and in fact even long-lived plants do perfectly well with their olfaction-like *R*-gene system (which uses genomic combinatorial rather than somatically recombinatorially generated diversity). The large number of *OR* genes and their intergenic diversity has typically been viewed (as in the *R*-gene system in plants) as having been selected to generate high amounts of odorant binding diversity. The general lack of pseudogenes has reinforced this idea of stringent natural selection for odorant-specific *OR* genes. Plants do appear to have pathogen-specific *R* genes. There seems to be a consistent pattern of a large number of ORs, with few pseudogenes, in species with high reliance on olfaction. The many *OR* genes in dogs and few if any in dolphins are good examples, but our own olfactory epithelium is more patchy than continuous, and we have only around 350 functional *OR* genes, with the rest of our *OR* genes being pseudogenes.

There is a high degree of polymorphism *within* human OR genes (see Mombaerts 2001), which might seem consistent with this evolutionary story, for example, if there has been relaxed selection in our ancestry, which is the usual inference. One upshot of this level of variation is that an *OR* may be "pseudo" in some people and functional in others. Among other changes, many human *OR* pseudogenes have had stop codons created by mutations in the coding region; these are either not fixed in our species or mutation may have recreated open reading frames by converting such stops back into amino acid codons. With a high degree of heterozygosity, each person may bear two different alleles at their roughly 350 functional *OR* genes, which effectively doubles the available repertoire of diversity, even if no two people have the same set of alleles or even the same set of functional genes.

Rather than viewing human olfaction as degenerate, one might make an alternative darwinian interpretation that, as in immunology and MHC specification, selection has favored olfactory diversity even in humans. This may sound fine in principle, but, unlike the immune system, we have to react cognitively to an odorant. If there is too much variation, we might not be able to detect any given odorant, and indeed odorant-specific anosmias are common and the perfume industry is kept busy because we more. Yet, with the amount and chaotic organization of variation, we should be than we are different vary in what we perceive or how we react.

The pattern of variation in the major *OR* gene cluster on human chromosome 17 is interesting in this regard (Gilad et al. 2000). As seen in 20 sequenced individuals, the functional *OR* genes in this cluster vary less than the pseudogenes, and there is evidence from comparison with orthologs in chimps that there has been weak positive selection at the genes (but not the pseudogenes), which may maintain *OR* diversity. This may be too weak to be attributed to odorant specificity, although for some odorants most people do react in a similar positive or negative way, as if there is some form of specificity. These facts need to be reconciled with the observation that at least many human *OR* genes are not pseudo in everyone.

At the same time, a comparison between humans and other primates found that humans have accumulated pseudogene-producing mutations (that is, that disrupt coding relative to functional orthologs) at a rate roughly four times that in other primates (Gilad et al. 2003; Rouquier, Blancher et al. 2000). This suggests excess loss of genes in the human lineage, though the other primates also have considerably higher fractions of pseudogenes than does the mouse. Whether what has been favored is a kind of variable pseudogene pattern remains to be shown.

Important information will come from the analysis of variation in the *OR* genes in animals that have few pseudogenes, which is assumed to be due to selection for function. Interestingly, based on the early indications, mice do have *OR* polymorphisms (P. Mombaerts, personal communication). What is preventing pseudogenes in these species? The question is cogent because even in mice the evidence suggests that odorant detection is combinatorial and open-ended rather than prescriptively specifying one set of receptors for each possible odorant. Why wouldn't mice benefit from a high mutational repertoire, even at the expense of making some genes pseudogenes in some individuals? Possibly, the determination of olfactory genetics from inbred mice (in most cases, probably from studies of a single strain) could obscure some of these questions until wild mice, or mice from multiple independently-derived strains, are examined.

A cautionary note in any such functional evolutionary speculation is that ORs are expressed in nonolfactory tissues like testis and heart and thus may have pleiotropic functions. One possibility is that there have been various types of balancing selection, but a simpler explanation may be that this just reflects "leaky" nonspecific gene expression; the testis expresses many genes that have no obvious germline function (Mombaerts 1999). Alternatively, it has been rather loosely speculated that distributed expression of highly variable OR genes can be used in development as a kind of "area code" to identify tissue-specific codes during development (Dryer 2000).

The answers to these many questions will be interesting. We need to remember also that it is after the fact that we evaluate the nature or importance of chemosensation to a given species. Chemosensation is a generic need of cellular life, but a given species uses what it has and has what it uses. Why, for example shouldn't birds or humans have a better sense of smell? There is no one chemosensory "need" for an organism or a chemosensory problem to "solve."

CONCLUSION

Chemosensation is widespread in the living world. As so typifies evolution, there is no single way to detect chemicals in the environment. Communication among cells is an extension of interaction within cells and is about chemical information exchange. The division of chemosensation into internal and external systems is a somewhat arbitrary distinction (as pheromones show). The various chemical senses employ the widespread *7TMR* class of signal receptors. Within clusters, the genes seem generally to be of similar origin, but between *types* (*OR*s, *VR*s, *GR*s, and other chemosensory receptors) there is only distant homology. This may indicate that, as in so many other systems, chemosensory functions have arisen multiple times independently and/or that the different aspects of chemosensation have evolved in their own ways. But this has happened from a common general starting mechanism.

The various sensory cells are located differently in different organisms, which may in part reflect the relatively less organized nature of chemical information in the world, compared with light and sound. Unlike the programmed ability to detect specific molecules such as hormones or pheromones, olfaction appears to be designed, like the immune system, as a general open-ended molecular detection system.

Olfaction is, however, more than the reception of a molecular "signal." It should come as no surprise that many other genes involved in the processing and use of the signal, locally as well as to and in the brain, as well as the *OR* genes themselves, are involved. A series of experiments in *Drosophila* that involved crosses between olfactory mutant flies have shown that olfactory response is a kind of complex trait, with allelic variation having quantitative effects on the trait, epistatic effects as well as complementarity or other interactions among the genes (Anholt et al. 2001; Anholt and Mackay 2001; Fedorowicz et al. 1998). This elegant experimental result shows what we know in so many ways to be generally true of complex traits in nature. It is in a sense also reassuring: not only are traits assembled over evolutionary time to involve many elements, but selection does not stipulate that there is one "wild type" way to make a trait. Instead, what we get is variation and to some extent complexity that buffers a multigenic system against mutation.

We have outlined the basic mechanical means by which the molecular detection is done, but the most important question perhaps has not been touched. It appears sufficient for the immune system to recognize and inactivate invading organisms. Odors, however, seem to require cognitive, information-integrating responses, which should, at least in some instances, depend on the nature of the compound. To a great extent, reactions are learned: avoidance of a substance is based on experience. Many bitter-tasting butterflies have to be sampled before being shunned. But is reaction all learned by experience, or are there undiscovered mechanisms by which the *type* of molecule being detected is interpreted? Do the different classes of odorant receptors carry, historically, some information in their ligand-binding properties that is not simply due to chance?

We know this is true for specific types of molecules: taste for sweet and bitter, for example, and pheromones. Are there others? Do these exist in mammals as well as invertebrates (which have typically been assumed to be "hard-wired" automotons)? Or are there cognitive processes to do this in ways unknown to us at present? These questions probably will be answered in the near future.

Chapter 14
Detecting Light

Light is the ultimate resource for all forms of life. Organisms use light in three basic ways, the most ubiquitous being in direct physiological processes, the capture of energy via photosynthesis. This is of course what plants do and, in terms of the ecology of the Earth, photosynthesis is probably the most active important use of light energy by living organisms. Second, many organisms simply detect the location of a light source, moving toward or away from it. Third is *vision*, the interpretive use of spatially arranged patterns of light by animals as a source of information about objects in their environment. Mobile organisms do this to avoid collision, avoid predators, pursue prey and other food or water sources, detect conspecifics, associate with mates, and otherwise participate in social behavior, among many other purposes. Information can also be *sent* by organisms through light transmission, as by flowers to attract pollinating insects or by behavioral displays (like leks, in which clouds of males aggregate to attract females). Many attributes of light are used in this way, including wavelength or frequency and intensity.

However, as important as light is in this latter sense, organisms do not need to perceive light or do so in any particular way, even for many of the uses just listed. This is shown, as we will see, by the great variation in what animals, even closely related ones, can see. Simple detection of light without regard for pattern can be sufficient for some, whereas others depend on fine resolution of specific objects (prey) or aspects of objects (berries on a tree). Some use light only to detect motion.

As we have asked in other contexts, how can organisms be genetically programmed to use light to resolve the huge diversity of conditions that they cannot specifically anticipate in a hard-wiring sense, unique conditions that have never occurred—not even once—in the entire 3 billion years of their ancestry?

LIGHT RECEPTION

Energy organized as electromagnetic waves, and/or streams of photons, is called radiant energy. Visible light comprises a small subset of the total spectrum of radiant energy (Figure 14-1).

At one end of the spectrum are cosmic rays and at the other are (for humans) electrical power waves (even lower frequency radiation is possible). The different kinds of radiant energy can be characterized by their wavelength (inversely related

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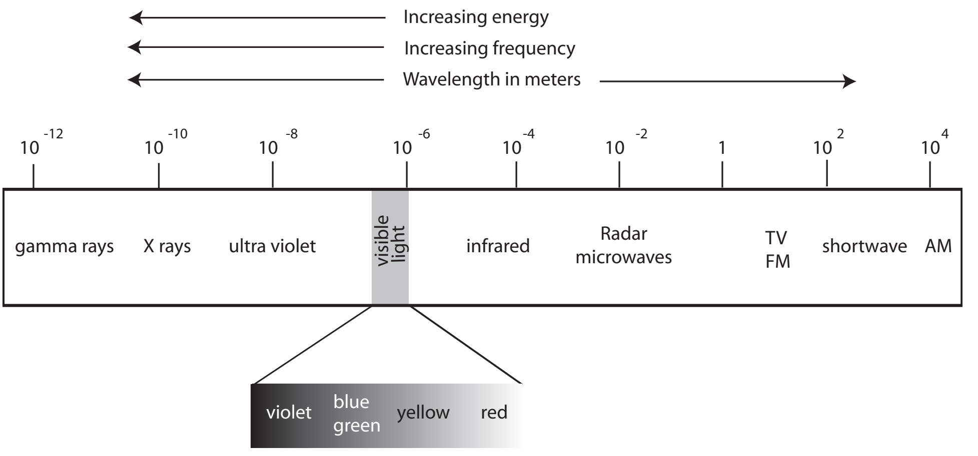


Figure 14-1. Radiant-energy spectrum.

to frequency because the energy travels at the speed of light). The average wavelength of cosmic rays, the shortest cycles of radiant energy known, is in the range 10^{-12} meters (million millionths of a meter). The wavelength of electric power waves is on the order of 10^6 (million) meters (50–60 Hz, or cycles per second). The radiant energy visible to humans comprises a narrow portion of this spectrum, ranging from 380 to 760 nm (nanometers or billionths of a meter). Other organisms can see somewhat more, or less, of this same range. Snakes, for example, have infrared (heat) receptors, and bees and birds have receptors for ultraviolet light.

Visible light is detected by *photoreceptors*. These can take many forms, and we can only describe some of them. For example, the entire cell of some single-celled organisms, such as amoebas, may be sensitive to light, moving toward or away from it. They use a membrane-bound photoreceptor that induces the release of cAMP, which in turn induces changes in the beating of their cilia and cellular motion. Some worms have photoreceptive cells, or eyespots, scattered throughout the epithelium on the surface of their bodies. In the earthworm, these serve to orient the organism directionally, as they prefer to live underground in darkness.

Direct sunlight includes energy in the ultraviolet (UV) part of the spectrum, but reflected light is dimmer and retains little UV, and many species are sensitive to both. Vertebrates have evolved two basic kinds of photoreceptor cells, the *rod* and *cone* cells, which are embedded in the retina at the back of the eye and attached to the axons of neurons in the optic nerve (Figure 14-2A shows the structure of the human eye and retina, and 14-2B depicts the structure of rods and cones with a retinal molecule bound to a 7TMR photoreceptor protein embedded in the surface of a rhodopsin disc). Based on its properties, a photoreceptor molecule responds to light of particular energy and frequency. Rods perceive light and dark (i.e., black, white, and shades of gray), and organisms use these mainly for perception in dim light; they are maximally sensitive in the middle of the visual spectrum and able to respond to a very small amount of light. Cones are more specialized for color perception and acuity of vision and are used to detect form and motion but do require more light than rods to be activated, hence do not function well in dim light (humans lose color

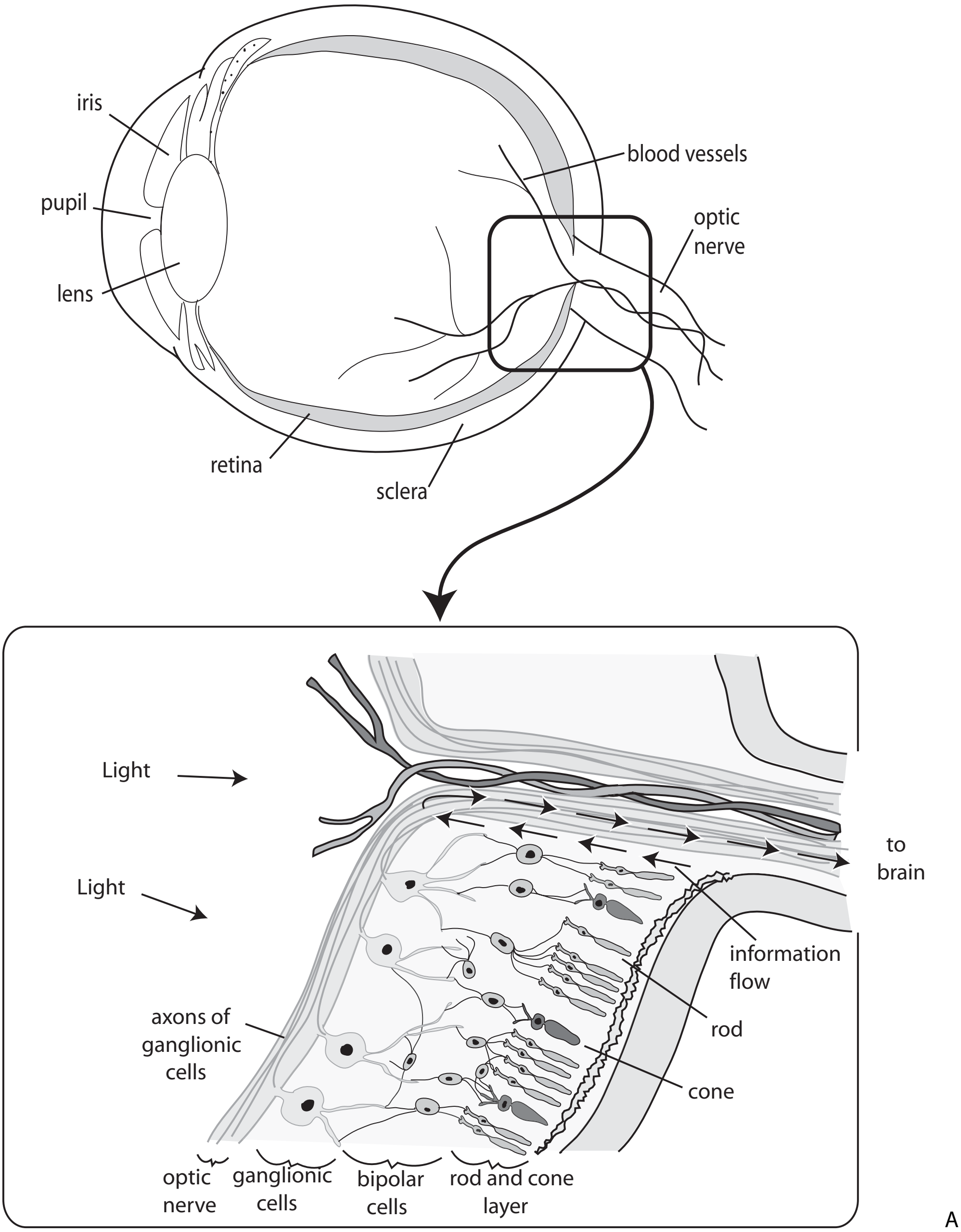


Figure 14-2. Parts of vertebrate visual system (human). (A) Structure of the eye and retina; (B) cellular structure of rods and cones, including a diagram of an opsin and its chromophore.

perception as light dims). The ratio of rods to cones varies across species as well as in different areas of their respective retinas and is related to their adaptations, but the cells function in a similar manner from species to species. UV-sensitive vision in vertebrates is generally a separate phenomenon, although it involves related genes and mechanism (Yokoyama 2000; Yokoyama and Yokoyama 2000).

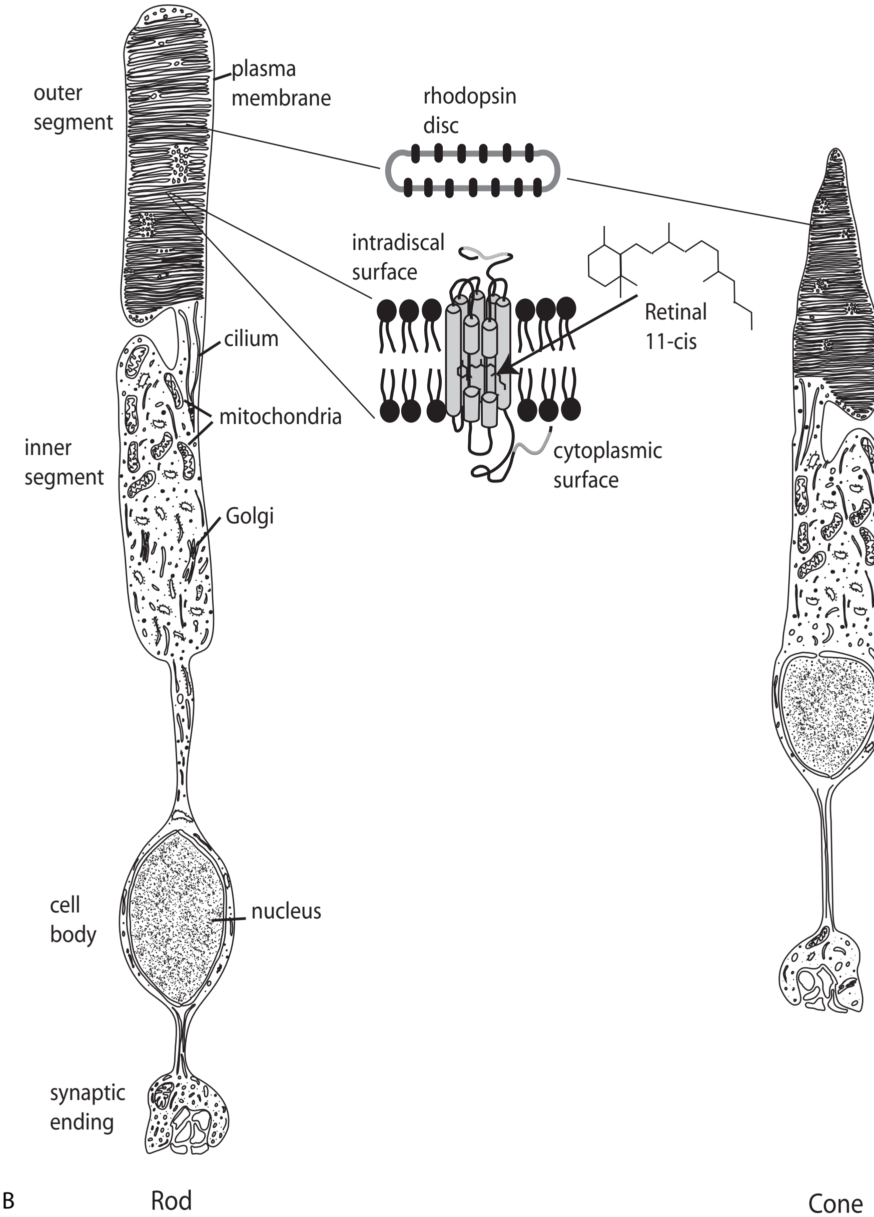


Figure 14-2. *Continued*

Rods and cones have inner and outer segments. The outer segments are composed of a series of stacked bilipid membranous disks that contain visual pigment called *photopigment*. There are two main ways in which these disks can be presented: at the apical end of the cell or in cilia formed on that end. These cellular arrangements have been thought to divide protostomes and deuterostomes, respectively, but the photoreceptor phylogeny turns out to be rather more complex and even raises questions about the nature of chordate-nonchordate eye homologies, or the ease with which similar arrangements of photoreceptors can re-evolve (Arendt and Wittbrodt 2001), a problem we have visited several times (e.g., sea urchin larval stages, Figure 8-9). Based on differences between message transduction pathways that are used, Arendt and Willbrodt suggest that the early bilateran ancestor species had both cellular types.

The photopigment is composed of an *opsin* protein and a *chromophore* called *retinal*, derived from vitamin A. The opsin and chromophore are bound together and embedded in transmembrane receptors on the outer disks. Light energy is captured in these disks and, through a series of chemical reactions that take place in the photopigment, is transduced into receptor potential and sent, via the optic nerve, to the brain where it is interpreted as light and color. Alternative chromophores used by some taxa modify the spectral response. Bird, amphibian, and reptile photoreceptor pigments have an overlying oil droplet that filters light and modifies their spectral sensitivities. As a result, different taxa can use different pigment and chromophore combinations to achieve a similar spectral response.

When light is received by a photoreceptor, the chromophore all-*trans* retinal converts to 11-*cis* retinal, which changes the conformation of the opsin protein. This in turn modifies its intracellular domain (the opsin must then be recharged by a new chromophore molecule).

In vertebrates this "bleaching" releases the chromophore from the opsin; the chromophore is then regenerated by neighboring cells reversing its conformation to the all-*trans* form so it can be reused. In invertebrates, the chromophore does not leave the opsin before being reversed.

Invertebrates and vertebrates use homologous *opsins*. These are coded by yet another branch of the *7TMR* gene family, although the two major animal groups use unrelated second messengers to relay the signal. Vertebrate photoreceptors are coupled to the G protein *transducin*. Absorption of light causes a conformational change in the receptor, leading to increased binding of GTP by the α-subunit of transducin. This activates cGMP phosphodiesterase, which degrades cGMP, causing cGMP-gated Na^+ channels to close, hyperpolarizing the cell and inhibiting neurotransmitter release. This in turn propagates signal to the brain. Thus, neurons are inhibited in the dark, when neurotransmitter is released at a high rate. By contrast, the invertebrate second messenger is inositol triphosphate, and their photoreceptors respond by depolarizing rather than hyperpolarizing (Hardie and Raghu 2001; Nilsson 1996; Ranganathan et al. 1995).

Invertebrate photoreceptor cells also have layers of membranes filled with *rhodopsin*, a photoreceptor pigment, and the photoreceptor collects light in the same way as in vertebrates (Yarfitz and Hurley 1994). Each photoreceptor cell in the *Drosophila* eye has a structure called a *rhabdomere* that contains ~60,000 microvilli. Millions of rhodopsin molecules associated with the downstream light transduction cascade reside in the microvilli.

Opsin Evolution and Function

The number of *opsin* genes varies among species (Pichaud et al. 1999). Although the basic structure of opsins is conserved, each gene has a distribution of wavelengths to which it responds, a distribution with a peak wavelength sensitivity, and diminishing sensitivity at surrounding wavelengths (Figure 14-3C). The spectral sensitivity can be evaluated experimentally and is determined by the amino acid

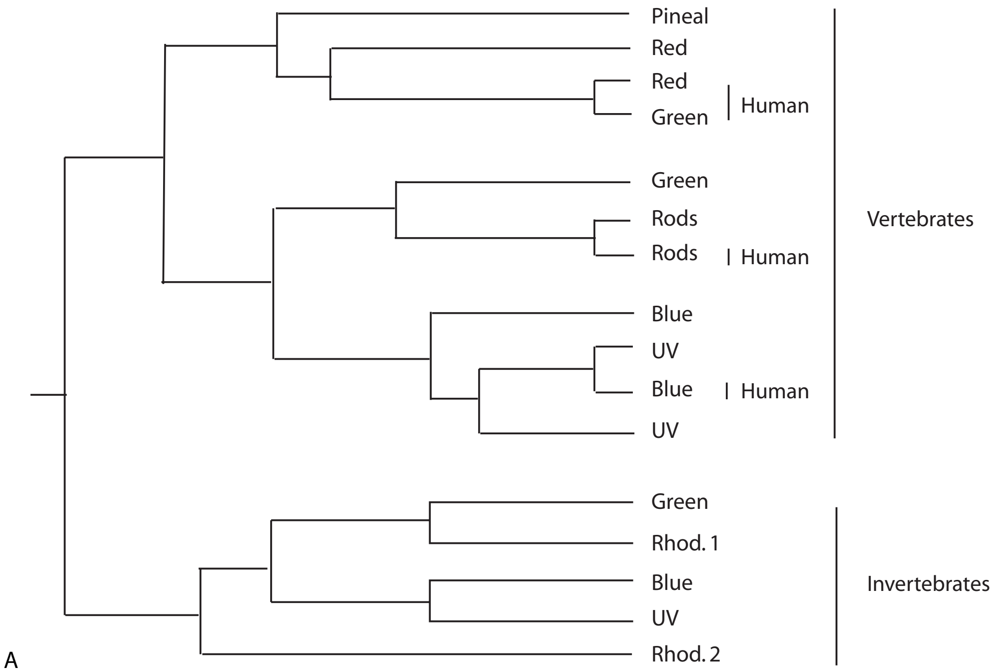
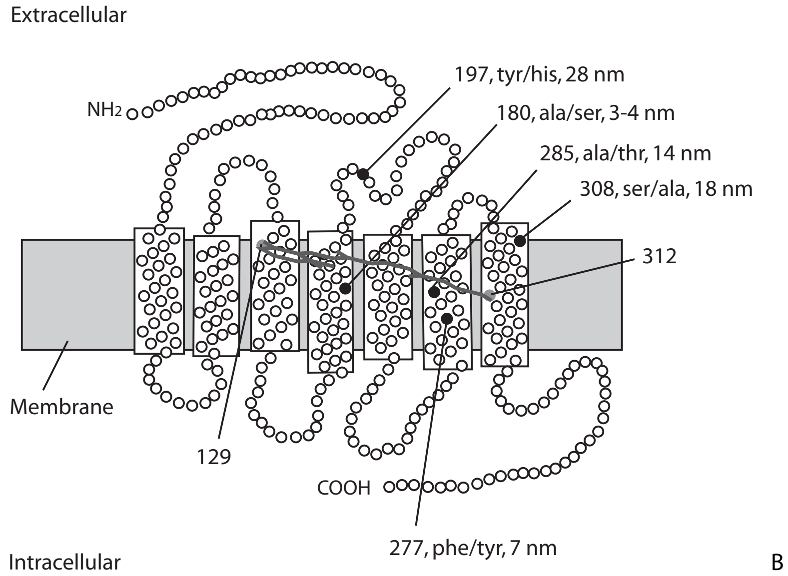


Figure 14-3. *Opsin* genes. (A) Simplified *opsin* gene tree; in many instances similar but not identical *opsins* (based on gene sequence and estimated peak sensitivity) are found within the groups tested, but shown here as one branch of similar genes; (B) location of the amino acids in the opsin protein structure that have the greatest effect in red-green spectral cone sensitivity among naturally observed opsins, showing their likely important relationship to the retinal molecule bound to the photoreceptor (e.g., Yokoyama 2000; Yokoyama and Yokoyama 2000); (C) distribution of wavelengths to which opsins respond (rhodopsins have peak sensitivities around the 500 nm range).

sequence of the opsin protein and its interaction with the chromophore. Modifications in the photoreceptor itself, such as the use of oil droplets by some species, mean that the actual response of the photoreceptor may not correspond to that of the visual pigment alone. Examples are the chicken *Rh2* gene that is paired with a green oil droplet and the goldfish *Rh2* that is green-shifted because of the modified chromophore it uses (Yokoyama and Yokoyama 2000). Because its different opsins respond differently, an organism can compare the signal strength of adjacent opsins to assess "color" and other attributes of incident light coming from the same part of the environment.

Insects have dual dim-bright and color vision capabilities that vary among the species that have been tested (Pichaud, Briscoe et al. 1999). Flies have independent receptors for the two types of vision, but the ability of other insects to distinguish between the two is less clear because the neural impulses to the brain from the two systems seem to merge. Honey bee and *Drosophila* eyes express three *opsins* that absorb maximally in the UV, blue, and green ranges. Some butterflies appear to have



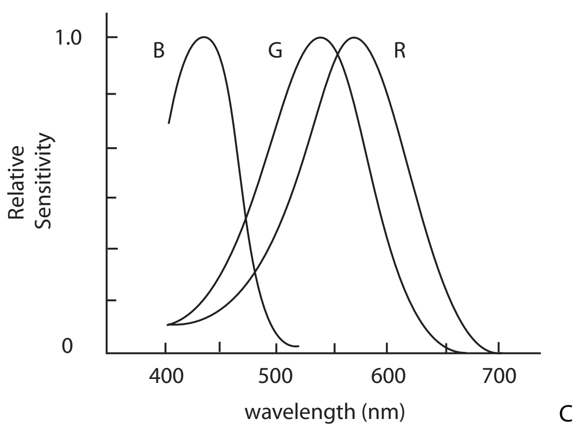


Figure 14-3. *Continued*

combinations of opsins and filtering pigments that allow for six classes of spectral peak sensitivities (Pichaud, Briscoe et al. 1999).

Opsins responding to similar light frequencies are typically closely related in protein structure and, presumably, evolutionary history (Figure 14-3A) (Yokoyama 2000). However, through gene duplication, a number of cone *opsins* evolved, and today there is great variety among vertebrates and invertebrates in their cone and rod cell arrangements, relative number, type number, and wiring to the central nervous system. Both vertebrate and invertebrate *opsin* sequence relationships suggest that rhodopsin evolved from green-sensitive ancestry.

Many amino acids vary among opsins, but a relatively small number of key amino acids appear to account for most of the variation in spectral sensitivity, at least in the repeated evolution of red-green sensitivity differences that have been studied in detail (Yokoyama 2000; Yokoyama and Yokoyama 2000). Current reconstructions based on teleost fish, amphibian, reptile, bird, and mammalian data suggest that the ancestral vertebrate had five *opsins*, *rhodopsin* plus four cone types, probably indicating tetrachromatic vision (Bowmaker 1998). The latter include, *opsins* sensitive to red, green, blue, and ultraviolet parts of the spectrum, along with a pineal *opsin* (see below) (Bowmaker 1998; Yokoyama 2000; Yokoyama and Yokoyama 2000). The gene phylogeny suggests that only the long (red-green) and the shortest (blue) wavelength *opsins* were present in stem mammals, giving them dichromatic vision, with a relative sparseness of cones in their retinas that may indicate that their early evolution was as nocturnal species. Invertebrates have rhodopsin-green and blue-UV gene classes and may only have had the two in their stem ancestors. Clearly, terms such as "tetrachromatic" are misnomers in that with four different peak sensitivities the organism can actually parse a wide range of colors, not just the four—just as we can see the entire color range with our three (red, blue, green) *opsins*.

For foraging and social signaling, some vertebrates such as birds use UV light as well as the broad spectrum of visible light. This is a potentially important fact in assessing the value of protective coloration as an adaptation against bird predation because the usual hypotheses have been based on what human investigators can see. A famous example is the case of industrial melanism, in which the rapid evolution of protection in peppered moths was said to have occurred because moths that matched the color of lichens on tree trunks on which the moths rested were not seen (or eaten!) by bird predators. Visible moths were eaten, and their unfortunate wing-color alleles disappeared along with them. However, the UV sensitivity of bird vision may have rendered moths less effectively disguised than they appear to our human eyes (Grant 1999; Weiss 2002c).

In the retina, cones tend to cluster in hexagonal groupings, surrounded by rods, but with a greater overall concentration of cones near the center and of rods near the periphery of the retina. Rod cells predominate in vertebrates that live in dim light. Interestingly, it appears that a given cone cell as a rule expresses only one type of *opsin*, although this may be less tightly regulated in species other than primates (Wang et al. 1999). Some primates, including humans, have two closely linked *opsins* on their X chromosome, that arose by gene duplication. Today, these respond to red and green ranges of the spectrum. The expression of only a single gene in a cluster is another example of *cis* allelic exclusion previously seen in immune, globin, and olfactory genes. Which of the tandem X-linked *opsin* genes is expressed in a given cone cell may involve competitive binding of regulatory proteins (Wang, Smallwood et al. 1999). This selection occurs probabilistically, so that both genes are expressed in sufficient numbers of cone cells.

X-inactivation in females ensures that only one chromosome's genes are used for red and green in any given cell and introduces some variation between each retina and among areas within the same retina. Exclusion does not occur in blue cones, which express both copies of the gene (blue *opsin* is autosomal and hence diploid). How blue vs. red-green *trans* exclusion occurs so that a given cone expresses only one or the other, when the two are on different chromosomes, is not known. In fact, some species do express both blue and green *opsins* in the same receptor cell (Glosmann and Ahnelt 2002).

Other genes are involved in a variety of "nonvisual" animal light perception phenomena. Among these are light-sensitive melanopsins found in frog skin and mammal retinal ganglia, which help regulate circadian (day-night) rhythm and pupil reflex and others. These genes are widespread in nature, and at least some are part of the *opsin* gene family. However, melanopsins in vertebrates appear more closely related to those in their invertebrate relatives; for example, in situ, their chromophore conformation is reversible and does not require helper cells as in vertebrate retinal photoreceptors. Vertebrates use genes in the *P* (*pineal*) subgroup to regulate diurnal cycling mechanism through the *retinohypothalamic tract*. A net of *melanopsin*-expressing cells in the inner mammalian retina may serve this function independent of the outer retinal photoreceptor cells (rods and cones) (e.g., Grant 1999; Provencio et al. 2000; Provencio et al. 2002). Blind subterranean mole rats appear to use this mechanism for circadian rhythms despite having degenerate eyes (Hannibal et al. 2002). Blind humans and experimentally blinded rodents may retain light-responsive nonvisual functionality. Insects with eyes removed can also perceive light intensity through a rudimentary "dermal" light sense (Steven 1963).

The Evolution of Color Vision

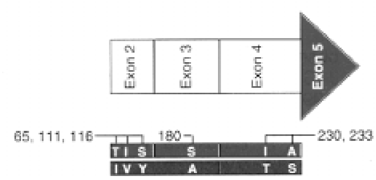
It is tempting to relate color sensitivities to aspects of the environment that may have provided the adaptive darwinian basis of organisms by selection; environmental lighting conditions, the color of food sources, mates and conspecifics, and the like have been suggested as the selective forces (e.g., Mollon 1989; Treisman 1999; Yokoyama 2000). For example, vertebrates living in dim light often mainly have long-wave (blue) sensitive cone pigments as well as rods. Selection based on visual cues, both to favor seeing ability and to favor being seen or not, depending on the circumstances, can be strong and rapid. Adaptation to wavelength sensitivity is often referred to as spectral "tuning" by natural selection (we also saw this notion in regard to olfaction). It is, however, more difficult to evaluate specifically the range and nature of what an organism can actually see, than to evaluate the specific nature of an opsin protein.

The tandem nature of human red and green pigment genes provides an interesting test case in regard to color vision. A male has only a single X, and all red- or green-expressing opsins use the respective alleles on that single chromosome. Because of X-inactivation, even a female, who has two X chromosomes, will only express one of her two red or green opsins in any given cone. This adds an element of stochastic variation among females (and between the left and right eyes of a given female) in their red-green sensitivity. Generally, this has subtle effects at most.

Mutant opsins are reasonably common. Because blue opsin is autosomal (chromosome 7), both males and females have two copies of the blue opsin, and blue color blindness is rare in either sex: even if one allele is defective, the chances are small that the other will be as well. However, red or green color blindness is not unusual in males; they only have a single red and green *opsin*, so that if either gene is defective there is no normal allele to "cover" for it. It is relatively unlikely that a female will inherit two dysfunctional red or green *opsins*. (In Hardy-Weinberg terms, if p is the defective allele frequency, her chance of having two such alleles is p^2, typically a very small value, and similar to the situation for the autosomal blue *opsin*. Even though a female only expresses genes from one of her X chromosomes in any specific retinal cell, roughly half of her cones will express her functional allele. This may affect her color sensitivity somewhat, but she will not be color blind.). By contrast, a fraction p of males will inherit the defective allele, but having only that single X have no covering protection.

Color blindness is considered a kind of disorder but that is a human subjective judgment. There is generally considerable variation in the color sensitivities of *opsin*s. The red and green *opsin* genes are closely linked, and mutation, recombination, and/or inaccurate replication produces a variety of copied, disrupted, or fused *opsins*, with substantial variation in the resulting peak sensitivities that can be explained by the nature of the mutation (Figure 14-4) (Neitz et al. 1996; Neitz et al.

A



	λ_{max} (nm) Ref. (15)	Ref. (16)
	556.7	563
	552.4	556
	553.0	559
	549.2	555
	544.8	551
	ND	554

B

	Participant	Gene arrangement from Southern blots	Codon 180 (A, Ala; S, Ser) 1st gene	Downstream Hybrid genes	Differences? (Y, yes; N, no) Exons 4	3	2	Gene arrangement (this study)	Predicted spectral separation	*D* of most difficult plate read
Dichromats	d2		A	-	-	-	N		0.0	>0.081
	d1		S	-	-	-	N		0.0	>0.081
Very severe	189		A	S	N	Y	Y		-0.6† to 3‡	>0.081
Severe	187		S	S	N	N	Y	65, 111, 116	3.7† to 4‡	0.081
	021		A	A	N	N	Y		1‡ to 3.6†	0.081
	012		A	A	N	N	Y		1‡ to 3.6†	0.072
	022		A	A	N	N	Y		1‡ to 3.6†	0.072
	188		S	S	N	N	Y		3.7† to 4‡	0.072
	176		A	A	N	N	Y		1‡ to 3.6†	0.072
Mild	191		A	A	Y	N	Y*		5‡ to 7.6†	0.032
	185		A	A	Y	N	Y*		5‡ to 7.6†	0.032
	027		A	A	Y	N	Y		5‡ to 7.6†	0.032
	028		S	A, S	N	Y	Y*		7.9† to 8‡	0.032
Very mild	009		S	A	Y	Y	Y*		11.9† to 12‡	0.022
	186		S	A	Y	Y	Y*		11.9† to 12‡	0.022
	017		S	A, S	Y	Y	Y*		11.9† to 12‡	0.022
	190		S	A, S	Y	Y	Y*		11.9† to 12‡	0.022
	182		S	S	Y	N	Y*		9‡	0.022

Figure 14-4. The type of red-green color blindness is determined by the arrangement and mutation of the X-linked *opsin* genes. This table shows phenotypes associated with mutant, deleted, or fused genes that have been observed. Spectral sensitivities were measured in two ways. Reprinted from Neitz et al. (Neitz et al.) with permission; see the original for details.

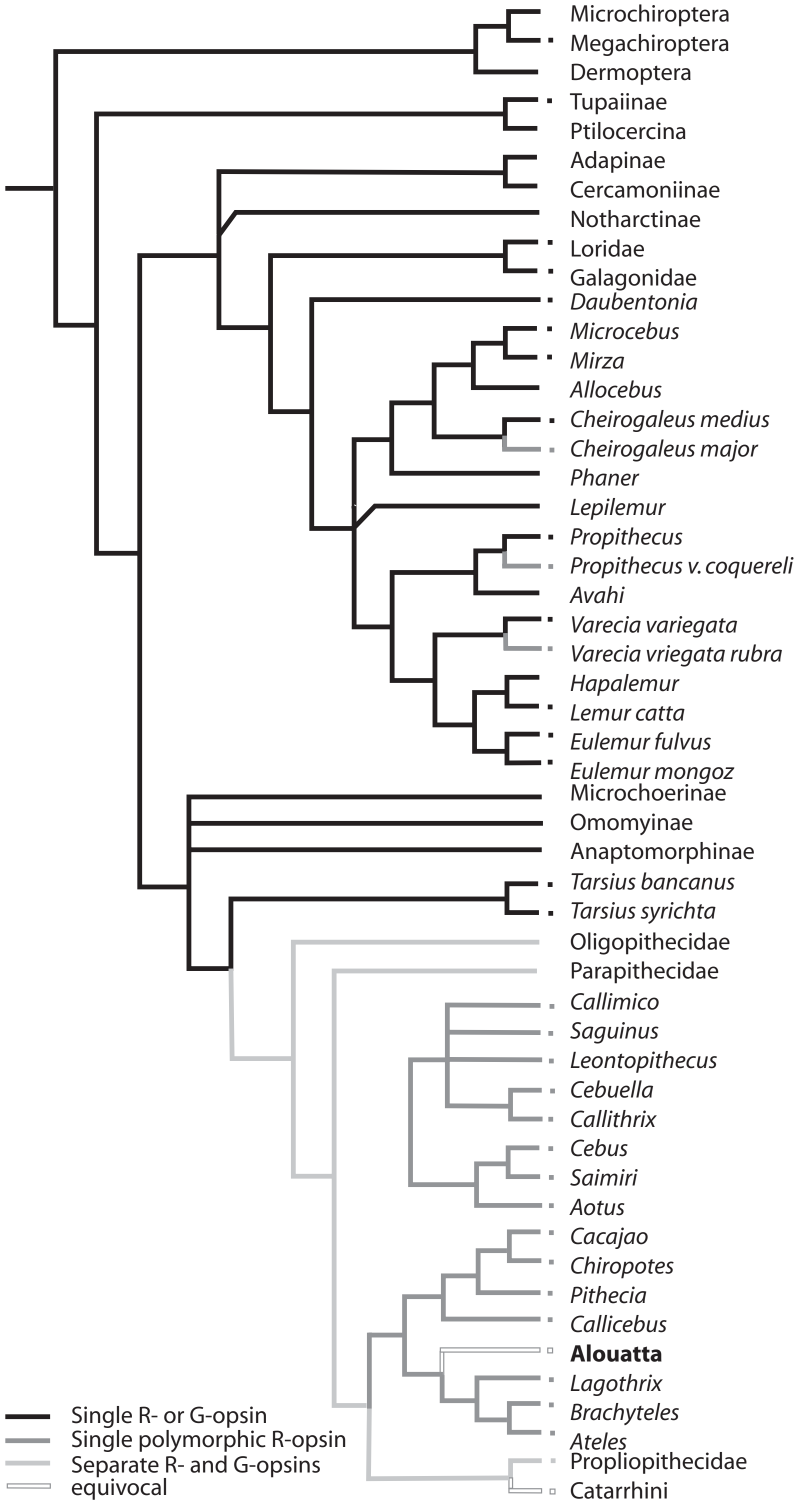
1995). The amount of variation probably should temper what we consider "normal" and is also relevant to the age-old philosophical question of whether all people see the same colors. Not only is there variation in the genes themselves, but there is a stochastic element in the *opsins* expressed by any given cone, meaning that not even the left and right retinas of the same person are exactly the same. Of course, even if they were identical, this cannot answer the question of whether people who detect the same thing *perceive* things similarly, which probably remains as philosophical as ever.

The patterns of red-green X-linked *opsins* found among primates has been used to infer whether their common ancestor had di- or trichromatic vision. Old World monkeys and apes have trichromatic color vision as we do, with both red- and green-sensitive *opsins*, suggesting that this was probably the condition or our shared ancestor. However, most New World monkeys have only one X-linked *opsin*, and there's a twist. The single X-linked *opsin* is polymorphic in many species that have been studied, and the observed alleles are variously sensitive to red- or green-range light. Depending on the alleles' frequencies, a female's genotype may be **RR**, **RG**, or **GG**, whereas a male is either **R** or **G** sensitive. The relative frequencies of these genotypes will be determined by the allele frequencies (e.g., $p_{\boldsymbol{R}}^2$, $2p_{\boldsymbol{R}}(1 - p_{\boldsymbol{R}})$, $(1 - p_{\boldsymbol{R}})^2$ in Hardy Weinberg Equilibrium, using $p_{\boldsymbol{R}}$ for the frequency of the **R** gene).

What accounts for this widespread variation in the peak sensitivities and polymorphism among monkeys with but a single X-linked *opsin*? The phenomenon provides an opportunity to examine the nature of darwinian explanations (e.g., Mollon 1989; Nathans 1999; Yokoyama 2000). The typical suggested scenario is that there has been selection for ability to detect fruit or other plant foods (for example, leaves whose color indicates taste or toxicity) in the dappled background of tropical forests. Animals who could see colors appropriate to their dietary staples had higher fitness, over time spectrally "tuning" their opsins' peak sensitivity frequencies. The polymorphic nature of the single X-linked opsin cannot be quite so easily explained. Why was it not "tuned" to an optimal frequency? Instead, as it is now, members of the same population are variably sensitive to color in a strange way: males are differently sensitive and to only one of the available colors, in proportion to the **R** and **G** allele frequencies. Females vary in proportion to the respective genotype frequencies (as above), the **RR**'s and **GG**'s being dichromatic and only the **RG**'s being trichromatic. How would selection produce this?

Assuming that trichromacy must be evolutionarily better and hence favored by selection (from a human-centered perspective), it has been suggested that the (at most) few trichromatic females in any small local monkey troop may have led their peers to food sources. Otherwise, only a single best allele would have been favored by selection. If the selective scenario is correct, one might expect a kind of balancing selection to have optimized the frequency of female heterozygotes, which will occur with allele frequencies close to 0.5 (which maximizes $2p_{\boldsymbol{R}}(1 - p_{\boldsymbol{R}})$); this does not seem generally to be the case (Cropp et al. 2002; Heesy and Ross 2001), although available samples are inadequate for a definitive answer.

Perhaps there is a simpler and more natural explanation that is also more parsimonious (though, as noted early in this book, there is no reason that evolution needs to have followed the simplest path). It is useful first to note that the placement of cones in the retina may be related to spatial as well as, or even rather than, color perception. Also, even fully trichromatic individuals use other features of light such as shading and lightness in image discrimination perception. Figure 14-5 shows the


Microchiroptera
Megachiroptera
Dermoptera
Tupaiinae
Ptilocercina
Adapinae
Cercamoniinae
Notharctinae
Loridae
Galagonidae
Daubentonia
Microcebus
Mirza
Allocebus
Cheirogaleus medius
Cheirogaleus major
Phaner
Lepilemur
Propithecus
Propithecus v. coquereli
Avahi
Varecia variegata
Varecia vriegata rubra
Hapalemur
Lemur catta
Eulemur fulvus
Eulemur mongoz
Microchoerinae
Omomyinae
Anaptomorphinae
Tarsius bancanus
Tarsius syrichta
Oligopithecidae
Parapithecidae
Callimico
Saguinus
Leontopithecus
Cebuella
Callithrix
Cebus
Saimiri
Aotus
Cacajao
Chiropotes
Pithecia
Callicebus
Alouatta
Lagothrix
Brachyteles
Ateles
Propliopithecidae
Catarrhini
Single R- or G-opsin
Single polymorphic R-opsin
Separate R- and G-opsins
equivocal

color sensitivities of the red/green *opsin* genes in primates (Heesy and Ross 2001), but the distribution of peak sensitivities among these alleles (Figure 14-6) does not suggest tight selection for red-centered and green-centered alleles but rather suggests that the alleles are spread across the entire red-green part of the spectrum. Selection might only have ensured that an animal would have opsins sensitive to *some* part of the red-green range rather than any *particular* one.

Because of codon redundancy, the polymorphisms in the key amino acids that have mainly been responsible for peak sensitivities in this range (Nathans 1999; Yokoyama 2000) require only a single nucleotide change; therefore, recurrent mutation to similar alleles in different species is not implausible on an evolutionary time scale. However, the appearance of similar mutations in multiple lineages might just reflect genetic drift on an ancient polymorphism in the ancestral lineage. Consistent with this, in both squirrel monkeys and humans, there is less variation in the single blue opsin than in the X-linked genes (Shimmin et al. 1998), suggesting that the blue-sensitive gene has been under stronger selective constraint than the polymorphic X-linked gene in squirrel monkeys (Cropp, Boinski et al. 2002).

The presumed need for precise spectral tuning may be less than is typically thought if one considers the many ways in which visual signals are interpreted, that opsins have sensitivity at wavelengths around their peak sensitivity, and that colors are not perceived strictly on the basis of peak sensitivities. For example, vertebrate retinas compare the relative strength of signal detected by different parts of the system (e.g., from two types of nearby cone cells) in various ways, even before signal is sent to the brain (e.g., Nathans 1999). In addition, what an animal "sees" is affected by its past experience.

In a species with *two* X-linked opsins, it is plausible that selection could keep their sensitivities separated as part of their normal repertoire (gene duplication of X-linked opsins occurred at least twice in primate lineages; New World howler monkeys have trichromacy similar to that in Old World monkeys and apes). But these systems, too, have considerable normal variation, suggesting that selection has, at most, not been too stringent.

The evolution of color vision can also illustrate the potential importance of organismal as opposed to classical natural selection. Some fish live deep in the sea where there is little light and what gets through is of short wavelength. Coelocanths use a combination of two rhodopsin-related opsins that are sensitive only to such light (Yokoyama et al. 1999). The fish are adapted to life in the depths. A fish will

Figure 14-5. Phylogeny of primate X-linked color vision capabilities. Phylogenetic tree of primates showing species that have alleles conferring only red (R) or green (G) sensitivity, a single gene with observed polymorphisms conferring alleles with R and G sensitivity, two X-linked genes (R- and G-sensitive), or some less-clear sensitivity. One can infer that the original primates may have been "dichromats" but with a single gene that was polymorphic for peak sensitivities in the red-green range. Gene duplication led to two X-linked genes, one specialized for red and the other for the green ends of this part of the spectrum, enabling "true" (human-like) trichromacy in the ancestor of Old World monkeys and apes. Gene phylogenies suggest that this occurred independently in the lineage of howler monkeys (*Alouatta*) in the New World. Taxa with no symbol indicate no data available. Redrawn from (Heesy and Ross 2001) with permission.

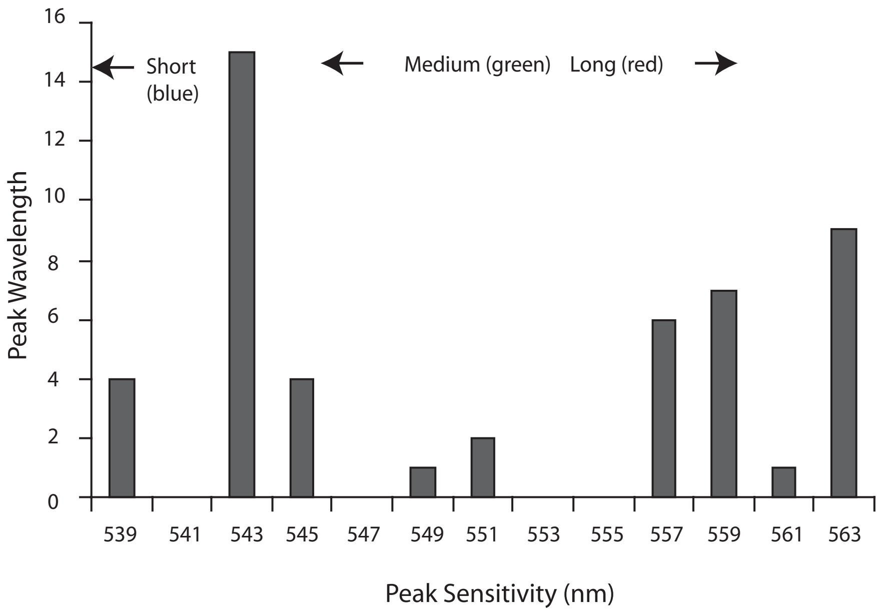


Figure 14-6. Peak spectral sensitivities in X-linked primate *opsin* genes. Height of each bar is the count of alleles of that peak sensitivity observed in tested primate species (that is, these are not allele frequencies: each observed allele of that sensitivity category incremented the category count by 1, regardless of the frequency of that allele in the species). Reprinted from (Weiss 2002c) with permission.

move around and seek food wherever it might be detected. Fish that by mutational chance were able to see in dim low wavelength light might gravitate toward greater sea depths and mate with others they find there. This scenario would require choice of available subenvironments, but not classical darwinian selection within a single population, even though color vision is important in animal ways of life (Weiss 2002c).

Cichlid fish have in some instances become highly speciose (numerous highly diverse species, perhaps including incipient species), even within what appears to be a continuous environment, including the large Lake Victoria in Africa (Terai et al. 2002) and crater lakes in Central America (Wilson et al. 2000). The various species are characterized largely by color variation and in some instances trophic variation (e.g., structural variation in their eating apparatus that suits their local subenvironment within the lake). This has presented the problem of sympatric speciation (Chapter 2), that is, how species can form without a physical mating barrier. One explanation is organismal selection, perhaps related to sexual selection; fish choose regions of the lake suited to their traits and mate with others in the local subenvironment.

Organismal selection can reflect existing genetically determined traits like color and color vision (on which, for example, color-based sexual selection may depend). Variation in the long-wavelength sensitive *opsin* gene in the Victoria cichlids is associated with fish color and/or local food species. The locations of at least some of the amino acid variants observed are those shown to have spectral sensitivity effects as described earlier (Figure 14-3B). This is at least suggestive evidence for color-based sympatric speciation (Terai, Mayer et al. 2002), although the pattern of polymorphisms, specific *opsin*-environment correlations, and means other than direct color sensation for picking mates and finding food might alter the explanation. In any case, there are many similarities between this variation and that in primates, that may provide an excellent "laboratory" for the challenge to interpret genetic patterns to the nature and extent of selection events that may be responsible.

The Origins of Photoreception

Some unicellular organisms, such as Euglena or algae such as *Chlamydomonas*, *Volvox*, or *Haematococcus*, have a light-sensitive *eyespot*, a region with a higher sensitivity to light than the rest of the cell. The eyespot is located between the cell's two flagella and its equator; it sends information about the quality and intensity of surrounding light to the flagella, enabling the organism to orient vis-à-vis the light, in a process called *phototaxis*. These eyes have optics, photoreceptors, and a primitive signal transduction mechanism, with layers of hexagonally packed lipid globules that reflect more or less light, depending on the number of layers (Hegemann 1997). Signal transduction uses retinal-containing photoreceptors, two rhodopsins in *Chlamydomonas* (Ebrey 2002; Sineshchekov et al. 2002).

Bacteria have rhodopsin-like molecules with seven transmembrane domains but little sequence resemblance to the corresponding eukaryote genes. Eubacteria and cyanobacteria are photosensitive or photosynthetic and produce phytochromes that work via phosphorylation signal transduction. At least some of these appear to be ancestral and related to mechanisms found in plants. Several classes of genes are involved, or possibly involved, some of which employ attached chromophore molecules (Herdman et al. 2000). This suggests that eukaryote and prokaryote photosensitivities have at least some common origins, although new mechanisms seem to have evolved in multicellular organisms.

PHOTORECEPTION IN PLANTS

Plants are always "seeking the light desiring the sunshine, . . . unconscious of either" (Joseph Conrad, *Almayer's Folly*, 1895). Plants evolved from organisms like simple photosynthesizing cyanobacteria into large organisms with many leaves that allow the plant to maximize photosynthetic activity. They depend on light for many aspects of their life cycle, including *photosynthesis*, *photoperiodism*, *photomovement*, and development, flowering, and seed germination. Although they are sessile, they have become exquisitely adept at responding to and exploiting variation in light duration, quality, quantity, and direction (Neff et al. 2000), and in the kind of intra-organismal independence mentioned in Chapter 8, different parts of a plant can respond independently (only some branches need move in a given direction).

Plants can respond with both positive phototropism, or movement toward the light by the leaves and stems, and negative *phototropism*, or movement away from light by the roots, responses that are mediated by the hormone auxin (light is only one factor in the direction of growth of a seedling however; shoots grow upward even when a plant is growing in complete darkness and roots downward, due to positive and negative *gravitropism*). Chloroplasts in the leaves move toward light to maximize the capture of photons, and the *stomata* (pores in leaves) respond to light by opening (see Figure 14-7A for a drawing of a chloroplast). Plants respond to more than simply the presence or absence of sunlight, but also to wavelength, intensity, and directionality of light as well as length of day. They use these cues to modulate their growth and development, in a process called *photomorphogenesis*.

A number of photoreceptors and photopigments have evolved to exploit whatever light they can get. These are coded by genes in three classes related to cryptochromes and phototropins (UV-B, UV-A, blue), and phytochromes (red/far-red), inducing expression of the appropriate response genes (e.g., Quail 2002).

Homologies have been found between some plant and animal photoreceptors. Among other light-induced responses in plants, cryptochromes are involved in the regulation of circadian rhythms and the timing of flowering. A diverse family of photoreceptors, *cryptochromes*, are found throughout higher eukaryotes, both plants and animals, including humans (Lin 2002), where they are involved in regulation of circadian rhythms. Cryptochromes share protein structure and composition of the chromophore with microbial DNA photolyases, or DNA repair enzymes. Phytochromes homologous to those in plants have been found in cyanobacteria (Quail 2002), and the "photosensory domain" is highly conserved between phyla. *Opsin*-like photoreceptors may also be found in plants; they are in green algae, and retinal has been purified from plants.

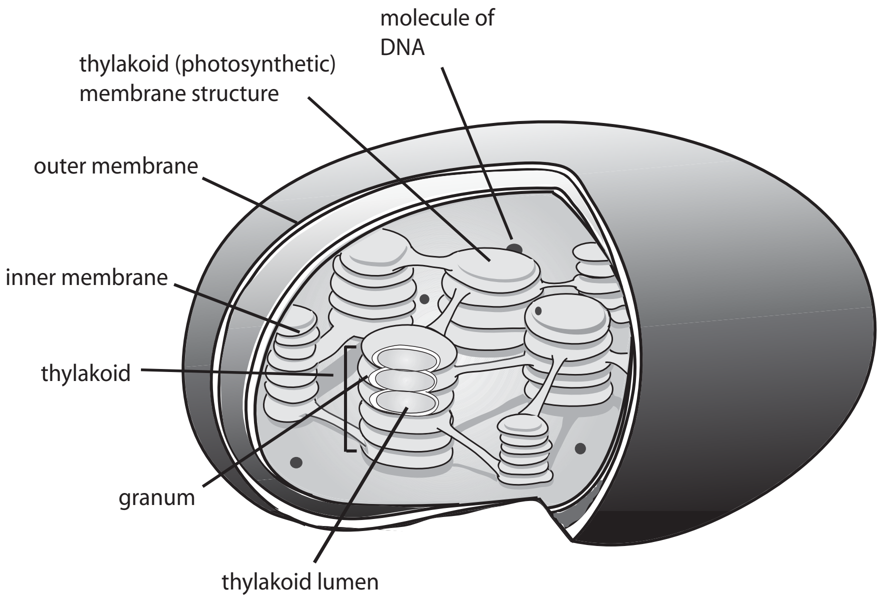
Plants build carbohydrates from light energy and raw materials from water and CO_2. In the natural cycle of the biosphere, they obtain hydrogen from water and carbon and oxygen from the CO_2 expelled into the atmosphere by animals. The carbon and O_2 are taken into leaves through stomata, and water is taken up by roots and circulated up through the vasculature to the leaves, where the light energy is used to break down the raw materials into hydrogen, carbon, and oxygen, which are then recombined into carbohydrates. The plant subsequently uses these carbohydrates as energy, stores them, or builds them into more complex molecules such as oils or proteins.

Photosynthesis is initiated by "visible" light in the 400–700 nm range, primarily red and blue light. This is a complicated electrochemical process and will only be sketched out lightly here. Light is absorbed by *chlorophyll a* and *b* and *carotenoids*, which are pigment molecules in the chloroplasts of green plants. These pigments absorb green light poorly—it is the reflection of the light at the green wavelengths that causes chlorophyll-laden plants to appear green.

There are 200 to 300 pigment molecules bound to protein complexes in the photosynthetic, or *thylakoid*, membrane of the chloroplast (Figure 14-7B), and they form an *antenna system* that absorbs light energy and transfers it in the form of excited electrons to a chlorophyll molecule that serves as a *reaction center*. The reaction center then transfers its high-energy electrons to an acceptor molecule in the electron transport chain. High-energy electrons are then transferred through several membranes, for synthesis of ATP and the electron-carrier NADPH, which is used later for the synthesis of carbohydrates from CO_2 and water.

Photoreceptors to UV-B light are important in the development of seedlings (Fankhauser and Chory 1999). *Cryptochromes* (CRY1 and CRY2) and *nonphototropic hypocotyls1* (NPH1) detect UV-A and blue light, respectively. The cryptochromes are important in seedling development as well as the transition to flowering, whereas NPH1 and other factors are important for phototropism. Phototropins, of which there are at least two in the laboratory model plant (the mustard family member *Arabidopsis*), control growth and movement toward the light source.

A.



B.

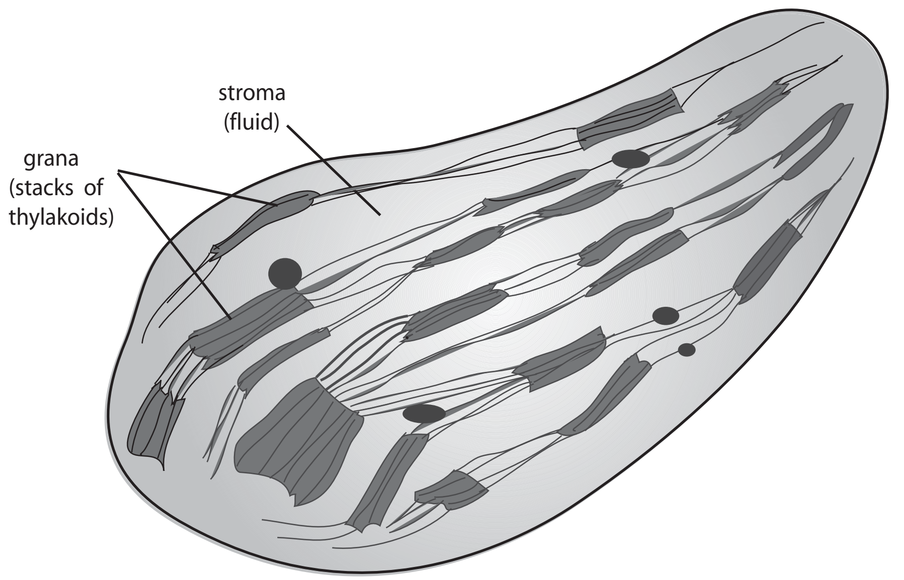


Figure 14-7. Plant chloroplasts. (A) Chloroplast structure; (B) cross section.

Phytochromes are important in many aspects of plant development (Fankhauser and Chory 1999). These are used in the detection of neighboring vegetation and the initiation of shade avoidance, the measurement of day length and light exposure and the consequent initiation of reactions to day length and duration of light exposure so that flowering occurs at the appropriate time and the like, and the control of the timing of reproduction.

Phytochromes are encoded by small and evolutionarily very old multigene families (Quail 2002), with structural similarity to prokaryotic histidine kinases, although the plant proteins phosphorylate the amino acids serine and threonine rather than histidine or aspartate. *Arabidopsis* has five known phytochromes, A to E, which play a role in plant development at all stages. Photoreceptors for light essential to growth apparently have multiple interactions that provide complementary or redundant ways through common transcriptional pathways. Phytochromes are pigmented proteins that can convert between two spectrally distinct forms, Pr, a red-absorbing form, and Pfr, a far-red-absorbing form. Photoconversion of Pr to Pfr induces a range of developmental responses in the plant, which have been shown to halt when Pfr converts back to Pr, showing that Pfr is the active form of the receptor.

Biological pathways in plant light response involve familiar elements of gene regulation (Quail 2002). For example, although plants have far fewer than animals, some G proteins have been identified as components in the phytochrome signaling pathway. They induce chlorophyll biosynthesis, among other functions, as well as the expression of many light-regulated genes. Light-induced changes in ion transport across the cell membrane, and other kinds of signal transduction, are apparent, but they have not yet been characterized in detail (Fankhauser and Chory 1999; Quail 2002). Essentially, many aspects of plant metabolism, growth and reproduction rely on light and plants have elaborate gene-regulation signal transduction mechanisms that respond to different aspects of received light, which involve several gene families specific to plants.

THE EVOLUTION OF EYES

Light can be detected in many ways, but light detection is not exactly what we mean when we think of *vision*. Vision is more complex because it implies the processing of intricate *spatial* patterns of incoming light along with its intensity and aspects of its spectral characteristics. In higher organisms, vision involves interpretation of complex signals by the central nervous system and translation of these into decision-making that usually includes a directed response, such as by muscular instructions directing movement. Before the response, the patterned aspect of the signal must be *detected*.

Because light travels in straight lines, an organism receives spatially organized light energy that is a map of the objects from which the energy comes. To see pattern, the organism must have some corresponding form of neurological spatial "map". This is in contrast to the much less spatially organized nature of olfaction, sound, or (sometimes) heat, the essentially nonspatial nature of immunological information, and the temporal arrangement of daylight or air temperature. For spatial processing, there must be a receptor map. Interestingly, as we will discuss in Chapter 16, even senses that do not depend on spatial perception use spatial maps in the brain to decode signal. This is probably a function of the fairly uniform histological, laminar, and synaptic organization of the segments of the brain that decode sensory input.

The hundreds of ommatidia that comprise the compound eyes of insects provide a fixed receptor array, each with its own neuronal connection to the brain. In camera-type eyes, such as those in mammals, the rods and cones in the retina are fixed in location and their corresponding neurons can be connected to the brain in a way that directly conserves spatial relationships, or at least the brain can reconstruct them because the signals from the same matrix position in the retina (or compound eye) are fixed. In mammals, unified stereoscopic vision further integrates the slightly different images from right and left eyes. An important part of this is that there need be no specific a priori aspect of each ommatidium or retinal cell, relative to images that are going to be detected: the organism is set up to detect *any* spatial light pattern. It does not need, for example, built-in diagrams of its predators, food, or landscape panoramas.

The Multiple Evolutionary Histories of Eyes

There are many ways in which eyes that are used for spatial inference can be constructed. Figure 14-8 shows some of them. The arrangement and number of eyes vary extensively, even when considering just bilateral cerebral eyes (e.g., see Arendt and Wittbrodt 2001). These include relatively simple larval eye spots in many branches of animal taxa, to more elaborate eyes formed by an array, usually a cup, of retinal cells. In some eyes, this has deepened and been closed except for a small opening, which like a pinhole camera allows a single image to be received in an array of photoreceptor cells on the inside. More precise (in human terms, at least), focused images are formed by eyes with lenses in this aperture, as found in mammals. Each ommatidium in an insect compound eye is a simple eye in itself, with spatially distributed neurons from its individual receptor cells receiving similar signals that are sent to similar regions of the brain.

Comparative anatomy and taxonomy have shown that eyes have evolved independently many times (a classic analysis suggests 40–65) (Arendt and Wittbrodt 2001; Salvini-Plawin and Mayr 1961). For that reason, eyes have long been used as exemplars of parallel evolution whose distribution is in the phylogeny of animals; that is, eyes are examples of analogy rather than homology because similar eyes appeared intermingled among taxa that do not appear to have shared one type of eye in their common ancestor. The evolutionary complexity of eyes perplexed Darwin (*Origin of Species*), who confessed that he could hardly imagine something as structurally complex as an eye evolving and reevolving independently so many times. A simpler explanation would be that animal eyes were homologous rather than analogous, but to rescue such a notion required a common ancestral eye that could have evolved into the diverse descendant forms seen today. Darwin hypothesized a primitive animal eye consisting of two cell types, a photoreceptor and a nonphotosensitive pigment cell with a substance that shades the light so that its direction could be detected, all perhaps encased in some kind of translucent covering. Primitive eyes resembling this structure exist in various taxa, but it was the remarkable discovery that genes fundamental to vision were widely shared among animals that made the idea credible.

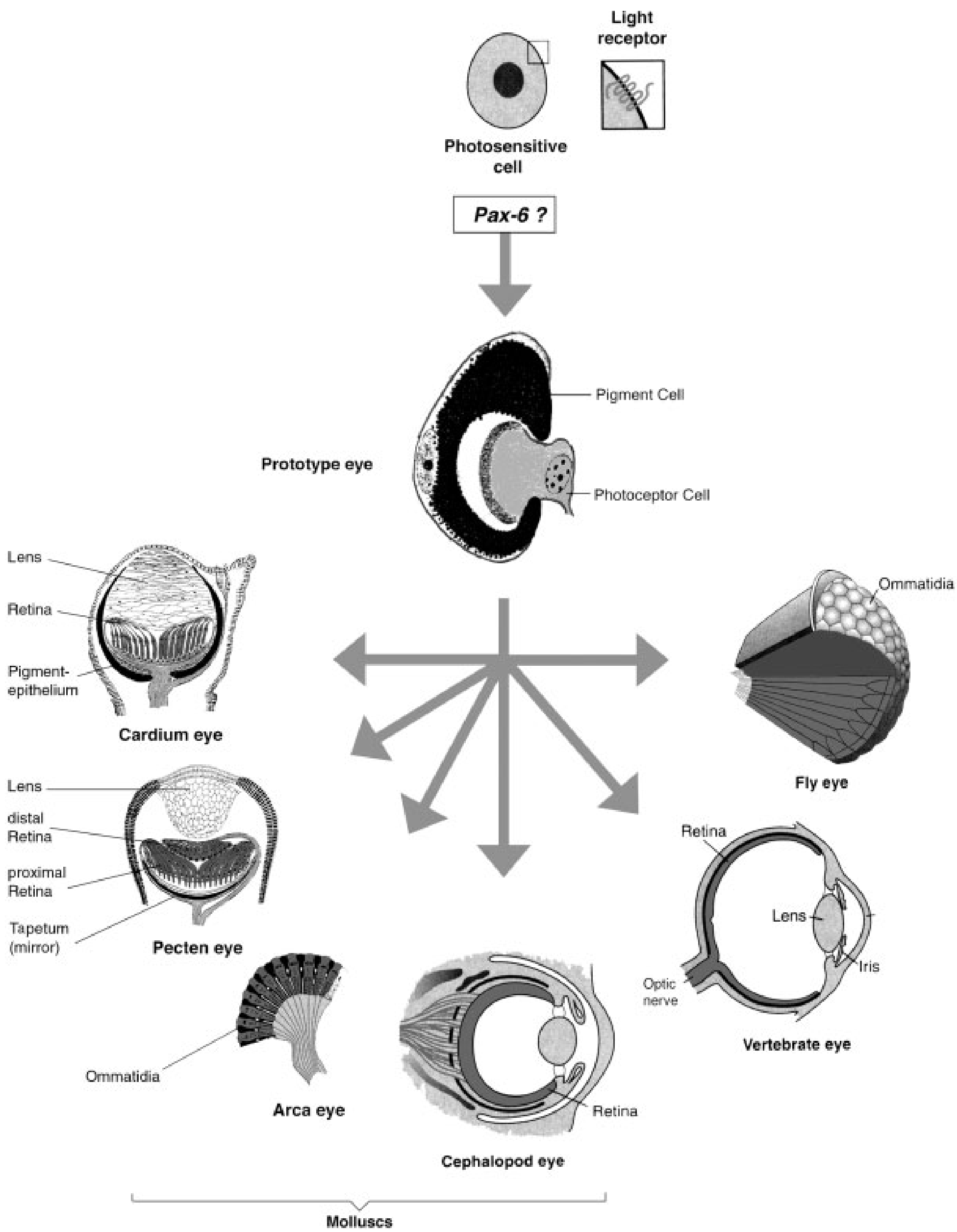


Figure 14-8. Evolution of diverse eyes from a common rudimentary photoreceptor, following Darwin's original notions; variation even within mollusks is shown, as is some of the recurrent evolution of eye types. From (Gehring 2002) with permission.

Pax6 and Conserved Homology Among Eyes

The transcription factor (TF) *Pax6* was one of these genes. Expressed early in development, *Pax6* appears to serve as a selector for the initiation of differentiation cascades involved in eye development in most metazoans tested. Evidence showing this has come from expression experiments, as well as from natural or artificial mutational studies in mouse, humans, flies, and other animal species (e.g., Gehring 2002; Gehring and Ikeo 1999; Punzo et al. 2001). Rhodopsins and intracellular light-sensitive mechanisms exist in bacteria, suggesting that light sensitivity evolved before multicellular life. Furthermore, *Pax6* is expressed in several other later-stage eye structures and even regulates the expression of at least some opsin genes, like insect rhodopsin, expressed in mature photoreceptors.

The phylogenetically deep sharing of *Pax6* and *rhodopsins* can be seen as developmentally bracketing the morphological and sensory aspects of light reception among animals. This might seem to provide Darwin with the elements of his suggested common Precambrian origin for animal eyes (Figure 14-8), (Gehring 2002; Gehring and Ikeo 1999). But the sharing of a couple of genes does not provide obvious ease for his concern as to how similar types of complex eye *morphologies* have evolved several times independently or how similar eyes can evolve through different *developmental* pathways even in related species and from very different tissue contexts (e.g., Hall 1999). It is these morphologies that account for how light is used in various taxa. One explanation is that developmental patterning mechanisms were recruited, or *intercalated*, between the initiation of visual systems by *Pax6* and the later induction of *opsin* gene expression (e.g., Gehring 2002; Gehring and Ikeo 1999).

Different types of eyes involve different morphogenic processes, such as placode formation, invagination (e.g., of an optic cup), and periodic patterning (e.g., ommatidia). As reviewed in Chapter 9, these are standard parts of the developmental repertoire of complex organisms, which during evolution could have been invoked in new contexts in between the first expression of *Pax6* and the later expression of photoreceptors. The various forms of eye could have evolved sometimes inserting shared but other times different morphogenic mechanisms. Because the developmental mechanisms are resident parts of the animal developmental toolkit, such evolution could be relatively rapid (Pichaud et al. 2001).

Relevant to this is that eyelike structures can be induced experimentally in insects and vertebrates by ectopic application of Pax6 protein (Gehring and Ikeo 1999), in locations where relevant signaling molecules are already expressed (e.g., Kumar and Moses 2001; Pichaud, Treisman et al. 2001), but in contexts that normally do not develop into eyes (e.g., other imaginal disks in insects). Pax6 from one species can even induce such development in another.

These results suggest that different eyes share genetic homologies and that they did not entirely evolve independently as analogous organs. This is interesting because the intercalated mechanisms are also used in other structures, so that eyes would in part be homologous to antennae, feathers, and teeth (which also use Hedgehog, Bmps, and the like). The basic elements of the logic of the rapid evolution of diverse eyes are also found in other systems, including olfaction and various aspects of developmental polarity, repeated structures, and the like that we have seen are so characteristic of evolution.

Crystallins and the Nonconserved "Homology"

Another interesting aspect of eye evolution concerns the *crystallins*, major proteins found in vertebrate lens and corneal tissue. An important characteristic of crystallins is their light transparency properties when desiccated and compacted. However, genes coding crystallins did not evolve to serve a visual purpose. Instead, different species have opportunistically recruited different, often wholly unrelated proteins to use as their lens or corneal proteins (Tomarev and Piatigorsky 1996). These proteins typically also serve other functions in the same organism, such as heat shock response or housekeeping enzymes. Lens and cornea develop embryologically from the same tissue, but in a given species the corneal crystallins are different from those used in the lens. Many crystallins include *Pax6* REs in their 5′ flanking regulatory regions, and experiments show that the enhancers drive appropriate eye-specific expression, often across species (e.g., the enhancer from a chick can induce lens expression in a mouse). *Sox2* and retinoic acid receptor pathways may be alternative or additional parts of the shared regulatory mechanism (see below). However, unlike the conservation of the *Pax6* regulatory pathway itself, in the case of crystallins, the *function* rather than the specific gene is conserved. Phenogenetic drift has replaced one gene with another in different lineages that have shared the form of their eyes since a common ancestor.

EYES EXEMPLIFY IMPORTANT LESSONS ABOUT RECONSTRUCTING THE EVOLUTION OF COMPLEX STRUCTURES

There has been enthusiastic acceptance of the *Pax6/opsin* solution to Darwin's problem about the repeated, apparently independent evolution of complex eyes. The intercalary evolution scenario has seemed quite convincing, and similar relationships are being identified for other traits—indeed, in some ways that is becoming the rule rather than the exception, as we have seen. However, the picture is more complex and shows some of the dangers inherent in accepting evolutionary reconstructions that are too tidy. It is instructive to see what some of these problems are.

The "Master"-y of *Pax6*

It is clear that *Pax6* is important in eye development. No one thinks that genes act in isolation, but the idea that *Pax6* is *the* master selector for vision has gained common currency. There are many reasons why we like simple, grand explanations and metaphors (and usually they fit other things in our culture—this may account for the reliance on competition as a mechanism and the notion of "master" genes—the CEOs of development; we have noted similar caveats in regard to terms like "organizer" and "selector"). However, oversimplification gives a misrepresentative picture of eye biology and of evolution in general. Remember the complexity of developmental and signal-transducing pathways, and the ways that they are often incompletely documented (Chapter 7). Few if any pathways act alone.

The original *Eyeless* mutant was one of the classic traits of early fly genetics and ultimately led to the isolation of the homeobox/paired-box *Eyeless* (*Ey*) TF—a homolog of *Pax6* in vertebrates. However, *Pax6* is only one of several (known) genes that are roughly comparable in their importance in eye development (*Ey*, *Toy*, *So*, *Eya*, *Dac*, *Eyg*, *Opt*, *Hth*, *Tsh*). These genes can induce ectopic eyes much as *Pax6* can (e.g., Bessa et al. 2002; Goudreau et al. 2002; Treisman 1999). The gene *Twin of eyeless (Toy)* is a *Pax6*-type gene closely related to *Ey* and perhaps evolutionarily closer to vertebrate *Pax6* than *Ey* is itself. Experimental ectopic expression of *Ey*, or of *Toy*, results in the formation of ommatidia on the legs, wings, and antennae, but *Toy* is expressed developmentally upstream of *Ey* and directly regulates the latter's eye-specific enhancer.

Other TFs, including *Sine oculis* (*So*), *Eyes absent* (*Eya*), and *Dachshund* (*Dac*), interact and are essential for development of compound eyes (Cutforth and Gaul 1997; Czerny et al. 1999; Punzo et al. 2002). Loss of *So* or *Dac* results in loss of eye structures. Ectopic expression of *Dac* induces expression of *Ey*, and the subsequent formation of an eye, although at lower efficiency than with ectopic expression of *Ey* alone. *So* probably acts downstream of both *Dac* and *Ey*, as it is not sufficient for eye formation. *So* is one of the class of TFs called *Six* genes (Pineda et al. 2000) whose members interact with *Pax6* at different stages of eye development in different species, including planarians, insects, and vertebrates, and can compensate to some extent for each other. The fly paralog *Optix* can generate eyes in an *Ey*-independent way (Seimiya and Gehring 2000).

Several homologs of these genes are also expressed in vertebrate eye development, but comparison is difficult because vertebrate eyes develop differently, and vertebrates also often have more members of the same gene families than do flies. For example, eyes develop from a single imaginal disk primordial tissue in flies and through interaction of two tissues (an ectodermal lens placode and the underlying optic vesicle of the brain) in mammals. *Pax6* is higher in the eye pathways in flies than in vertebrates; for example, optic vesicles form in *Pax6*-mutant (*Sey*) mice (Harris 1997). *Pax6* and *Six3* interact in a quantitative, mutually inductive way in eye development, suggesting that neither acts as "the" control gene.

In mice, Pax6 and Sox2 protein bind together to induce lens development, and cooperative enhancers for these genes are found flanking the δ-*crystallin* gene, which is subsequently expressed in the lens in mice and chickens (Furuta and Hogan 1998; Kamachi et al. 2001). Similar interactions between *Pax* and *Six* homologs are involved early in fly eye development. *Pax*-coded protein may provide a DNA-binding domain and Six a domain that facilitates transcriptional activation. Expression of the signaling factor *Bmp4* in the optic vesicle is also involved, but this appears to function independently of *Pax6* (Furuta and Hogan 1998). *Pax6* also interacts with *Pax2* and at least one other related gene in eye development (Papatsenko et al. 2001).

The homeobox gene *Rx* is neurally expressed earlier than the eye in vertebrates, and experiments show that the development of retinal tissue (a late stage) depends on it (Mathers et al. 1997). In at least some fish, a TF *Rx3* is needed to initiate the early stages of eye development (Loosli et al. 2001). There is a homolog in flies (*DRx*), but it is not expressed in eye imaginal disks.

It is probably better to view these various genes, including *Pax6* but perhaps others not yet known, as forming a horizontal network of control rather than a vertical Master-driven hierarchy (Kumar and Moses 2001; Pichaud, Briscoe et al. 1999; Pichaud, Treisman et al. 2001). Inactivating this gene can leave eyes unimpaired. This is consistent with the general nature of gene activation, which requires the binding of complexes of multiple regulatory proteins. A selector ("master" regulatory) gene needs a tissue environment previously made ready, which in the case of eyes is due to expression of growth factors such as *Bmp*s (in flies, the homolog *Dpp*), *Egf*s, and *Hedgehog*, or *Notch* signaling, and the absence of inhibiting signals such as *Wnt* (in flies, the homolog *Wg*), among other factors. Similar factors are needed to specify eye vs. antenna fate in flies, for example, from a common starting primordium, and their expression is not restricted to eye primordia. Experimentally induced ectopic *Ey* expression only induces eyes in prepared soil, so to speak. *Pax6* is an early neural gene in many ways that would be atop more than just eyes in any case. A gene *Tsh* (*Teashirt*) is not needed in normal fly eye development but can trigger ectopic eyes experimentally, presumably by interacting with other genes.

The Necessity for *Pax6*

There are other ways in which the universality or necessity for genes like *Pax6* has perhaps been overstated or oversimplified (Gehring 2002; Gehring and Ikeo 1999; Kumar and Moses 1997; Kumar 2001). *Pax6* is clearly important to eyes, but it is important to *us* that we have a realistic understanding of the role of genes in evolution, which often requires sorting out dauntingly complex or seemingly contradictory facts. Even in the classical fly *Ey* mutations, eyes were not entirely or always missing, and over some generations regeneration could occur, and this involved different genes in different lines. Although the fly is used as the classic example of the universal master nature of the *Pax6* pathway in visual systems, even in the fly the ocelli (central, simple eyes) do not require *Ey* but *Toy* instead (Punzo, Seimiya et al. 2002); ocelli use *opsin* genes, sometimes different from those used in ommatidia, as the *Rh2* gene is in *Drosophila* (Pollock and Benzer 1988; Smith et al. 1993). *Pax6* knockout flies have ocelli, but at least some knockouts of the related gene *Eya* lose both compound and medial eyes (T. Oakley, personal communication). The apparent independence of *melanopsin* pathways from the classical ones used in eyes also suggests that there is more to this story.

Pax6 is a member of the "paired" class of TF, and like some of the members of that family has two DNA binding domains, a homeodomain related to that in the homeobox TFs and another called the paired domain. *Pax6* binding sites have been found flanking crystallin and perhaps other eye-related genes, a fact used to support the master gene idea, but the story is not so simple. The paired domain seems to be needed, but evidence about the homeodomain is to date unclear. For example, it has been shown that *Pax6* can regulate the expression of invertebrate rhodopsins, but this regulation is complex (Papatsenko, Nazina et al. 2001), and eyes can develop normally, including rhodopsin expression, in the absence of the homeodomain (Punzo, Kurata et al. 2001).

Pax6 has complex binding domain and splice variants, so that the natural pattern of eye-related effects is not simple (Duncan et al. 2000) and may not be easily inferable from a knowledge of RE sequences alone. How this relates to the other genes in the eye-related regulatory hierarchies is unclear. The fact that *Ey* regulates *Rh1* in flies, a late-stage event, shows that the role of *Pax6* is not just as the master that sets off the eye cascade (this illustrates the potentially misleading metaphoric use of the word "master"—CEOs are not found on the assembly line, and generals no longer lead the charge into battle). There are occasions when the "master" gene has been substituted over evolutionary time and occasions when the later events came first, which were then regulated by a higher-level "master" gene (e.g., Graham et al. 2003).

The presence of *Pax6* homeodomain enhancers in opsin genes has been used to support the master-gene argument, but the gene is not expressed in differentiated vertebrate rod or cone cells that do express *opsins*. Perhaps, *Pax6* is involved only earlier in photoreceptor development. Planarians have two *Pax6* homologs, but experimentally neither is needed for the regeneration or maintenance of eyes, although a *Six* homolog is essential (Goudreau, Petrou et al. 2002; Harris 1997; Pineda et al. 2002). A comparable role has been demonstrated for *Six* class genes, including *Sine oculis*, in flies; as noted above, *Six* and *Pax* genes seem to interact.

The Nature and Role of Crystallins

Many lens and corneal crystallin genes have *Pax6* enhancers, suggesting that this is how they were intercalated opportunistically by natural selection into the eye development cascade in different lineages. *Pax6* seems to be generally involved in early neural development; however, crystallins are nonneural in nature. Crystallin genes have acquired *Pax6* enhancers, but this gene is not required by all crystallins, for example, duck crystallin (which may also not require *Sox2*) (Brunekreef et al. 1996). In some vertebrates (e.g., chick and mouse), the optic vesicle in the forebrain plays a necessary role in lens induction, and one study has shown that *Bmp4* is necessary for this to occur, but this pathway may be independent of *Pax6* (Furuta and Hogan 1998).

With regard to crystallins, one argument includes the notion that *some* such transparent lens and corneal proteins are needed—and recruited, for example, by the selection-favored appearance of *Pax* enhancers in the future crystallin genes' regulatory regions—in eye assembly. The idea is that this was an opportunistic pathway in that it did not matter to natural selection *which* genes were recruited. However, gene knockout experiments have generally found that there is little if anything wrong with eye development when the crystallins are missing. Experimental overexpression of Pax6 protein in mice has induced cataract, but apparently due to cytostructure genes rather than crystallin anomalies (Duncan, Kozmik et al. 2000).

One possible interpretation that requires less invocation of directional adaptive selection is that *Pax6* and other eye-related regulatory genes bind enhancers flanking hundreds of genes that become expressed in eyes, and hence in lens and cornea. Among proteins expressed in a tissue, there will be a distribution of abundance, and distributions of that sort are usually skewed—many elements will be relatively rare, only a few will be common. Many factors, including chance or the presence of other REs, can affect the relative amount of the gene product found in any tissue, including eyes. One can always extract proteins from lens or cornea, rank them by expression level, and *define* those with the highest level to be "crystallins." Crystallins are water soluble and densely packed in these tissues, but this is true of many other proteins with lower expression levels. Nothing prevents us from crediting the presence of these relatively abundant proteins to their *Pax6* enhancers as if driven by selection for visual purposes. There may also be other important related genes that serve as, or with, the known crystallins (Chauhan et al. 2002; Chauhan et al. 2002, and A. Cvekl, personal communication).

As far as selection goes, the key fact may simply be the compaction of sufficient water-soluble proteins, and among the hundreds of proteins expressed in a lens or cornea, there may be little in the way of selection involved in their evolution to "be" crystallins. It is important to be aware of the possible arbitrariness of our labeling based on strong darwinian assumptions and to be sure that we have evidence that selection is in fact responsible. Other mechanisms can also result in abundant lens expression. Duck crystallin appears not to be regulated by *Pax6* in the lens; instead, the ubiquitous protein Sp1 seems to suffice (Brunekreef, Kraft et al. 1996).

Blind mole rats use their eyes for circadian rhythm. They, and moles, have little if any real vision, yet crystallins are still expressed in their lenses (Avivi et al. 2001; Hendriks et al. 1987; Quax-Jeuken et al. 1985; Smulders et al. 2002). Mutations have accumulated in the mole rat lens crystallins—but not as rapidly as in a pseudogene. These "crystallins" may still have some function. But are they really crystallins? There is also a report of *alphaA-crystallin* that is expressed in the eyes of a species of blind cave fish (Behrens et al. 1997; Behrens et al. 1998) in the expected developmental stages and places based on closely related sighted fish.

Some of these data may turn out to be experimental artifacts, and selection explanations can of course be correct. These examples do illustrate the general point that evolutionary stories are usually not simple, and phylogenetically deep conservation is usually not complete. Nonetheless, there *are* clear genetic homologies among eyes that *do* suggest the unity of animal evolution. Thus, the differences among animal eyes may show that the common origins were at a much simpler, older molecular level than that of a common "eye" (e.g., Arendt and Wittbrodt 2001), just as Darwin suggested. His speculation that this could not have happened by chance was prescient. That we can now see this in genetic terms is not surprising, when we think of the highly conserved developmental processes that produce complex structures in multicellular organisms.

The discovery of the genes added new and convincing evidence, but it should not be surprising that paired anterior cerebral light-sensing structures found in diverse animals would not be entirely unrelated. An ironic note is that, despite playing a major role in showing these connections, and their prediction by Darwin, Gehring (like Darwin) suggests that the original eye evolved by *chance* (Gehring and Ikeo 1999). However, that is a strange invocation of chance from a darwinian point of view because the first start toward eyes in animals may *not* have required a "hopeful monster" generated by chance, given the primitive photosensors in single-celled organisms.

A number of "natural experiments" with eyes show that mutation can inactivate vision in many ways. Blind fish have evolved in various cave lakes in Mexico, although this process has been relatively recent in evolutionary terms, and vision can sometimes be restored in crosses between species (or subspecies). The reason is that different genes have been mutated in different species of fish, and crosses allow the functional genes in each blind parental strain to complement each other in the offspring, restoring function. Other animals not using vision, such as those in areas where it would be of no use, have lost much or most of the sight that was presumably found in their ancestors (moles are a favorite example). The star-nosed mole has an unusual variant on these themes—its nose provides a special kind of tactile rather than light-based spatial sense, as will be described in Chapter 16.

Ants and bees vary greatly in the size, components, and acuity of their eyes, and this is generally correlated with their role in life and the amount of time they spend outside their hive. In an historically well-known observation, Henry Bates' (Bates 1862) attention was drawn to Amazonian leaf-cutter, fungus-culturing *Sauba* ants (genus *Atta*). He observed three main groups of worker ants, one being a large-headed class that another famous Amazonian adventurer H. M. Tomlinson noted keep "moving hither and thither about the main body; having an eye on matters generally, I suppose" (Tomlinson 1912). Ironically, Bates identified a third group with "an eye"—a single, central eye—but he only observed this Cyclopean caste deep underground where there is no light. Actually, there seems to be a continuum of size in these workers, who vary in the jobs that they can do (Hölldobler and Wilson 1990); there are overall correlations between head and eye size, so whether the variable nature of these particular eyes represents an anomaly of any kind is unclear.

CONCLUSION: "IF THE ÆGLE WERE JUDGE"

Organisms use light directly as an energy source or as a source of information about their surroundings. The ability to detect and characterize objects at a distance is important in the way of life of many or even most animals. A variety of attributes of light are used, including direction, brightness, image, distance, and motion of objects. The evolution of vision in turn affected the evolution of the organisms being *detected* by their light images. This occurs in many ways, not least being the evolution of mating rituals and display, protective camouflage and deceptive coloration (e.g., caudal eye spots that disguise a tail as a head to trick predators, and Batesian mimicry in which tasty species mimic bitter ones), light-based attractants in flowers and fruit, motion behavior to escape predators. Visual cues are also used to set traps, by disguising a predator. Fireflies have species-specific flash signals to attract each other (and their predators who mimic them to lure a lusty but careless fly to its death). These phenomena variously use color alone and/or coloration pattern.

As important as it has been, however, seeing is not a specific problem to be solved by organisms, and no environment demands any particular type of vision. Indeed, the presence of various primitive photoreception mechanisms shows that organisms can use light information without having a brain (e.g., primitive organellar vision in unicellular organisms; jellyfish). Vision probably also illustrates the role organismal selection may play in nature. Organisms use what they have, and this may sort them out as well as classical darwinian selection does, but without fitness differences.

The homologies indicated by *Pax6* were dramatic findings. But this was only among the first of many similar findings of deeply conserved genetic mechanisms involving regulatory, enzymatic, and structural genes in many different structures and organ systems. The eye story, along with a few others like the *Hox* axial patterning system, led the way in these discoveries, but findings like this are rather general. Indeed, we have now come to expect to find such conservation. This unifies the animal world and may suggest that life is younger than we thought, relative to the evolution of genetic mechanisms—there has not been enough time for the sharing of these genetic mechanisms to be erased, replaced by selection, or subjected to phenogenetic drift, as sequestration among species almost guarantees will eventually occur if the Earth remains hospitable to life for a long enough time. However, the sharing that has persisted so far has perhaps tempted overstatements of homology because the conservation has been far from complete or simple, as eyes and vision illustrate.

Photoreception is evolutionarily related in interesting ways to olfaction and has diverged from a common ancestral mechanism for cells to sense external conditions. They use related members of the *7TMR* family (and *Pax6* is involved in olfactory development). This homology can be overstated because there are so many genes in this family and they serve such diverse purposes. However, like olfactory receptors, photoreceptors are G protein-linked receptors that activate cyclic nucleotide-gated channels using cGMP to affect cAMP concentration (it is synthesized by guanylyl cyclase and degraded by cGMP phosphodiesterase). A major step was the evolution of the different binding properties of these related genes—one for chemical shape and the other for light energy.

As is true of hearing and olfaction, seeing is not just a molecular trick played by the brain with opsin firing. The eyes and head can be moved relative to the environment, and the brain must integrate that movement with the neural information it receives from the eyes or other sensory neurons. Although this book is about genetic mechanisms and how they have been used to receive diverse information from the environment—especially unique or unpredictable information—we should not forget that morphology is also a major aspect of organisms. Morphology (and its coordinated use) is responsible for the mobility of external ears, eyes, nostrils, heads, and bodies, and this is all an integral part of how organisms detect and respond to their environment.

Most animals can respond to the world in ways that are not prespecified. Some merely need to know where a light source is. Others need a spatially arranged map of some aspects of the external environment and have evolved complex means to integrate spatially arrayed input, as we will see in Chapter 16. There is a correspondence between the properties of optics and the structures found in the eyes in nature, strongly suggesting that adaptation by natural selection has been at work.

But in nature, there are many ways to see, and no organism has them all. Animal brains have evolved to be able to receive organized light signals and resolve them in one way or another. There isn't one species that might not do better with additional visual ability, but all make do with what they have, and that has been good enough.

In Greek mythology, Juno jealously suspected that her husband Jupiter had a mistress who took the form of a heifer to disguise her identity (Juno was right). She hired the herdsman Argus to keep an eye—actually, to keep his *hundred* eyes—on the heifer to prevent further mischief. Argus never slept with more than two of his eyes closed and so was ever-vigilant. This evolutionary experiment ended suddenly, however, when Jupiter commissioned Mercury to get rid of Argus. Mercury lulled Argus to sleep with story-telling and then killed him in a stroke. Juno's revenge included bedecking the peacock's tail with Argus' eyes, which are now seen but no longer see. Being covered with 100 eyes always on the alert might have been selected for, but it wasn't (except, perhaps, in the scallop, whose hundred or so eyes are arrayed, beadlike, along its mantle).

Aristotle and his peers through Classical times had many ideas about what species could see. Aristotle, for example, thought that moles were blind. But in a huge compendium to debunk long-held mistaken ideas published in 1646, Thomas Browne (Browne 1646) reported that he observed mole embryos to develop eyes and that in fact moles could see. That a species could not see as well as others do, he notes, is a value judgment of which we need beware: "if the Ægle were judge, wee might be blinde our selves."

Chapter 15
The Development and Structure of Nervous Systems

In previous chapters, we have discussed the open-ended environmental variability that organisms encounter and *detect* with sight, taste, smell, feel, and so on. In Chapters 15 and 16, we will describe in basic terms what is known about the central problem of how organisms *perceive* this infinite and constantly changing variety of sensory signals.

As humans, we tend to relate these issues to our own personal experience, leading us to equate perception with consciousness. But the two are highly or perhaps even entirely different. In fact, we only have indirect indicators—at best—of how different the mental experience of other organisms may be from those that we understand. The nature and even the definition of consciousness are still debated and unclear, as is whether nonhuman organisms experience it at all. Many organisms to whom we would not attribute consciousness clearly do have integrated, organized responses to complex environmental input. Is that so different from what humans experience? How would we know one way or the other?

An organism does not have to be complex and multicellular to respond to light, vibration, temperature change, and other environmental signals. Some do so locally, with no centralized processing, and we noted that immune response was like this and that in plants each part can in many ways respond independently of the rest of the plant. Other species, however, have a hierarchical and highly specialized central nervous system (CNS) that serves to create an internal representation of environmental cues and to organize responses. Even plants and single-celled organisms integrate and respond to multiple environmental cues; therefore, a neural mechanism for receiving and responding to unpredictable environmental signals is clearly not required for survival, or even for adapting to changing external conditions. Only a small subset of living organisms does so through a CNS.

Charles Darwin himself, in his 1872 *Descent of Man*, goes to great lengths to connect human behavioral components with those of other species, to show that in fact we share common origins with other animals. He was appropriately

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impressed even by ants: “the wonderfully diversified instincts, mental powers, and affections of ants are notorious, yet their cerebral ganglia are not so large as the quarter of a small pin’s head. Under this view, the brain of an ant is one of the most marvelous atoms of matter in the world, perhaps more so than the brain of a man.” Do we have a rational basis for asserting that simple organisms do not have a form of “awareness” or organismal level percept and that their reactions to inputs are as unknowing as, for example, that of a motion detector that turns on lights in a house?

In organisms with a CNS, sensory input may undergo some processing locally at the site where it is received, but ultimately raw light, sound, taste, touch, and other signals all proceed to the nervous system in essentially the same way, via action potentials, and the signals are decoded by modality in the brain. The pathway the signal travels in the brain determines how it is ultimately perceived, and the pathway is determined by the destination of the particular neuron through which the signal has traveled to reach the brain. Light signals are sent from the eye to the visual centers in the brain, where they are translated into perception of color, shape, motion, and the like. Sound waves are collected in the ear and shunted along to the auditory centers in the brain to be translated into perceived danger signals, music, communication, and so forth. However, *percept*—whatever it actually turns out to be—is not entirely location dependent because, as we will see, the same kinds of input (e.g., sound, light) can occur in different parts of the brains in different people or even in the same person at different times in his/her life.

In most sensory systems, neurons project their axons *topographically*, that is, in an orderly fashion that provides a precise spatial “map” or representation of the location of a particular class of receptors on the surface of the body, whether the retina, the olfactory epithelium, the cochlea, or the skin. The axons terminate in distinct sensory centers of the brain, their parametric representation of the physical world still conserved. That is, neurons that are adjacent in the receptor surface terminate adjacent to each other in the sensory area of the brain to which they project. The central processing system detects in detail where the signal came from, and this may represent the external world in a literal sense, as in vision and touch, but need not as in olfaction or hearing. These ensembles are called *neural maps.*

Signals are processed through multiple levels in the CNS before final perception. In most sensory systems, both parallel and hierarchical (serial, with one step preceding the next in order) processing are involved in the anatomic connections that send signals to the appropriate centers of the brain for interpretation. Auditory systems, for example, can simultaneously process many sounds as well as many aspects of each sound. Light is processed into increasing complexity by higher cortical areas, and different qualitative aspects of light are processed simultaneously. The various sensory areas in the brain have many functional and structural commonalities, and, despite segregation of the sensory areas of the CNS, in the end the brain integrates the different signals to create what is experienced as a unitary but multidimensional representation of the external world.

It might be rather subjective to discuss which organ system is the most complex in its structure or function, but the nervous system would certainly be a candidate, especially in vertebrates where the developmental dynamics are far from a simple hierarchy and the connection between structure and function not always clear. In Chapter 15, we will describe in broad terms the various types of nervous systems known, their architecture, how they work, and some of what is known about the genetics of their development. In Chapter 16, we will discuss different sensory modalities and how they are decoded and perceived in the brain.

INVERTEBRATE NERVOUS SYSTEMS

Action potentials probably existed in the first single-celled organisms, predating neuronal activity and the evolution of the nervous system, evolved for some other purpose, perhaps having to do with response to external chemical conditions. Some modern single-celled organisms can effect a kind of preneuronal electrical signaling and are capable of producing action potentials that regulate, for example, the direction of the beating of cilia and thus movement of the cell.

Colonies of multicelled organisms that do not have a central nervous system, such as the cnidarian *Obelia*, may also produce action potentials. *Obelia* colonies share a digestive system, and electrical signals can spread through the common epithelial lining of the digestive cavity. Other cnidarians have nervous systems of a different sort. Hydra, for example, have *nerve nets*, collections of nonpolar neurons that send electrical impulses in any direction such that the whole body responds to any stimulus, although there can be a concentration of these nerves around the mouth for enhanced detection of food. There is no hierarchy of control, nor any specialization of neuronal function. The nerve nets also control the organism's movement (Matthews 2001) as well as acquisition of food.

Echinoderms like starfish typically have a locally organized nervous system, with no central *ganglion* (see Figure 15-1). A ganglion is a knotlike cluster of nerve cells. The neurons of these radially symmetric organisms connect into a *nerve ring*, with branches extending into each of the five arms to innervate them. Jelly fishes have a neural net with ganglia, which allows some organization of movement such as swimming by pulsation. Wormlike hemichordates have a nerve net in the epidermis, which thickens to form several solid nerve cords, and a neurocord, a collection of giant nerve cells, which connects the nerve cords formed from the nerve net. The neurocord probably plays a role in rapid motor responses (Butler and Hodos 1996). Flatworms have a collection of enlarged anterior ganglia, analogous to a simple brain. Longitudinal nerve cords allow for some control of body movements.

Although insects have brains, their nervous systems are much less centralized than those of vertebrates. Generally, an insect CNS is organized into a series of ganglia, strung together along a *commissure* (a junction at which corresponding structures join), that runs along the ventral nerve cord, the latter being linked to the brain (see Figure 15-2A). The brain itself is a collection of six fused ganglia (three pairs) that each control specifically circumscribed activities. The first pair, the *protocerebrum*, innervates the compound eyes and the ocelli. The second, the *deutocerebrum*, integrates signals from the antennae. The third, the *tritocerebrum*, innervates the *labrum* (roof of the mouth) and integrates input from the protocerebrum and the deutocerebrum. It also links the brain to the rest of the ventral nerve cord and the ganglia along the cord that control other organs and behaviors, such as feeding, mating, locomotion, and sensory reception. As in the vertebrate CNS, there are three kinds of neurons in the insect nervous system: *sensory afferents*, *motoneurons*, and *interneurons*, nerve cells in the CNS that link the sensory and motoneurons.

The major structures of the insect brain include the *mushroom bodies*, *central body*, the *optic lobes*, and *the antennal lobes* (Figure 15-2B). The mushroom bodies

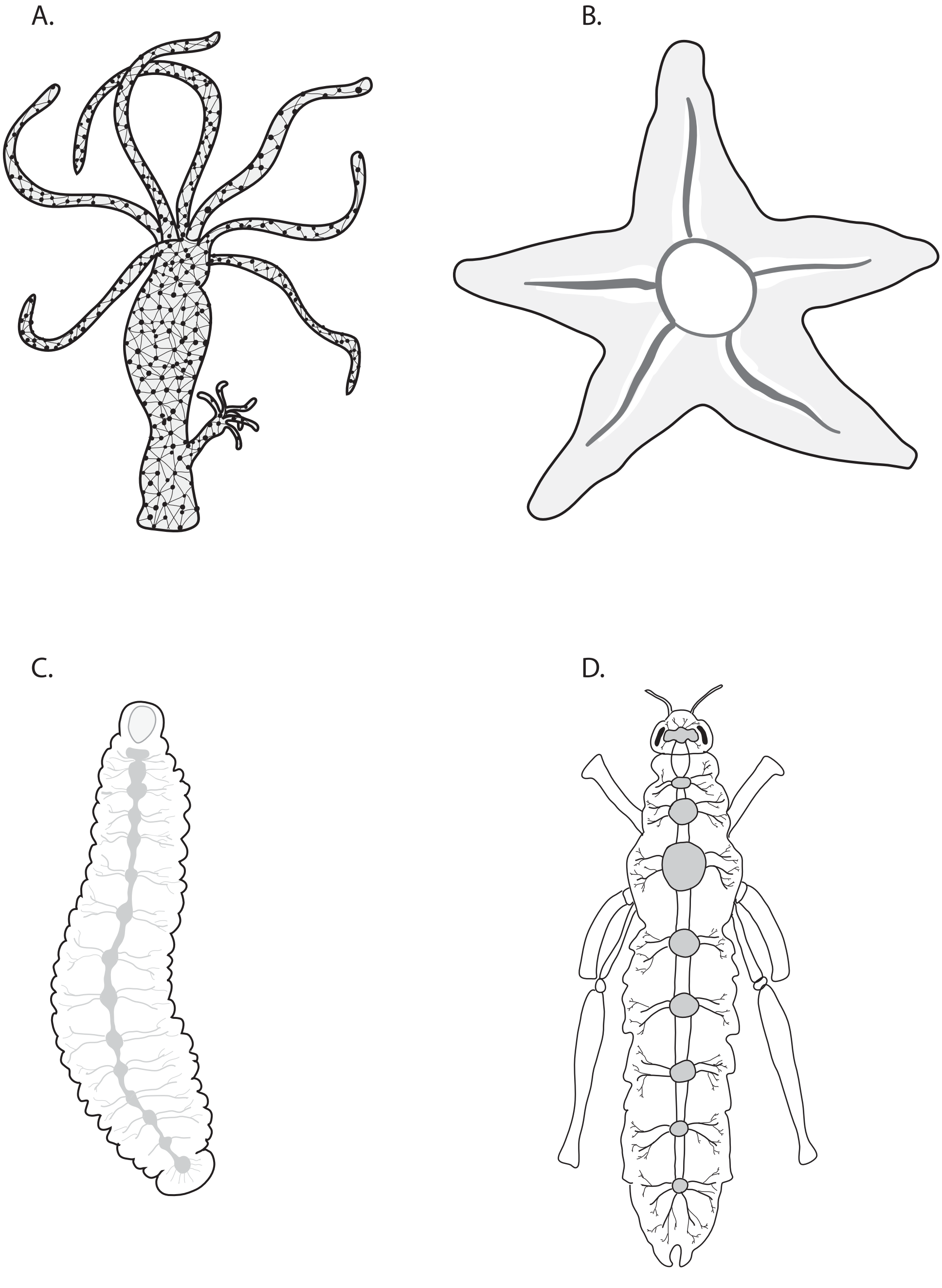


Figure 15-1. Organization of the nervous system. (A) The nerve net of the hydra; (B) the starfish nerve ring; (C) the leech ganglia and; (D) the cephalized nervous system of the grasshopper. Redrawn from (Matthews 2001).

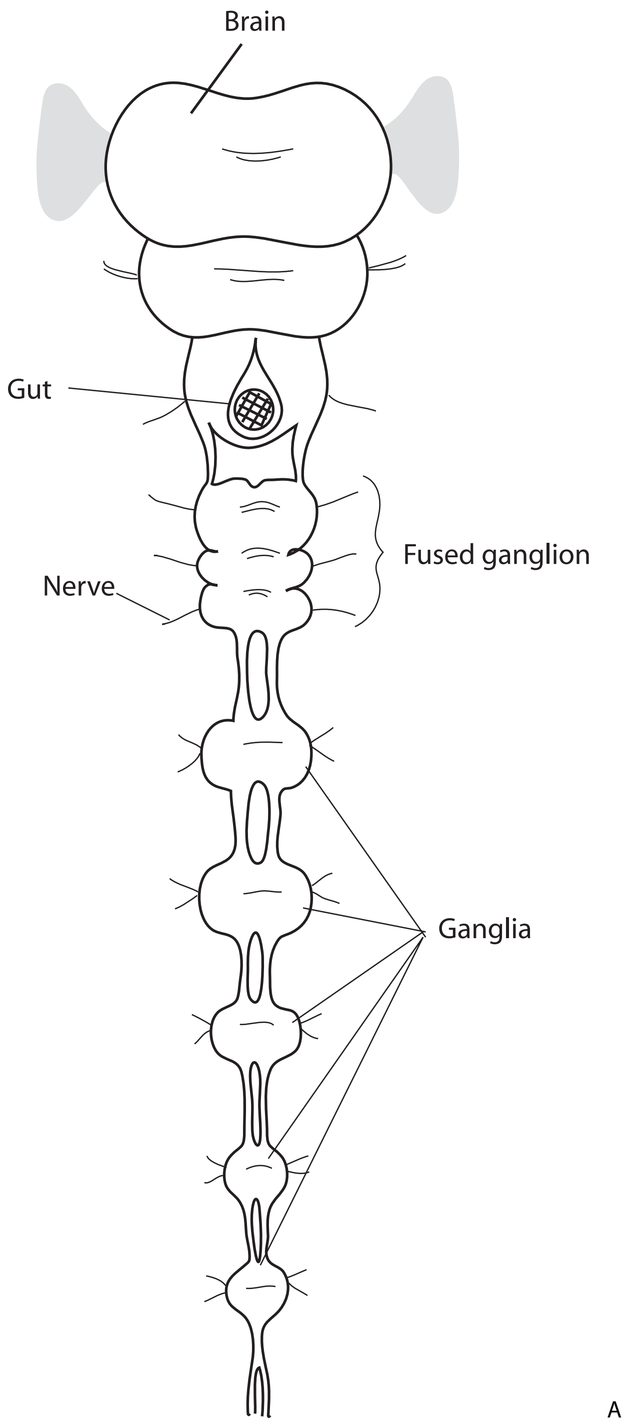
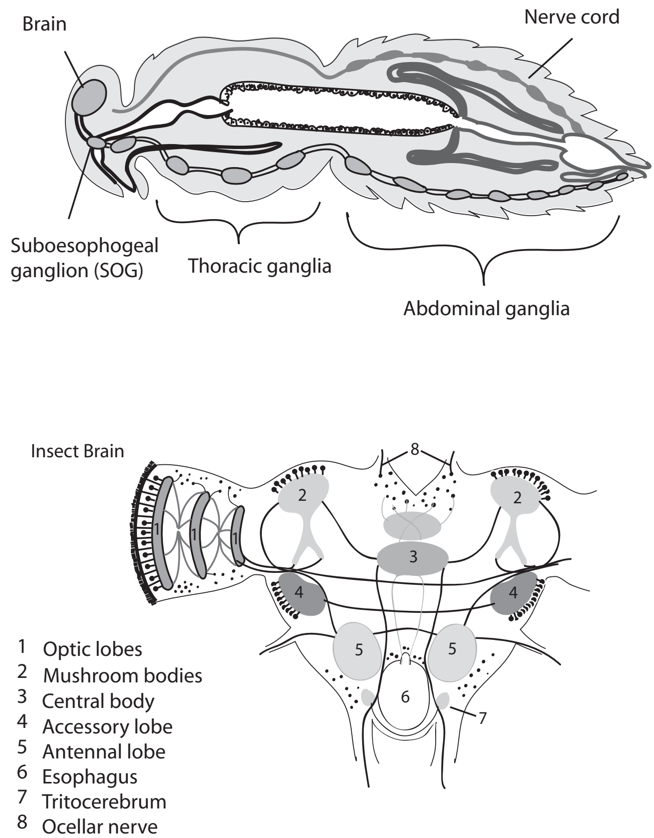


Figure 15-2. Schematics of the insect nervous system (A) and major structures of the insect brain (B).



B

Figure 15-2. *Continued*

are involved in olfaction. The central body is a way-station through which nerve fibers pass from one hemisphere to the other. The nature of the signal processing that takes place here is unclear; it may be involved in visual processing (Burrows 1996). The optic lobes are laminar, that is, organized in layers, and process information from the insect's compound eyes through the layers in a way that, as in vertebrates, retains the retinotopic (spatial) map of the image. The antennal lobes process olfactory signals coming from the antennae.

This may be an appropriate place for us again to note the many parallels between vertebrate and invertebrate nervous systems. Biologists long noted the similarity of organization (eyes, mouth, digestive system, limbs, and so forth) between invertebrates and vertebrates, although they were considered analogous rather than homologous. As we have already seen in many ways, recent studies have shown similarities—sometimes close similarities—system by system, at the gene, developmental, and cytological levels. We noted in Chapter 9 the suggestion by Geoffroy St. Hilaire in the early 1800s that insects were inverted vertebrates, in that the dorsal nerve cord in vertebrates is homologous to the ventral dorsal nerve in invertebrates (and the digestive systems are in the corresponding inverted locations). There are some corresponding genetic similarities as well, but it is important to resist the temptation to extrapolate too far or to expect even true homology to be too precise at the trait level.

Overall, invertebrate nervous systems range from very simple and diffuse to relatively complex and centralized. As a result, the organisms react in ways that seem interpretable to us. Darwin went through a list of ways. Insects exhibit what appears to be fear, which can be elicited by various signals, panic (when they are upside down, for example), and exploratory behaviors, not to mention courtship (including males fighting for mates). There are many examples of learned behavior in insects as well, and they can be trained in the laboratory (Waddell and Quinn 2001). This may seem surprising, but their brains do have room for the integration of new experiences and behaviors (Meinertzhagen 2001). None of this explains how information is integrated, however, much less how the organism experiences the information. Another thing to consider is whether we are anthropomorphizing with terms like "panic" and "courtship," especially if consciousness is not involved (but is it?).

VERTEBRATE NERVOUS SYSTEMS

As chordate body plans evolved, neuronal connections tended to become centralized and cephalized with a knot of neurons organized into a ganglion at the head of the animal. Neuronal function became more specialized and hierarchically organized. This has been much more elaborated in vertebrates than in the somewhat similar path taken by some arthropod lineages. Vertebrate CNS development involves most of the signaling, transduction, transcription, receptor, patterning, and combinatorial expression phenomena that we have seen many times, as well as mechanisms for highly controlled migration of cells through areas already inhabited by other cells.

Early in the evolution of the vertebrate CNS, the brain stem was the only "cerebral ganglion" that could receive and process sensory information, and this is still the way that the nervous systems of some vertebrates are organized. The lancelet, or *amphioxus*, is an example of a modern "lower" chordate with a notochord and a nerve cord above it, but no brain, no eyes, and, in fact, no head. Over evolutionary time, although the more "advanced" anterior parts of the brain grew larger, much of the synaptic organization of the brain stem was conserved and modified. Some sensory nerves still make synapses (connections) in the brain stem on their way to processing centers in higher, more recently evolved regions of the brain (Matthews 2001).

Why did an organized nervous system that is centralized at one end evolve, and why at the front? One obvious answer is that "front" is basically defined by the direction the animal moves. This centralized end confronts the environment first—if the main sensors responsible for finding food and mates and avoiding predators are close together and localized in an area that first contacts the approaching environment neural connections can be short and the response can be quick and more easily integrated. This is a rather circular explanation, in part because of how we define "front" (and we ignore as exceptions due to phylogenetic inertia those animals, like some crabs, that generally move "backward"). The evolutionary explanations could be countered by saying that the tail end should have the sensors because this is the direction from which the greatest dangers may come or that animals should have eyes in the back *and* front of their heads (or, like Argus, 100 eyes). Perhaps the brain should be closest to the endocrine organs, to facilitate quick response, or buried deep in the middle of the organism to protect the vital "nerve center," much as military nerve centers are protected in deep underground bunkers. After all, this is how we explain the fact that veins are more superficial than arteries (which, when severed, cause more immediate threat to survival than veins that are severed).

Vertebrate nervous systems are segmented and modular at morphological, histological, cytological, and physiological levels (e.g., Carlson 1999; Redies and Puelles 2001). In some ways, the brain can be thought of as a set of Matryoshka dolls, with major segments composed of smaller units, themselves formed of still smaller ones, in turn made of even smaller ones, and so on, defined in different ways at each step, down to the level of the neuron itself. The brain is physically as well as "virtually" (functionally) segmented, and the distinction is important because under some conditions the physical location of a function can vary or move (and hence, percept, whatever it is experientially, need not be physically localized in the brain). However, not all segments in an adult vertebrate brain correspond to the initial segment structure, and indeed the segments are not always neatly nested; they are generally interdigitated structurally and even functionally.

Segmented Development and Neural Differentiation in Vertebrates

The nervous system is segmented along its anterior-posterior axis from the rostral (face) to the caudal (tail) ends and transversely, that is, dorsoventrally, and also from medial or mesial (inside, central) to lateral or distal (outside, peripheral). In a hierarchical or nested way, segments at an early stage give rise to segmental substages later on. The system initially forms along the dorsal midline, from a layer of cells called the *neural plate* that overlies an anterior-posterior supporting rod called the *notochord* (e.g., Carlson 1999; Gilbert 2003; Matthews 2001). The notochord is a source of Shh (*Hedgehog* class) SF (Gavalas and Krumlauf 2000; Lumsden and Krumlauf 1996; Wurst and Bally-Cuif 2001), which also induces *Shh* expression in the adjacent overlying neural plate cells. Laterally (on either side of the Shh source) cells express *Bmp* SF proteins (*Bmp4* and *Bmp7*). In a process called *neurulation*, this *Bmp*-expressing flanking tissue grows up and around on either side, closing dorsally to form the top, or *roof plate*, of a hollow *neural tube*. The ventral and medial part, the *Shh*-expressing cells that overlie the notochord, is known as the *floor plate*. (Recall that Bmp proteins have dorsal effects.)

The *Bmp*-signaling area is now dorsal, atop the closed tube, and diffuses down both sides, inducing various transcription factors (TFs), including *Pax3* and *7* and *Msx1* and *2*, whose expression defines the dorsolateral *alar plate*. Shh diffuses upward on either side from the floor plate, inducing genes including *Pax6* and *Nkx* TFs to define the ventrolateral *basal plate*. These expression patterns extend to the spinal cord caudally and are important in establishing regional identity in the forebrain.

The Bmp signal diffusing from lateral ectoderm also induces expression of TFs, including the zinc-finger gene *Slug*, in cells at the crest of the juncture that formed the roof plate. These cells take on a special role generally considered to be a defining characteristic of vertebrates—perhaps even qualifying as a fourth primary germ layer (Hall 2000)—by becoming the migrating cells of the neural crest (NC), described in Chapter 8. NC cells are involved in peripheral nervous system and cranial neural crest (CNC) in craniofacial bone, tooth, and other tissues.

The extent to which CNC cells are regionally prepatterned before migration away from their source at the neural tube is unclear. CNC cells have plasticity such that to some extent their role is determined by response to signals they meet, possibly including from endoderm, on their migratory way (e.g., Trainor and Krumlauf 2000). In Chapter 8, we described their interaction with overlying epithelial cells in the process of epithelial-mesenchymal interaction that generates various structures. There is evidence that the overlying tissue has already been prepatterned to produce signaling molecules that induce a response from in-migrating CNC cells, and postcranial NC cells cannot be induced to form head structures (Graveson et al. 1997).

The spinal cord predominates in the control of functions in lower vertebrates, with the brain gaining dominance with the evolution of increasing size of the forebrain in later vertebrates. Figure 15-3 shows the relative sizes of the major sections of various vertebrate brains. This evolutionary differentiation with its complex behavioral and sensory consequences can thus be viewed as having come about, in part at least, through relatively simple allometric modifications of existing structures that were easy to achieve. Some of the evolutionarily "early" functions remain largely controlled by their original parts of the CNS.

The spinal cord is a collection of nerve fibers and nerve cells in vertebrates that extends from the *medulla oblongata* at the base of the brain through the spinal column. The number of nerves comprising the spinal cord differs somewhat among vertebrate species, but there is remarkable stability in their position and function. In primates, 31 pairs of nerves, both afferent (incoming sensory) and efferent (outgoing motor), travel along the length of the cord, to emerge at various points between the vertebrae to relay information to and from the brain and the rest of the body. Twelve nerve pairs, the *cranial nerves*, extend directly from the ventral surface of the brain itself to affect functions in the head and face, as well as some functions in the trunk (e.g., diaphragm). Most of these are both sensory and motor, although the cranial nerves involved in olfaction, vision, and hearing are exclusively sensory.

The brain itself develops as a complex convoluted layer of cells called the *cortex* that surrounds an inner fluid space (the core of the original neural tube) known as the *ventricle*. Longitudinally, the brain forms three main regions: the *hindbrain* (*rhombencephalon*), the *midbrain* (*mesencephalon*), and the *forebrain* (*prosencephalon*). The hindbrain functionally divides into the *metencephalon*, which includes the *cerebellum*, and the *pons* and the myelencephalon, which includes the medulla oblongata. A constriction called an *isthmus* separates the midbrain and hindbrain.

Neurons from the medulla oblongata help to control functions such as breathing, swallowing, cardiovascular function, digestion, and some body movement. The pons, at the junction between the hindbrain and the midbrain, is also involved in regulating breathing. The cerebellum receives sensory information from many parts of the body as well as signals from motor control areas of the forebrain, which it helps to coordinate into motor commands. In popular vernacular, we do not have to "think" about these functions for them to work. They work during sleep and even during coma, and hence do not require consciousness. The rest of the brain *is* aware

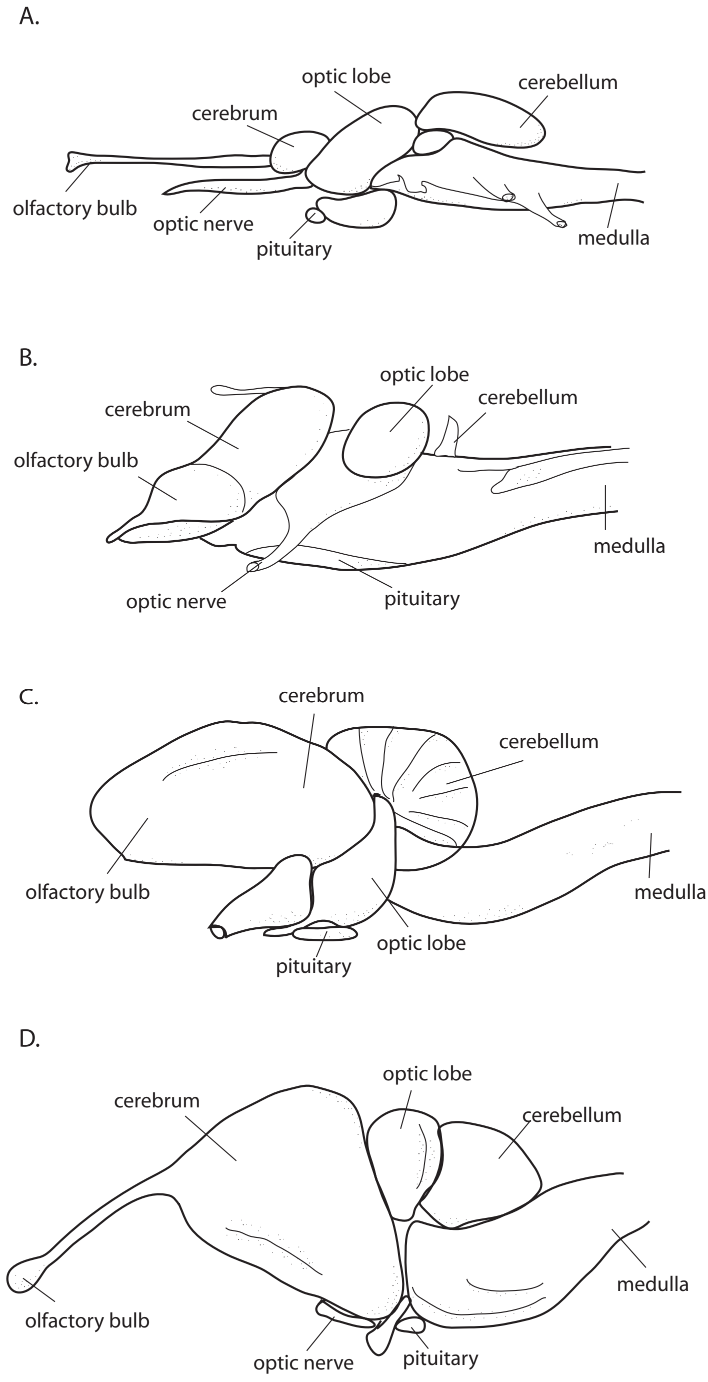


Figure 15-3. Major sections of selected vertebrate brains. (A) fish; (B) amphibian (frog); (C) bird; (D) reptile (alligator).

of these functions; for example, they can change circumstantially and can even be influenced by thought, as the racing heart rate triggered by fear.

The hindbrain and midbrain constitute the *brain stem*. The midbrain contributes to the control of movement and receives sensory information that it passes along to higher areas for interpretation and response. Its major subdivisions are the *inferior* and the *superior colliculus*, the first a relay and processing center for auditory signals and the second for visual signals. The *reticular formation* comprises some of the rest of the midbrain. This section controls the organism's state of arousal and plays a role in various sensory and motor systems.

Figure 15-4 shows the expression boundaries in the hindbrain-midbrain region of many genes that have been examined. Keep in mind the mixed and incomplete dartboard nature of such expression or pathway catalogs. Regional differentiation establishes the isthmus, which then becomes a boundary relative to expression of signaling factors *Fgf8* posteriorly and *Wnt1* anteriorly. The isthmus is a similarly sharp expression boundary of two TFs: *Otx2* anteriorly and *Gbx2* posteriorly. Not all genes important in patterning in this area are so restricted, however; *En1* and *En2* are expressed on either side of the isthmus.

The hindbrain develops into a series of segments called *rhombomeres* that are reflected in expression boundaries. For example, *Krox20* expression differentiates odd from even rhombomeres, and combinatorial expression patterns of genes from the four *Hox* clusters along with *Krox20*, *Follistatin*, *Kreisler*, and others identify each rhombomere (Voiculescu et al. 2001). This is the anterior-most part of the famous AP combinatorial patterning effect of the *Hox* clusters presented in Chapter 9 and elsewhere. Consistent with this, experiments that individually inactivate these genes have homeotic (segment-shifting) effects to varying degrees. Functional studies have also shown that these various segment-defining genes interact. Signaling by *Fgf8* and diffusible retinoic acid (a morphogen related to vitamin A, not a gene product) and its receptors and binding proteins (which are gene products), affect *Hox* expression domains and hence rhombomere identity (e.g., Trainor and Krumlauf 2000). Rhombomere-specific gene expression patterns make it possible to trace the migration of NC cells from specific rhombomeres into structures like the pharyngeal (gill) arches (Gavalas and Krumlauf 2000).

A series of genes and their receptors (*Eph* and *Ephrin* in Figure 15-4) affect cellular differentiation in the region, and most of the genes shown, which are important to hindbrain specification, are also involved in other developmental processes; for example, Follistatin is a secreted molecule involved later in life in controlling reproductive hormone levels.

The isthmus is frequently referred to as an "organizer" because, like Spemann's, the apical ectodermal ridge (AER) in the developing limb, enamel knots in teeth, the focal Dll spot on butterfly wings, and numerous others, it is a source of signaling molecules that help induce subsequent development in receiving cells around it. Thus, Fgf8 signal emanating from the isthmus inhibits expression of *HoxA2* that helps define the anterior hindbrain subregion and in turn is part of the specification of NC cells migrating from the hindbrain to affect skeletal development in the jaws. *Hox* genes are expressed in the mandibular arch of embryonic agnaths, suggesting that the loss of this expression enabled the anterior pharyngeal arch to evolve into a jaw (Cohn 2002).

However, this is a useful place to restate the caveat that "organizer," like "master" and "selector," is a concept with complex meanings in the culture of the scientists

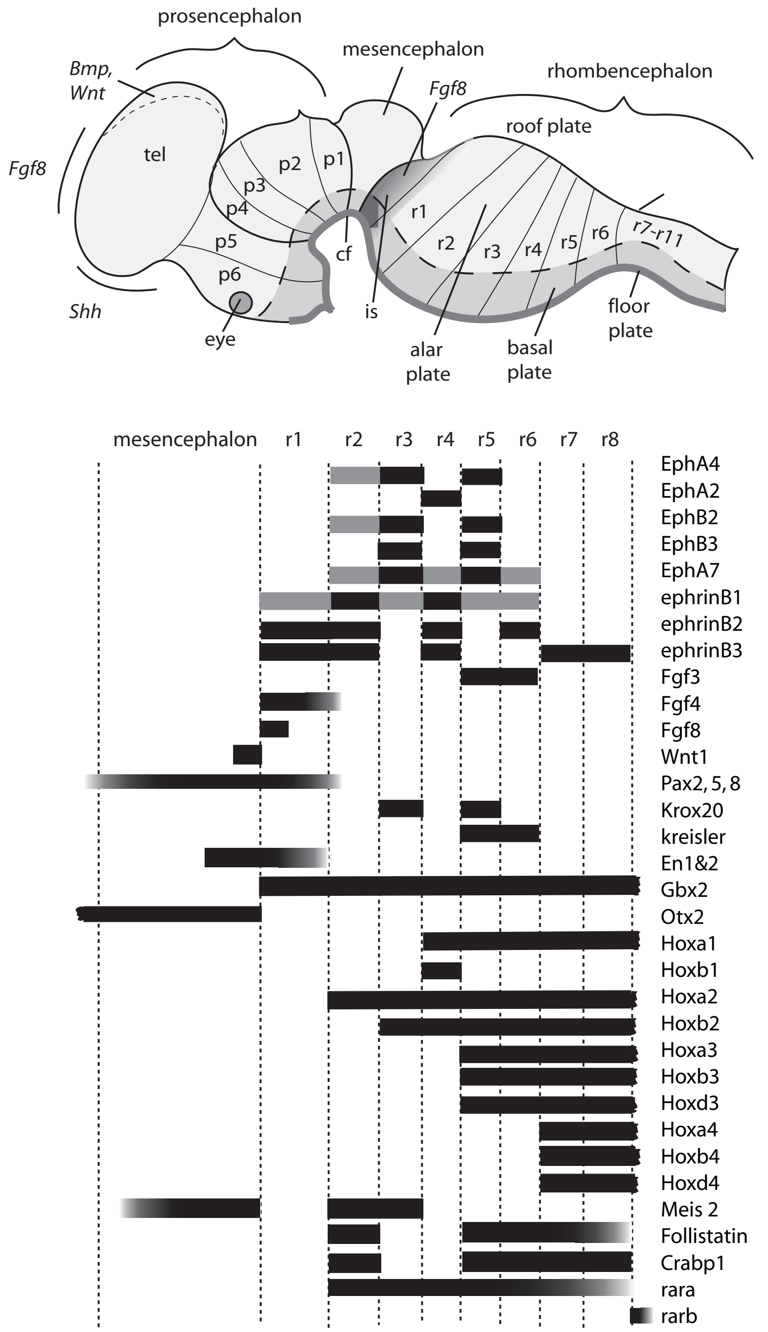


Figure 15-4. Longitudinal segmentation and associated gene expression in the early vertebrate brain. Schematic and selected (and many genes are not yet known). "Is" denotes the isthmus; p1-p6 prosomeres, and r1-411 the rhombomeres. Genes shown are *Hox* cluster genes, other TFs (*Drox20*, *Kreisler*, *En1&2*, *Gbx2*, *Otx2*, *Pax*, *Meis2*); SFs and receptors (*Eph*, *Ephrin*, *Fgf*, *Wnt*); factors related to retinoic acid and its receptors and binding proteins (*rara*, *rarb*, *Crabp1*). Composite based on (Gavalas and Krumlauf 2000; Lumsden and Krumlauf 1996; Marin and Rubenstein 2002; Pasini and Wilkinson 2002; Simeone 2000; Wilson and Rubenstein 2000; Wurst and Bally-Cuif 2001).

coining such usages in genetics. As such, the term can be misperceived to suggest a discrete, externally imposed purposive entity, rather than a basically quantitative region vs. qualitative entity, which itself arises from prior signaling interactions (Wilson and Rubenstein 2000). It is called an organizer because it is a regional source of signal that affects the development—sometimes but not always with sharp boundaries—of subsequent structures. In fact, things are not so simple and the defining genes are not the only important factors in phenomena affected by their production in the isthmus (e.g., Chambers and McGonnell 2002), nor is the isthmus the only place they are important even, even in neural development.

Vertebrate Forebrain Development

The forebrain, or prosencephalon, is generally thought of (anthropocentrically) as the place where the really important biological traits reside: where vertebrates interpret, integrate, and respond to sensory and somatic input and motor commands (e.g., Matthews 2001). The relative size of the forebrain varies greatly in vertebrates, from being among the smallest sections of the brain of fish and reptiles, to constituting most of the brain in primates. Many of the functions carried out in this part of more complex brains, such as visual processing, occur in the midbrain of animals with small forebrains, although some of the "higher" functions are missing from the latter. (Does this mean they "perceive" vision differently?) It is perhaps worth considering that, from the point of view of members of species without forebrains—species that are doing perfectly well in the world—the forebrain might seem a grotesquely exaggerated tissue that requires a lot of extra DNA, metabolic energy, and maintenance, and thus a lot of food. This belies any notion that evolution is an energetically parsimonious phenomenon.

Each part of the brain is further subdivided into sections with their own specialized functions. Our understanding of the degree to which specific functions, other than the major obvious ones like vision and olfaction, are precise and replicable is still incomplete but scans of brain activity under various conditions and in experimental animals whose brains have been modified, or affected by certain diseases, show that functions can be remarkably tightly located. That is, brain activity changes only in a very localized region of the forebrain during the experimental or test experience.

The forebrain develops from an outpocketing of anterior neural tissues (first to form the optic vesicle) and is affected by signaling from several major regions (for a detailed discussion of these complex processes see Marin and Rubenstein 2002); these are generally indicated in Figure 15-4. Early dorsal signals involve *Bmp* and *Wnt* genes. A second signaling area of importance in forebrain development is the *anterior neural plate* (ANP). This most rostral area is another source of Fgf8 signal in the developing brain and is also referred to as a forebrain organizer. Ventrally, forebrain development is affected by *Shh* to which we have already referred.

The outgrowth forms a hollow structure on each side, with a dorsal groove or sulcus between them that separates the future left and right cerebral hemispheres. Also forming are areas of thickening, including the ventral septum and the medial and lateral ganglionic eminences (MGE and LGE), as indicated by the transverse sections shown in Figure 15-5. Ventral to these areas are the *anterior entopeduncular area* and associated *anterior preoptic area*, where major tracts of neural fibers go into or come out of the telencephalon. Darker shading indicates the *ventricular zone*

(VZ) in which neural stem cells produce differentiated glial and neuronal cells that move into the adjoining *subventricular zones* (lighter shading). The dorsal region (dorsal and medial pallium in the figure) becomes the cortex.

The prosencephalon can be viewed as being divided into six segments called *prosomeres*, by analogy to rhombomeres, though the most anterior three of these are less clearly defined. Prosomeres are characterized by differential gene expression patterns (e.g., Marin and Rubenstein 2002). The *diencephalon* and the *telencephalon* are the major subsequent divisions of the forebrain. These, however, are further divided—the diencephalon into the *thalamus* and the *hypothalamus* and the telen-

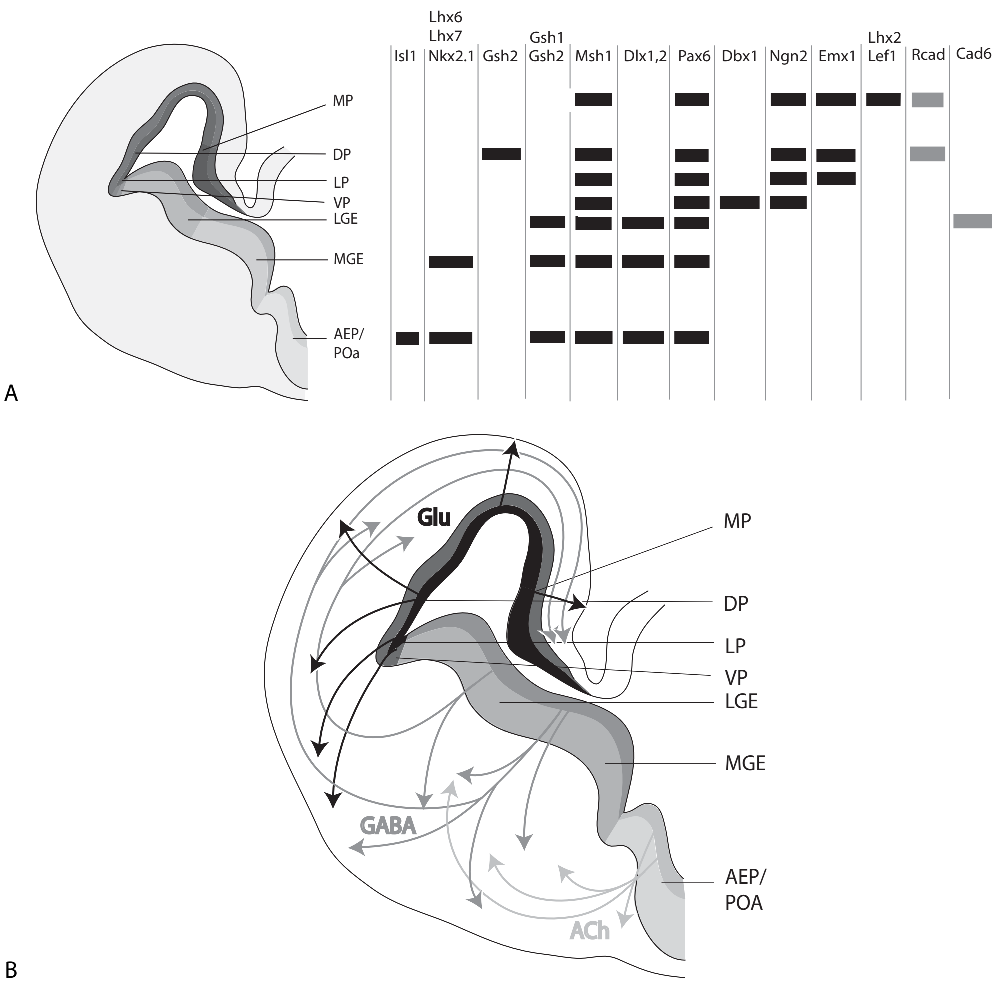


Figure 15-5. Gene expression in forebrain development. (A) Schematic of genes expressed in different regions. Bars show zones of expression for each gene. These are TFs except the two Cadherins on the right; (B) neurons migrate from the same areas but are at least partly prespecified by the time of migration, as shown here in regard to type of neurotransmitter expressed (Glutamate, GABA, and Acetylcholine). Highly schematic, roughly at embryonic day 14.5 or midgestation in the mouse. For anatomical terms, see text. Greyshade: darker, ventricular zone; lighter, subventricular zone. Composite modified after (Marin and Rubenstein 2002; O'Leary and Nakagawa 2002; Pasini and Wilkinson 2002; Redies and Puelles 2001; Schuurmans and Guillemot 2002; Wurst and Bally-Cuif 2001).

cephalon into the *cerebrum* and the *basal ganglia*, which are involved in control of movement. As will be seen below, the thalamus is central to lower processing and relay of signals from sensory organs to higher centers of the cerebrum, where they are further processed and integrated.

Figure 15-5A indicates some of the genes whose regional expression is correlated with these morphological areas. Overall dorsoventral differentiation is established or affected by mutually inhibitory interactions between dorsal Bmp4 and ventral Foxg1 (also called BF1), whereas the cortical LGE-MGE distinction reflects mutual inhibitory interactions between genes activated via *Nkx2.1* and *Pax6*; *Dlx*, *Msh*, *Gsh*, and *Lhx* gene expression also differentiates these regions. These expression patterns also reflect and are markers for the histological, and cytological differentiation of the region, and are generally conserved among vertebrates that have been tested (e.g., see Monuki et al. 2001; Puelles et al. 1999; Redies and Puelles 2001). The regional expression pattern is also consistent with known signal-transduction interactions, in that the interacting genes are expressed in the same region.

Although these expression patterns still mainly serve as regional markers, over- and under-expression studies have confirmed their general functional importance and relevance, and some of the temporal and regulatory hierarchies are being identified (Schuurmans and Guillemot 2002). Further, their importance in indicating the orderly and segmental, hierarchical nature of brain development is that it makes the evolutionary origin and development of an organ of this complex nature easier to understand.

As the size and complexity of vertebrates increased, the cerebrum began to dominate the forebrain. The cerebrum is filled with neurons and is where complex processing of sensory information (for example, auditory, visual, olfactory), the intentional planning and initiation of movement, and many forms of learning, among other activities, take place. The *cerebral cortex* is the 1- to 4-mm outer layer of the cerebrum and formed of three subdivisions, sometimes called the *neocortex* (the new cortex, unique to mammals), the *paleocortex* (old cortex), and the *archicortex*. (These names are intended to reflect the phylogenetic origins of the regions, but may not in fact do so, as there is not a clear consensus about the origins of the neocortex (Northcutt and Kaas 1995)). The paleocortex is basically the olfactory region of the brain and receives direct input from olfactory neurons, as specified in Chapter 13. The archicortex is primarily the *hippocampus*, or memory area, an inner fold of the temporal lobe. These regions differ cytoarchitecturally—the archicortex is a single cell layer, the paleocortex is two layers of cells, and the neocortex is formed of six layers, described below.

Several ideas regarding the origins of the neocortex have been proposed. It may be an enlargement of the reptilian dorsal *pallium* (roof, literally "cloak" or outermost layer) or the migration of two sources of cells from lower parts of the neuraxis (see Karten 1997 and Northcutt and Kaas 1995 for discussion of primitive mammalian cortexes). Since the evolved function is what counts, this might be largely an academic question. But it could turn out to have implications if, for example, there are as yet unknown vestiges of the original functions that may constrain the current ones in some way, be related to particular diseases, and so on. Searches for expression of cortex-specific gene combinations across the spectrum of living vertebrates will undoubtedly provide many clues about how this evolution occurred. Full documentation of the patterning processes will help.

The neocortex comprises about 95 percent of the cerebral cortex and is somewhat arbitrarily subdivided into four or five major regions (Buxhoeveden and Casanova 2002): the *frontal*, *parietal*, *temporal* and *occipital lobes* and sometimes including the *limbic system* (Figure 15-6). Although sensory processing is integrated throughout the brain, each of these lobes has specialized functions. The frontal lobes are involved in emotion, thinking, planning, personality, language, and sexual and social behavior, among other functions. Of course, some of these functions—as we humans choose to label them—are specific to humans (e.g., language, planning), and some (e.g., pheromone responses) to nonhuman (including some Old World monkeys and apes) vertebrates. But we must again guard against unwarranted anthropocentrism. Squirrels plan for the distant future when they bury nuts in the summertime; can we say they do not know why, nor envision a remembered meal on a snow-covered branch?

The cortical lobes are not symmetric, and, as a general rule, the human left side predominates in control of language, whereas the right lobe controls nonverbal activities. This is by no means invariant, nor is it clear how these separate functions are established genetically, although some of the molecular mechanisms are beginning to be elucidated (Essner et al. 2000; Morgan et al. 2002). The parietal lobe is involved in mechanosensory perception; touch, temperature, pain, pressure, as well as taste. It also helps integrate visual input. The temporal lobe is involved in auditory perception and integration, memory, and language abilities, among other functions such as the sense of "self" and "other," generally on the left and rights sides, respectively (this perceptual self-nonself distinction is not to be confused with a circulating, molecular, immunologically based distinction). The occipital lobe is responsible for some aspects of the processing and interpretation of visual input. The limbic system controls aspects of the autonomic and endocrine systems, including feeding behavior, homeostasis, and emotion.

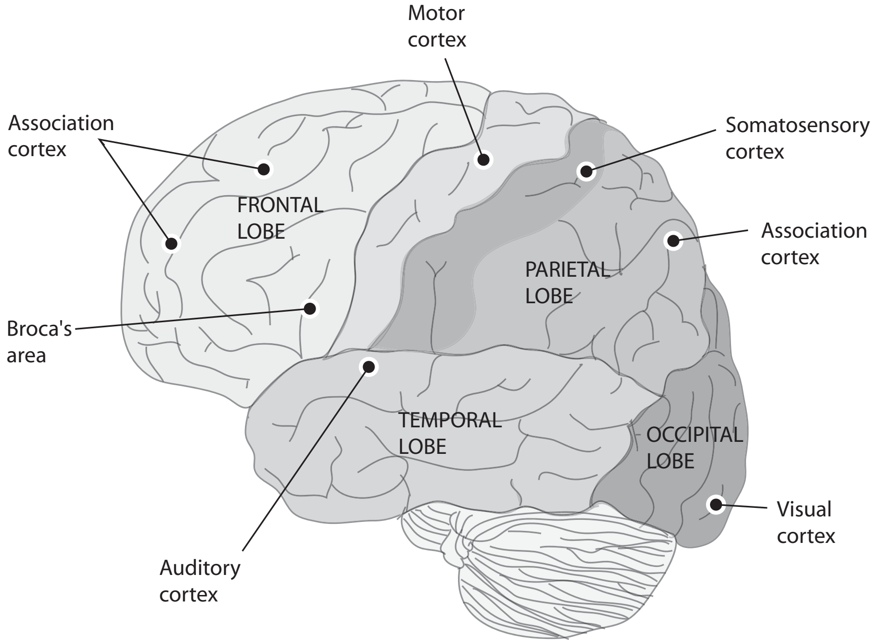
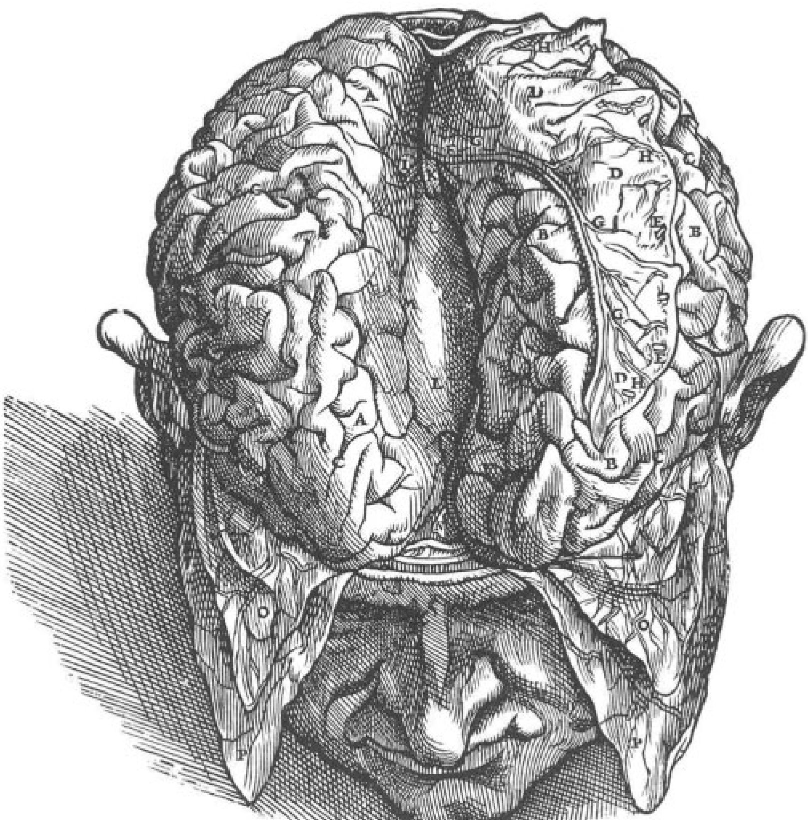


Figure 15-6. Lobes of the adult cerebral cortex and major sensory areas.

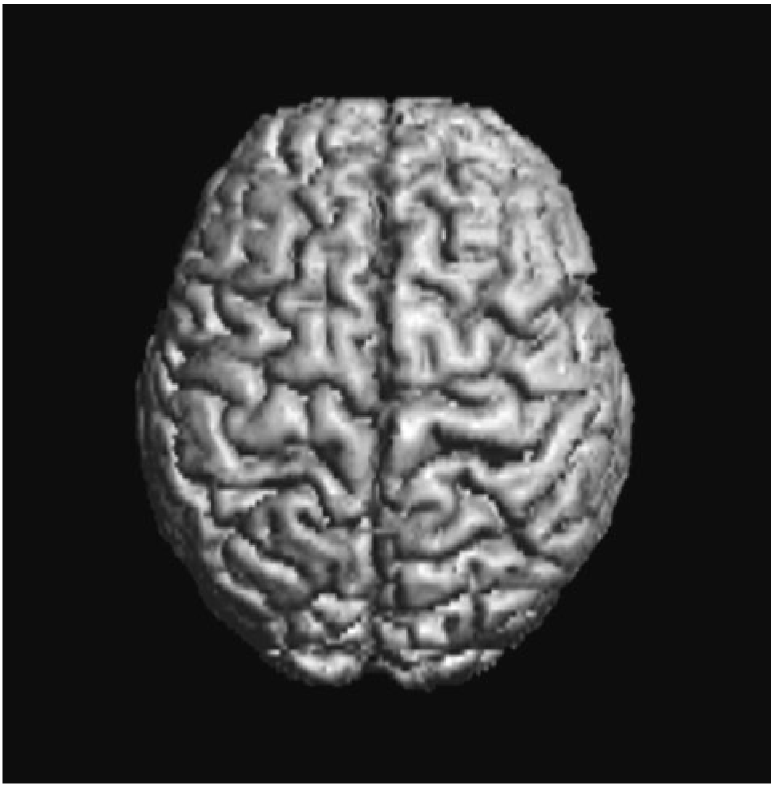
The external surface of the forebrain is characterized by species-specific patterns of fissures, *sulci* (grooves), and *gyri* (ridges) of tissue. The human brain is densely covered with such fissures. The degree of fissuring varies among species, however, with some like rodents having little such structure and primates much more. The deepest sulci reflect the early, major, primary division of the forebrain into its various lobes, which are not only phylogenetically conserved but reflect the kind of basic segmenting developmental processes we have discussed. However, the meaning of the rest of the fissuring is less clear, but is interesting from developmental, evolutionary, and functional points of view. Genetic evidence is currently weak and mainly suggests that overall pattern, size, and shape (plus the major fissural structures) are genetically influenced, at least to some general extent.

We have earlier seen drawings by the anatomist Andreas Vesalius. In his 1543 work, he also depicted the fissures of the human brain only *schematically* to indicate their nature. Today, we tend to assume things should be more literally represented to be "scientific." However, sulcal and gyral patterns vary even among twins and from left to right side in the same individual, and it is not yet clear how much



A

Figure 15-7. Brain surface patterns in humans. (A) Vesalius' drawing in 1543. (B) sulcal patterns vary even between left and right sides of the same brain. (A) Vesalius reprints are from (Vesalius 1543); (B) MRI data from Pearlson and Bata compiled into a figure by K. Aldridge and reprinted with her kind permission.



B

Figure 15-7. *Continued*

of this is purely random (and we don't understand the processes responsible). How much of the pattern is genetic, in what way it is genetic (specific sulci? the overall complexity of sulci?), and what functions are associated? For example, to what extent does the asymmetry reflect—or determine—the existence of hemispheric dominance, handedness, and so forth (Weiss and Aldridge 2003)?

These questions are not easy to answer but could be important to our understanding of brain evolution, particularly of the "higher" primates whose brains are particularly characterized by such patterns. If the fissure pattern has a large chance component, and if this affects function because, say, of the relative amount of cortex available for some particular function like literary or athletic ability—or Einstein's unique physics ability—then evolution might be viewed as providing that randomness as a kind of generator of variation, much as we see in our immune system. Alternatively, the details of fissure patterning could have no particular meaning. It is a very difficult question to answer, or even to state *how* to answer it.

The cortex is internally organized in a segmental way. The prevailing view is that the basic structure is columnar, with each column (or "minicolumn") being a fundamental functional processing unit about 50 μm wide and containing 80–100 neurons. All neurons within a column may serve the same sensory modality, although this is not a universally accepted interpretation (Nieuwenhuys et al. 1998).

The human cortex has six basic transverse cellular layers, each with distinct neuronal organization. The number of layers in the cortex differs among vertebrates, and the primordial number was probably two or three, as evinced by the number in contemporary lower vertebrates as well as in the evolutionarily earlier sections of the mammalian brain. Some of the layers contain neurons that project to subcortical structures; others contain neurons that extend only locally within the column. Groups of 60–80 minicolumns are, in turn, bound together by short neuronal connections into "macrocolumns," (Buxhoeveden and Casanova 2002). The laminar and columnar organization of nonmammalian brains is less visibly obvious, however. Columns are essentially the same size in most brains; apparently, the cortex expanded in size over evolutionary time with increases in column number, not column width or basic structure.

The six radial layers that form the primate neocortex are depicted in Figure 15-8. The cellular composition of the layers differs somewhat even among mammals (Northcutt and Kaas 1995), but, in primates, layer I is acellular, primarily an area in which incoming axons form synapses with dendrites from neurons from deeper layers. Layers II and III contain numerous pyramidally shaped neurons, *pyramid cells*, and these cells make synapses with many of the incoming axons from layer I. Layer IV contains star-shaped *stellate* cells, which have axons that remain local and receive input from axons carrying sensory signals from other regions of the cortex. Layer V is full of pyramidal cells with the largest cell bodies in the cortex. The axons from these neurons extend long distances, to the brain stem and spinal cord. Axons from pyramidal cells in layer VI extend to noncortical regions of the CNS, including the thalamus.

Neuronal Development and Migration

The cortex is thus a complex structure highly differentiated at the cell level and much more "refined" than the general regional segmentation reflected in the expression patterns described above. The ventricular zone is the source of differentiation of a variety of neurons and their associated *glial* (supporting) cells. In cortical development in mammals, neurons arise some distance away from their eventual destination. Neurons that will migrate to the cerebral cortex are formed early in gestation deep within the brain, in the ventricular zone. The VZ is developmentally transient; neurons form in this region, and when cells in the VZ stop dividing, neurons generally no longer proliferate.

In the forebrain, neurons migrate outward along transverse ("radial," inside-out, or ventricular-pallial) glial supporting cells, perhaps the way ivy climbs a trellis, though relying more on molecular than physical motive force, to form first a *preplate* above the ventricular zone. The next set of cells to migrate form a *subplate*, and these separate to form a region called the *cortical plate*. The outer layer of cells is then known as the *marginal zone*, and this differentiates into *Cajal-Retzius* cells. Neurons migrate into the cortical plate where they begin to differentiate, some becoming excitatory neurotransmitters, for example, and others inhibitory (the two provide a balance to activate and inactivate signal transmission at a given juncture, policing the neurotransmitter's ion channel states).

Neurons migrate along radial glial cells as guides. They enter the cortical plate and keep moving past other cells until they reach the Cajal-Retzius cells. Each new generation ends up passing previous generations to come to rest on the then-outermost layer. In this way, all six layers form, with layer VI deeper but forming earlier than layer II.

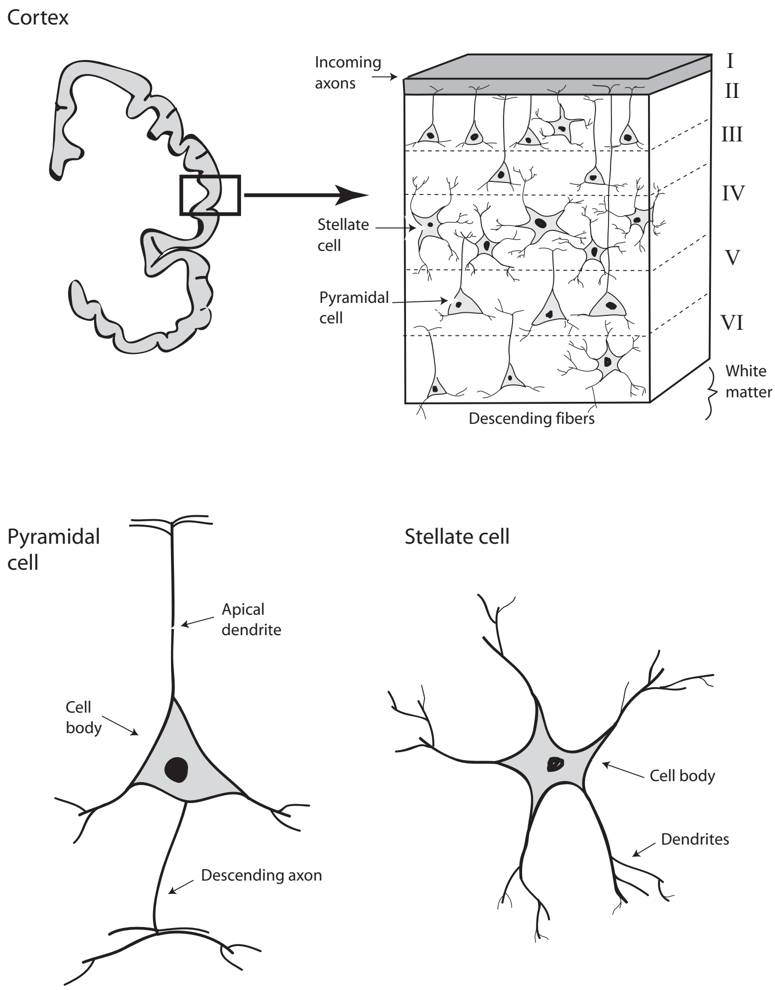


Figure 15-8. Cellular layers of the adult cortex; pyramidal and stellate cells, the principal neuronal cells of the cerebral cortex.

Some of these aspects can change over evolutionary time, perhaps by phenotypic substitution in which essentially the same functional result is achieved in developmentally different ways, even in related animals. In reptiles, the cortex develops in the opposite direction, from the outside in (Butler and Hodos 1996). The dorsal cortex is considered by some to be a homolog of the mammalian neocortex, and, unlike the neocortex, the dorsal cortex develops in such a way that the cells that form later are located below the earlier cells.

In an adult mammalian cortex, the cells on each layer share characteristic properties, such as the extent of axonal projection beyond or beneath the cortical column they form. But cellular identity seems to be plastic at least in part, determined by environmental cues, not genetic programming at first mitosis. Transplant experiments have shown that the cell's laminar position, when it undergoes its final differentiation, determines its fate. Arealization, however, does seem to be genetically programmed by mechanisms that maintain architecture or segmental boundaries, although neurons can also be redifferentiated to a new functional context.

The minicolumns of the cortex are homogeneous in appearance, despite the diversity of the information they process. This is in contrast with the specialized neurons in the different areas of the subcortical regions, which have an appearance specific to their task. The staged development of the mammalian cortex, with the cells that proliferate first forming the deepest cortical layer, and subsequent generations layering above them to form the layers closest to the surface, is called the "radial unit hypothesis" because it is believed to explain the development of the columnar organization of the cortex; related neurons develop by layering on each other and subsequently developing synaptic connections among themselves and in neighboring columns, to form the specialized regions of the cortex (Mountcastle 1978; Rakic 1988; Rakic and Lombroso 1998). Again, the functional significance of the columnar organization of the cortex has still not been substantiated, and, in fact, it is still not completely accepted that the entire mammalian neocortex indeed *is* structurally columnar (Nieuwenhuys, Donkelaar et al. 1998).

The molecular and genetic basis of these migrating pathways is beginning to be understood (Gupta et al. 2002; Hatten 2002; Lambert de Rouvroit and Goffinet 2001; Marin and Rubenstein 2002; Rice and Curran 2001). Much of the knowledge derives originally from naturally occurring human or mouse diseases, which led to the discovery of the genes involved. Here, we present a simplified account of what is currently understood. The Cajal-Retzius cells are a source of the *Reln* (or *Reelin*) gene product. A TF called *Tbr1* seems to be important in the formation of Cajal-Retzius cells and hence of Reln production. This secreted Reln protein attaches to the extracellular matrix and forms a concentration gradient from the marginal zone downward. Migrating neurons express various receptors on their surface, including the VLDR and ApoER2 lipoprotein receptors. These bind to Reln, triggering an intracellular messenger cascade that involves various factors (e.g., *Dab1*, *Cdk5*, *Cdk5r*) that, in ways not yet fully understood, may cause the neuron to lose cell adhesion cell-surface properties required to stay attached to guide glial cells. In any case, the neurons appear to drop off the glia at this point—right under the Cajal-Retzius layer that produced the Reln gradient to come to rest.

Radial migration retains the relative spatial location between the precursor cells in the VZ and their corresponding cells in the cortex; therefore, the cells involved have been called *projection neurons*. By contrast, other neurons migrate tangentially (laterally to the columns) and travel considerable distance within a layer to become *interneurons* and are thought to be involved in local circuitry integration (Anderson et al. 2001; Marin and Rubenstein 2001; 2003; Monuki, Porter et al. 2001; Monuki and Walsh 2001). In terms of what is known, the two types of neurons arise from different precursor populations and can be distinguished in part by expression of TFs that characterize the source and destination regions (Figure 15-5B).

Neurons can also be characterized by their major neurotransmitter expression, the particular ion channels, or other mechanisms used to receive or trigger the transmission of a signal impulse. The three main classes are functionally relevant as well as useful in identifying developmental lineages and aspects of brain regionalization. Their migration patterns are shown schematically in Figure 15-5B. Radially developing projection neurons generally are in the glutamatergic class, and tangentially developing interneurons are in the GABAergic class. Cholinergic neurons mainly use acetylcholine. Other neurotransmitters, such as dopamine, are used by other types of neurons.

Glutamatergic neurons predominate in the cortex, but the LGE and MGE send a substantial number of GABAergic neurons into the cortex (Marin and Rubenstein 2001; 2003; Puelles, Kuwana et al. 1999) (Figure 15-5B). GABAergic neurons migrate tangentially out of the MGE and LGE (e.g., Hatten 2002; Marin and Rubenstein 2001; 2002). The mechanisms of the extensive and highly patterned neural migration are varied. Some of the migrating cells express *neuropilins*, which are receptors for semaphorin 3A and 3F. The latter are expressed in the *striatum* region (below the cortex, including the LGE), and this appears to guide or trap these interneurons for a striatal destination, whereas other cells that do not express neuorpilins migrate to the cortex. Guidance of this migration seems to be affected by extracellular matrix molecules, including gene products *Slit* and its ligand *Robo*, and *Netrin* and its ligand *Dcc*, but details are not well known.

Olfactory neurons are a special class because they are continually replaced by new olfactory neurons, which originate in the subventricular zones and move to the olfactory bulb in a chain of cells ensheathed by protective, guiding glial cells. Two axon guidance repellents include *EphB* and *EphA* RTK receptors and their *Ephrin* ligands, and surface cell adhesion gene products are also important (see below).

How Are Boundaries and Segments Maintained?

The migration of neurons is but one of several cell migrations in vertebrate development (and such things occur to a lesser extent in other animal groups); NC cells are a prominent example. Cells that migrate need to find a path, to have a mechanism for following it, and to associate only with the appropriate cells in the neighborhood as they pass through. Migration involves both intracellular mechanisms and external chemotactic (attractant or repellant) clues and involves the production of signaling molecules that may diffuse or may be attached, as in the Reln mechanism for projection neuron migration. Also, the migrating cells must express the appropriate receptor.

Experimentally mingled neural cells may aggregate if they are genetically similar or repel if they are different. Alternatively, introduced cells may redifferentiate to take on the expression patterns of their surrounding cells.

Several cytological mechanisms are involved in boundary maintenance. Cells may bundle physically together based on identity of cell surface adhesion molecules. One family of such molecules is the *Cadherins* (e.g., Pasini and Wilkinson 2002). Cells that express the same set of Cadherins on their surface adhere to each other but will not adhere to cells expressing other members of the gene family. In this way, Cadherin expression "codes" can subdivide the developing brain either regionally early in development and/or functionally later on. Fiber tracts, or bundles of cells of common origin, can stay together, retaining morphological or functional coherence, while crossing through regions.

For example, as indicated in Figure 15-5A, the cortex expresses *Rcad*, whereas the adjacent LGE expresses *Cad6*. That the two sets of cells are distinct is also reflected by their expression of *Pax6* and *Dlx1*, respectively. Combinatorial *Cadherin* expression codes appear to play a role in maintaining prosomere integrity from the early to the mature brain (e.g., Redies and Puelles 2001). Fiber tracts from the motor area of the cortex, or to the cortex from auditory and visual sensory organs, pass through several brain regions and must thus maintain coherence. Thus, areas in the cortex express *Cad6*, *8*, and *11*, as do the incoming sensory thalamic nuclei (see below).

Once appropriately attached, neurons are physically guided by glial cells, hence keeping the right neurons going to the right place even when surrounded by other types of cells. Olfactory neurons may retain (or obtain) their specificity in terms of the olfactory receptor gene they express. This may serve as a kind of guide mechanism between the parts of brain and olfactory epithelium appropriate for that olfactory receptor gene (of course, this does not address the question of what determines spatial expression of different types of genes in the first place). Two subsets of *Ephrin* genes for membrane-bound tyrosine kinase receptors and their ligands, designated *EphA* and *B*, appear to be involved. Class specificity appears to be involved in axon guidance by separating A-expressing from B-expressing cells. *Ephrin* expression differences help separate the hindbrain into odd- and even-numbered rhombomeres. Other cell adhesion molecules like *Integrins* and diffusible gradients (*Netrins*, *Slit*, and others) also help guide axon migration (Redies and Puelles 2001).

The familiar *Notch* receptors, their ligands *Delta* and *Serrate*, and the modifying gene *Fringe* play compartmentalizing roles in *Drosophila* patterning, and the homologous mechanism appears to be involved in vertebrate neural development. Fringe modifies Notch receptor structure, suppressing its activation by Serrate, and together these genes can establish patterns of affinity or repulsion. The same system seems to be involved in the specification of certain neuronal precursor cells in the VZ to differentiate into neurons, while surrounded by inhibition zones of nondeveloping cells. *Notch*-related patterning mechanisms also appear to be involved in thalamus development and in dendrite morphology in cortical neurons and perhaps gliogenesis.

Neurons can also be reprogrammed when they contact differently programmed neurons. Reprogramming of intermingling neurons by their local context has been shown experimentally in the hindbrain (Pasini and Wilkinson 2002). As noted earlier, the isthmus separates anterior and posterior cell types, which we can identify by their morphology and expression of indicator genes *Fgf8* posteriorly and *Wnt1* anteriorly. Transplanted cells become reprogrammed to express indicator genes appropriate to their new context. The initial boundaries are sometimes "fuzzy" but subsequently sharpen. A model suggested by Pasini and Wilkinson is that cells respond to quantitative gradient thresholds of SF molecules like Fgf8. This is not a precise mechanism, but response to SF levels exceeding the threshold induces appropriate *Hox* and associated *Krox20* and *Kr* expression. In turn, this induces *Ephrin*, which, being a qualitative cell-binding system, sharpens boundaries between cell types. Consistent with these ideas is that inactivation of genes mentioned in this chapter often has quantitative rather than clear, simple, or qualitative effects.

Not only must a cell have a path, it has to have a means for moving along it, and there are several (Gupta, Tsai et al. 2002). In one, the cell creeps along following a projection that detects extracellular signals or works its way along a glial guide cell. In another, the cell sends out a filamental projection (*filopodium*) toward a guiding signal, which fills with cytoskeletal tubules. The nucleus and rest of the cell are then pulled into the projection, and the process begins anew. Appropriately, cytoskeletal and adhesion genes (*Lis1*, *Dcx*, *Filamin1*, *Cdk5*, and *Astn1*, *Integrin3*, respectively) have been found to be mutant in natural disorders of cortical layering (or the latter produced by experimental mutation in these genes (Gupta, Tsai et al. 2002; Hatten 2002; Marin and Rubenstein 2002).

Function and Cortical Areas

We can relate these various developmental process notions to the way that the neocortex becomes functionally organized, peri- or postnatally depending on the species, into *areas* of the brain specific to different sensory inputs, such as optical, auditory, or olfactory areas. The thalamus is a relay center for most incoming sensory signals as we will see in Chapter 16 in regard to particular sensory systems. At present, it is not readily apparent whether the role of thalamocortical axons (TCAs) in arealization is molecular or due to external cues like sensory input relayed to the brain, and there has been longstanding debate about this issue.

The *protomap* hypothesis suggests that the arealization is controlled intrinsically, when the neurons are generated (Rakic 1988). This would imply a kind of prepatterning in the brain, which would not be without precedent in vertebrate development, as we have seen. The alternative *protocortex* hypothesis suggests that arealization instead takes place when the neurons are in place and receiving incoming sensory signal (O'Leary 1989). For example, "rewiring" experiments in which afferents are redirected into areas to which they do not normally project—for example, optic afferents sent to the olfactory area—result in functional remapping of the reassigned cortical area (e.g., Scalia et al. 1995), suggest that the arealization of the cortex is driven by experience rather than by preprogramming. This view would be consistent with the brain as a flexible instrument of organismal survival rather than a form of response automaton. It would also be consistent with examples we have seen, including in neurally relevant situations, such as epithelial-mesenchymal interaction involving migrating NC cells.

Both kinds of patterning are probably important and act separately and synergistically, at different developmental stages (e.g., O'Leary and Nakagawa 2002; Sur and Leamey 2001). There are ample mechanisms available in the developmental repertoire. Before their differentiation, prenatally, differential expression of a number of familiar gene families and pathways is seen in numerous neocortical regions, as discussed above. The areas are then differentiated cytoarchitecturally and chemoarchitecturally, as well as by their distinct efferent and afferent pathways.

The borders between many functional areas in adults are distinct, and easily discernible, but the borders of these regions of gene expression during development are not. The borders subsequently become distinct at about the time that TCAs have extended to cortical layer IV, around the time of birth. The temporal patterning, coupled with many negative results of searches for genes confined to functional cortical areas before thalamic innervation, suggests that perhaps prenatal regionalization is genetically controlled but that arealization is controlled by thalamic inputs (Pallas 2001).

A FEW STRAY THOUGHTS

Much remains to be learned about the role of specific genes in brain development, but what we know so far leaves us with a kind of paradox. The brain and its intricate connections with the entire rest of the body might seem to be a system of a different kind from others. Yet we have seen most of the genes and signaling mechanisms in brain development, like *Fgf* or *Bmp* or *Shh* signaling, or *Hox* combinations, elsewhere in the body.

Patterns of gene expression during brain development generally suggest roles for genes that are confirmed by gene inactivation or mutation experiments (e.g., Gupta, Tsai et al. 2002; Marin and Rubenstein 2002), and these genes seem to be working in ways expected of them. For example, an experiment in which expression of *Fgf8* at embryonic day 11 in mice was altered or blocked in the developing neocortex before the formation of thalamocortical connections, yielded animals with dramatically reorganized cortical maps including cortical pattern shifts (Fukuchi-Shimogori and Grove 2001). *Bmp4* overexpression had segmental cytodifferentiation effects (Gomes et al. 2003). In both cases, the effects were confined to the expected expression of these interacting SFs (consistent, for example, with their playing roles in activation-inhibition kinds of patterning). Knockouts of the regulatory genes *Emx2* or *Pax6* have also altered cortical organization, with loss of portions of the cortical epithelium (Pallas 2001). *Holoprosencephaly* is the failure of the two hemispheres to separate, and several genes are known that can produce this; these have relevant expression domains. *Reelin* was discovered in a behaviorally anomalous animal whose cortical layers are aberrantly developed.

These experiments and many others suggest that the brain is regionally patterned through familiar mechanisms. However, even after extensive work, only a few genes have been found to be expressed solely within defined cortical areas *before* thalamocortical innervation, suggesting that definition of areal borders requires input from thalamocortical neurons relaying signal from sensory receptors themselves (Pallas 2001). Perhaps there is a kind of reverse prepatterning, in which the central location awaits information returning there from distal locations.

The nervous system is not only highly integrated at morphological and cytohistological levels itself, but must be integrated in relatively precise ways with all other body systems. This has to occur in real time, because of the intricate innervation of all structures. It may be that neurological development involves larger or more complex sets of genes than other systems, and the brain is an organ that expresses a higher fraction of all genes than most (however, the liver is another expression-rich organ).

Presently, experimental gene manipulation involves only one or at most two or three genes. Such experiments are vulnerable to oversimplified interpretation. For example, *Shh* mutations lead to holoprosencephaly, but many other genes or chromosomal regions have also been implicated in this trait, showing that a set or network of pathways rather than a single gene is responsible. Tens if not hundreds of genes, when mutated, can affect very specific aspects of human brain function or cognition—learning, language, or personality—or very general traits like intelligence or motor control. Genes may affect such traits through morphological development or subsequent function (e.g., neurotransmission).

Perhaps it is a human conceit by which we view the brain as more intricate than other systems or having uniquely complex higher-level organization. A lesson in humility is to dissect a limb, or hand. Still, the higher integrative functions *are* interesting in their own right. Major success in understanding them will come when we learn how expressing particular networks of genes leads to the end phenomena—smell, vision, problem solving, and the like—but this may have to await better methods for understanding emergent traits.

There are other interesting differences from nonneural traits that may be important here. In structures like ribs, eyes, and gut, the differentiated cells stay where they are and function locally. In the nervous system, local differentiation results in distant function. An electrochemical impulse in the nose leads to a "smell" in the brain. Indeed, neurons develop not only from local signals that start them off on their journey, but they respond to local context as they move or extend into different parts of the brain, or extend into distant parts of the body. Cognitive and integrative functions seem to be different sorts of phenomena from digestion, muscle contraction, and making bone. We have very few ideas about the ultimate nature of these fundamental brain functions.

Nonetheless, a prediction borne of experience is that these systems are not as unique as they seem to be but once again involve familiar elements. For example, the traditional notion of a prepattern is that prepared cells remain developmentally quiescent until they encounter an inductive signal. An example may be the role of neural crest cells in epithelial-mesenchymal interactions that develop structures like feathers. However, these cells follow signals along the way and there may be more direct contact or connection as both the NC cells and the tissues through which they move both grow out from the midline. Neurons also follow signals as they grow to innervate developing tissues like skin, muscle, and gut. This kind of prepared-track may be responsible for the migration of olfactory neurons, and perhaps in a way not yet known, for that of interneurons into the cortical layers (as shown in Figure 15-5B). Neurons also follow the growing structures they will innervate.

Some if not most of the signaling and cell-movement mechanisms also involve familiar cytoarchitectural and cytoskeletal genes and interactions. *Reelin*, for example, is expressed in cells elsewhere than the brain. *Tbr1* that may help induce *Reelin* expression may be largely confined to the forebrain, but is part of a larger family (the *T-box* TFs) that is, like other such families, old and has more diverse expression. Most of the architectural and extracellular matrix genes referred to in this chapter, that help guide axon migration, have multiple expression sites.

There are homologies between invertebrates and vertebrates in many of the pathways and mechanisms for neural cell-type specification and brain regionalization (Hirth and Reichert 1999). *Hox* genes are expressed in the embryonic vertebrate hindbrains and caudal nervous system of insects (based on studies in *Drosophila*) in a spatiotemporally conserved way and are involved in neuronal identity. *Otd/Otx* genes are expressed anteriorly, in rostral brain development in insects, and are required for vertebrate mid- and forebrain formation. A mechanism identified in vertebrates—for example, by a human disease—can often, perhaps typically, be found to have a homolog in *Drosophila* and many other animal species (even yeast or fungi). This is useful for learning about the mechanics and origins, but does not tell us what the brain is actually doing.

Nonetheless, we have at least a general sense of the overall organization of central nervous systems in different classes of animals. This makes it possible to make more sense of the way information received from an external source by the various sensory systems is processed, to which we now turn.

Chapter 16
Perceiving: Integrating Signals from the Environment

> . . . every sense has something peculiar, and also something common; peculiar, as, e.g., seeing is to the sense of sight, hearing to the auditory sense, and so on with the other senses severally; while all are accompanied by a common power, in virtue whereof a person perceives that he sees or hears (for, assuredly, it is not by the special sense of sight that one sees that he sees; and it is not by mere taste, or sight, or both together that one discerns, and has the faculty of discerning, that sweet things are different from white things, but by a faculty connected in common with all the organs of sense; for there is one sensory function, and the controlling sensory faculty is one . . .)
>
> (Aristotle, *On Sleep and Sleeplessness*)

In Chapter 16, we will provide a general overview of the ways organisms collect and make use of different signals from their environment. We will describe the peculiarities of each sense, as well as what is known of the "common power" that the senses share. Intriguingly, the answer to the question of how "a person perceives that he sees or hears" may be as elusive to us today as it was to Aristotle. We know many more details about the senses and how they function than did Aristotle, particularly at the molecular and cellular levels where he had no knowledge at all; however, we are still pondering how "that faculty connected in common with all the organs of sense" interprets the information the senses collect.

For some biological functions, there is a relatively direct correspondence between genes and function. For example, the protein code specified by the genes for the hemoglobin molecule relates to that molecule's ability to bind heme (an iron group) that relates to its capacity to bind oxygen. And similarly, the protein structure of collagen is related directly to its physical role as a connective tissue. But is there any such connection between protein structure and the "common power" of percept? If so, it seems totally to have eluded us. Instead, what seems much

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more likely is that the power lies strictly in the *organization* of neural processing systems.

How this works is not clearly related to genes per se but to their expression patterns (including during development). Similarly, as Darwin's and Bates' comments on insect behavior mentioned in previous chapters show, we cannot attribute function just to a numbers of cells game. Once again we can cite Aristotle. Although Francis Bacon and others who came along nearly two millennia later are usually credited with inaugurating the reductionist worldview of science, in his *Physics*, Aristotle asserts that "we do not think that we know a thing until we are acquainted with its primary causes or first principles, and have carried our analysis as far as its elements." This was not exactly modern molecular reductionism, but it is curious to ponder how apt this view may be in regard to understanding cognitive function, an "emergent" trait if there ever was one. Once again, and especially for traits determined by interaction among elements rather than the inherent properties of the elements, the specific genes used to lay down the wiring network are functionally arbitrary when it is within that network itself that the function lies. As a result, the survey in this chapter is not strictly "genetic," even if perception ultimately depends on gene expression.

In Chapter 15, we laid the basic groundwork for understanding how "higher" organisms integrate environmental cues by describing the development and structure of central nervous systems (CNS). Again, it is important to remember that only a small subset of life collects and makes sense of environmental signals in this way. Despite our own very understandable romance with the importance of the head in "higher" organisms, many organisms with noncentralized signal collection and integration systems, from single-celled bacteria to bacterial colonies to cnidarians and echinoderms and plants, have been supremely successful with no head.

LIGHT

As we saw in Chapter 14, most organisms in this world, single-celled bacteria, plants, insects, mammals and fish, make use of light energy in some way. Some convert solar into chemical energy, whereas others have receptors that trigger responses ranging from movement toward or away from a light source to conversion of the light into a perception of the surroundings.

Plant Light Reception

Plants use light in more ways than do animals and have evolved more pigments to collect and transduce it. They use light for photosynthesis, photoperiodism, photomovement, and photomorphogenesis. We might say plants have organismal responses that require some forms of perception because they respond to light in multiple ways (e.g., positive phototropism in stems and leaves and negative phototropism in roots), and these responses are based on signals beyond light itself (e.g., gravity, hormones). Furthermore, responses vary from intracellular ones, such as chloroplast movement, to higher level ones, such as stomata opening or differential growth of different sides of branches in the process of photomorphogenesis. Plants are also responding to more than simply the presence or absence of sunlight: they also respond to wavelength, intensity, and directionality of the light, as well as diurnal length.

The plant's local, noncentralized photoreception means that, although the photosynthesizing leaf cannot survive without the roots and circulatory system of the plant, the plant itself can survive without a specific leaf or even without many of its leaves. Unlike an organism with a cephalized nervous system, which cannot perceive light without its central ganglion (the head), a plant, with its many loci of efficient photosynthesis, can lose a lot of leaves before it isn't taking in enough solar energy to survive or to make local or organismal responses to light. We can think of this as automatic or purely molecular rather than "intelligent" response—so far as we know (reputed response to music and kind words notwithstanding), plants are unaware that they are reaching toward the skies—but, unless we agree to reserve the term for brain activity, that distinction seems somewhat artificial. Indeed, it may be rather anthropocentric and consciousness-focused to make such a distinction.

We also noted in Chapter 14 that algae can sense light and respond with phototactic motion directed in relation to its source. This involves complex signaling and response mechanisms, although of course less complex than in, say, organisms with a CNS that must respond by coordinated locomotion involving multiple parts. It is important not to forget that the result is not just the activation of a molecular pathway but an organismal response, even in the lowly algae.

Insect Vision

Drosophila have two visual systems: the compound eye for image formation and ocelli for light detection. The compound eye is formed of 700 or so ommatidia, each of which contains photoreceptor cells. The ommatidia are arranged in a honeycomb-like fashion, and the entire eye connects to three relay points in the optic lobes of the fly's CNS: the *lamina, medulla,* and *lobula complex.* The simple central *ocelli*, of which *Drosophila* have three, synapse with a single point in the CNS, the ocellar ganglion.

The ommatidial neurons terminate in one of the three layers. The projections from a single ommatidium extend to the lamina, and from there some axons continue on to the medulla. As in vertebrate vision, these projections are topographic. That is, the image is relayed to the brain in a way that preserves its physical layout or "map" in the eye which, because of the linear propagation of light itself, comes directly from the perceived image.

Some insects have an extraocular or *dermal light sense.* This has been shown with experiments in which the eyes have been made nonfunctional. This sense involves single neurons in the brain and/or ventral nerve cord responding to light.

In Chapter 14, we described some key aspects of the cascade of regulatory gene expression that is used to form the insect eyes themselves (Cutforth and Gaul 1997; Czerny et al. 1999; Punzo et al. 2002). Some of the regulatory genes that are involved in development of the optic lobes of the adult *Drosophila* nervous system have been identified and include *Wingless* (*Wg*) and *Decapentaplegic* (*Dpp*), genes already familiar to us because of their involvement in many other structures. *Minibrain,* a protein serine/threonine kinase gene, and *Division abnormally delayed* (*Dally*) are also involved in cell proliferation in the visual lobes of the fly brain. The final differentiation of cells in the developing lamina depends on the arrival of axons from the ommatidial photoreceptors, in a way homologous to the development of vertebrate olfaction (Cutforth and Gaul 1997), which will be described below. The process involves differential signal transduction, in this case among adjacent cells.

How an insect "perceives" light images is, however, still unknown, especially in terms of its experience. For example, how does a particular image generate a response such as flight or avoidance? Can these be called "emotional" responses, and if so how are they "felt" by the fly?

Vertebrate Vision

Light sensation typically requires detection of direction and strength of the signal. Some organisms do more with light, and it is for this that we use the term "vision." Except perhaps for touch, vision is the one sense that specifically requires the (sometimes detailed) *spatial* characterization of the signal. It is this that allows us to interpret an *image* that comes essentially in the form of pixels, to give an integrated assessment of the spatial relationships of objects distant from us, and we might say this is the value of vision. Eyes such as ours perceive many other aspects of light in addition to spatial ones, providing an even more nuanced sense of our environment.

We tend to think of the vertebrate retina as a simple analog of a camera's passive film or other kind of pixel sensor. Rods detect brightness and cones the various primary colors. But the interpretation of light absorbed by the rods and cones begins before the signal leaves the eye. Retinal neurons are specialized for different aspects of light, and the image that is sent to the brain is not that of absolute levels of illumination but instead a retinal map of spatial patterns of regions of relative light and dark, color, intensity, depth, and so forth.

The spatial orientation of the light that hits the retina is precisely maintained, although somewhat distorted, in the form of a *retinotopic map*, sent to each visual processing center in the brain; the map is intact in that neighboring cells in the retina project to groups of neighboring cells in the visual area of the thalamus, which in turn project to neighboring regions of the *striate cortex* (Kandel et al. 2000). "Somewhat distorted" means that the *relative* position of the signals is retained but the translation of a three-dimensional image received by a curved retina into a two-dimensional one necessarily alters it somewhat, and the absolute distances between receipt centers of two signals in the brain and between their receiving photoreceptors is not maintained. The *percept* provided by the brain compensates for this kind of "stretch" distortion with mechanisms at least partially understood (see below). In other words, the correction appears to be such that the percept more closely maps to the image striking the retina than to the way that image enters the brain. In fact, most sensory systems project their receptive surface to the appropriate brain centers in a similar way, but the spatial correction is most important in visual and tactile perception.

Two predominant pathways carry the signal from eye to brain in vertebrates: the *retinotectal* and the *retinogeniculate* pathways. In vertebrates with a small forebrain, such as birds, amphibians, and reptiles, the retinotectal projection is the most prominent. Here, the photoreceptors in the retina project onto bipolar nerve cells (cells with two processes extending from the cell body, the axon and the dendrite), with limited dendritic branching on one end and a short axon on the other, a structure that allows fast and precise conductance of the signal. These neurons in turn connect with retinal ganglion cells that form the optic nerve and go primarily to a part of the midbrain called the *optic tectum* ("roof"). This structure coordinates orienting responses—turning toward light or sensing prey or danger—rather than analysis of form or shape or the like, as in mammals. That is, retinotectal vision is predominantly linked with motor control and is representationally rather crude.

In mammals, the homolog to the optic tectum is the *superior colliculus* (a small moundlike region in the midbrain), which receives light signals, as well as other kinds of sensory input. The superior colliculus helps orient the head and eyes in relation to these other kinds of sensory information.

The second pathway, the retinogeniculate, is the predominant visual signal relay pathway in mammals but is only barely evident in vertebrates with small forebrains. As in the retinotectal pathway, light signals are conveyed along retinal ganglion cells to the visual processing centers of the brain. The retinal ganglion cells are called X or Y cells (M or P cells in primates) and comprise two different major routes for visual information to reach the brain. The X and Y cells transmit slightly different visual information to slightly different areas of the brain, with the X cells projecting to the magnocellular, or large-celled layers, and the Y cells projecting to the parvocellular, or small-celled layers of the laminar *lateral geniculate nucleus* (LGN) in the thalamus.

The most important difference between the X and Y cells is their response to color contrasts; X cells are essential for color vision. But X cells are also important in distinguishing images that require high spatial and low temporal resolution (i.e., they specialize in sustained responses and are best at analysis of stationary objects), whereas Y cells are important in vision that requires the opposite: low spatial and high temporal resolution (i.e., they have fast and transient responses and detect movement, basic shape, depth, brightness, texture, etc., of objects but are poor at analysis when objects are stationary) (Kandel, Schwartz et al. 2000).

Retinal ganglion cells exit each eye bunched together into the *optic nerve.* The optic nerve from each eye meets in the *optic chiasma* (see Figure 16-1), where signals from each side cross to the other brain hemisphere, to travel on to the lateral geniculate to be relayed from there to the primary cortical visual center. This crossover is essential for coordination of the images from both eyes to create stereoscopic vision.

Each neuron in the LGN is responsive to a single spot of light in a region of the visual field, and each layer of the LGN is monocular, that is, responsive to only one eye. The retinotopic map is maintained by the neurons of the LGN, which are primarily projection or *relay* neurons, that in turn project to different layers of the *primary visual cortex* (also called *Brodmann's area 17*, *V1*, or the *striate cortex*) of the forebrain. The X and Y pathways still project to different sublayers of the cortical layer IV, maintaining the separation of input from each eye.

The primary visual cortex, as much of the nervous system, is modularly organized into sets of columns. This "multiple columnar system," with hypercolumns for different tasks, includes *orientation* and *ocular dominance columns.* The neurons in the ocular dominance columns receive input from a small spot on the retina of a particular eye, and the orientation columns receive input from light that hits the retina on a particular plane, that is, of lines and edges all tilted at essentially the same angle to the vertical. To represent all orientations for both eyes, 18–20 columns are required. Neighboring hypercolumns represent neighboring sections of the retina. The neighboring hypercolumns communicate through horizontal connections that link cells with the same specific tasks in the same layers and in this way integrate information over many millimeters of cortex. Information from outside a cell's

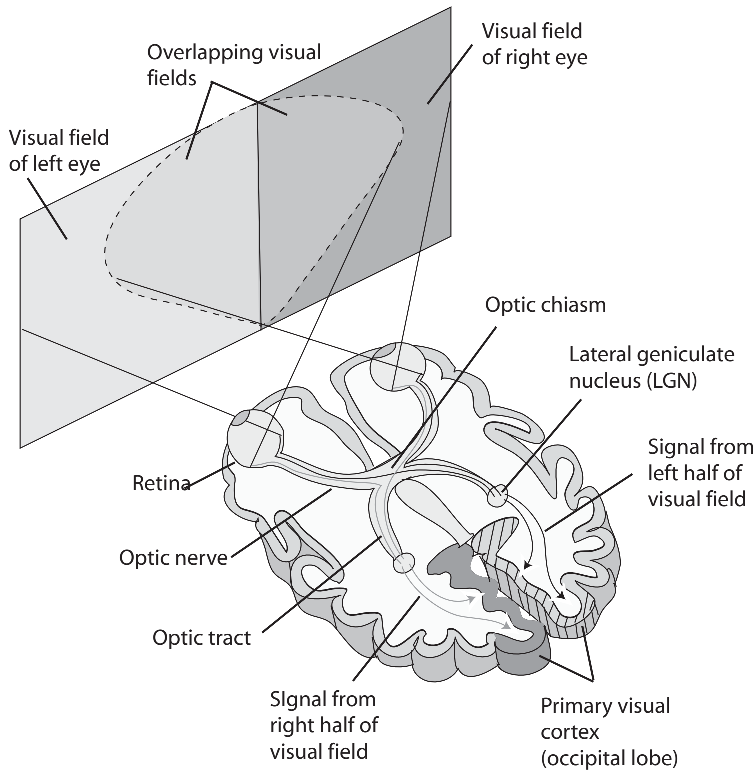

Figure 16-1. Transverse section showing pathway of visual signal to the human primary visual cortex. Redrawn from (Driesen 2003).

immediate environs may influence the way it processes information and thus influence the way we evaluate in context what we see.

After visual information leaves the primary visual cortex (V1), the signals go to 30 or more secondary visual processing centers in the occipital lobe and parts of the parietal and temporal lobes (Figure 15-6). In the end, perhaps 50 percent of the neocortex is involved in visual processing. Beyond the striate cortex, signal is processed with respect to color, motion, intensity, depth, and form.

The X and Y pathways remain segregated as the information leaves the primary visual cortex. The X pathway extends into the inferior temporal cortex, as the ventral cortical pathway and the Y pathway extend to the posterior parietal cortex. The dorsal pathway is largely responsible for the perceptions of motion, depth, and form, and the perception of contrast and contours takes place in the ventral pathway, although there is a good deal of overlap. Put simply, the dorsal pathway determines *where* an object is, and the ventral pathway is involved in recognizing *what* the object is (Kandel, Schwartz et al. 2000).

Visual images are passed to our brain inverted and backward, as they are represented on the retina, a curved but essentially two-dimensional surface. Bilateral symmetry allows organisms to see on both sides of themselves. One eye detects what is to the left of the organism, the other what is to the right. In some organisms, both eyes face forward enough that the fields of vision (images) of the two eyes overlap;

it then becomes possible to characterize the outside world more precisely in three dimensions, that is, additional information about relative distances from the eyes is available. The three dimensional aspects of vision are reproduced in the higher cortical centers, using both monocular and binocular cues. Binocular resolution allows the kinds of "triangulation" of differences that enable us to perceive depth or distance. Our brains apparently learn from experience to integrate the many cues to determine an object's distance, its size relative to other objects in the frame, motion, and other characteristics that give an image its three dimensionality, as well as to correct the inversion and reversal of the image as it enters the brain.

The retinal image is represented numerous times in the cortex, with many cells processing the image in different ways at once, in at least two different pathways. How do all the separate representations of what we see come together into a single image? This is known as the Binding Problem and is one of the central questions in cognitive psychology. The answer is still not clear. The brain apparently constructs a visual image in layers, by putting together the numerous interpretations produced at each level of processing, but whether there is a common pathway that integrates all this or whether the various afferent pathways interact in some way is not yet known.

Extraocular Vertebrate Photoreception

We referred briefly in Chapter 14 to the fact that organisms perceive light in many ways that do not involve eyes. We referred to vertebrate extraretinal photoreceptors that are important in regulation of circadian rhythms and photoperiodicity. The pineal and parapineal glands within the brain are the most important such photoreceptors. Brains are more or less permeable to light, depending on the circumference of the head: although the light becomes somewhat refracted and filtered, in many animals sufficient light penetrates the skull and brain to reach photoreceptors in the pineal and parapineal glands. In larger animals, light input to the pineal gland is part of the visual pathway. The pineal gland regulates synthesis of the hormone melatonin, which is derived from the neurotransmitter serotonin and is secreted at night. The neurochemical basis for its regulation of circadian rhythms is not known. Although this response to light is in some senses behavioral, it appears to be simple in that it does not involve detecting other aspects of light such as direction or, especially, image.

Many animals also use light in nonperceptive or certainly nonbehavioral ways. Some require sunlight in the ultraviolet range, for example, for the conversion of cholesterol into vitamin D or to induce the production of melanin by melanocytes to protect cells from the damage ultraviolet light can cause to DNA. Interestingly, melanocytes are derived from NC cells and in that sense are neural structures.

VIBRATION SENSING

Vibration, the wavelike movement of air or other objects in which an organism comes into contact, can be created by traveling sound waves or by mechanical disturbance (such as water currents or wind). By itself, vibration detection is a form of the sense of touch. However, like light, there is additional information carried by the frequency, amplitude, complexity, and location of vibratory motion. It is this that we refer to as hearing, but as usual there are subtleties and gradations. And it is not only animals that sense and respond to vibration.

Plants

Some plants, such as mimosa, are able to perceive vibration and respond by folding the leaves that have sensed the motion. This is presumably a protective response, to shield delicate leaves. Response to vibration is called *seismonasty*; the plant may have the same response to touch, and this is called *thigmonasty*. The response is caused by changes in turgor pressure, which is driven electrochemically, through depolarization of cell membranes. The cell membranes of specialized *motor cells* become more permeable to potassium ions, resulting in the outflow of ions and osmotic loss of water from the cell. This results in the shrinking of the cells that sensed the vibration, which causes the leaves to droop. A whole plant may respond to loud sound or shaking, but this is not thought to be an organismal response, but one that takes place independently in each leaf.

Lateral Line

The lateral line is a vibration detection mechanism of fish and aquatic amphibians (see Chapter 12 and Figure 16-2) and is used for many vital individual as well as social behaviors. The spatial distribution of its mechanoreceptive organs, known as neuromasts, determines the receptive field, which can vary among species, and the innervation patterns determine how sensitive the system is going to be. Neuromast morphology also varies and is related to the types of fluid motion that will be detected, whether velocity or acceleration (Maruska and Tricas 1998).

The lateral line system comprises a series of neuromasts located in tunnel-like canals in the dermis of the head and along the midlateral flank of the fish and in pit organs throughout the body. The tunnels open at intervals at the body surface to expose the mechanosensory cells to the exterior. Neuromasts are the basic functional organ of the lateral line and are composed of groups of about 30 sensory hair cells and 60 supporting cells and are covered by a gelatinous cupula. The hair cells of the lateral line are essentially identical morphologically and functionally to those in the vertebrate inner ear and detect movement via the displacement of stereocilia in the same way. Most fish have both canal neuromasts and superficial neuromasts, which detect water current. Neuromasts are innervated by branches of the posterior (PLL) and anterior (ALL) lateral line nerves—those on the head by the ALL and those on the sides and tail by the PLL. Projections from the ALL and PLL extend to two locations in the hindbrain in two neighboring columns, preserving a somatotopic representation of neuromast order and thus the location in the body where a vibration was perceived.

The vertebrate homologs of the *Drosophila* proneural gene *Atonal* (*Ato*) include *Math1* in mouse, *Zath1* in zebra fish, and *Atoh* in humans; these bHLH-class TFs are all essential for the development of inner ear hair cells and promote neuroblast differentiation and subsequent differentiation of the peripheral and CNS, including the brain (Itoh and Chitnis 2001).

There is great interspecific diversity in brain morphology among fish, with the area or size of a particular functional part of the brain correlated with modal specialization of the species. Deep-water fish have poor color vision, for example, and

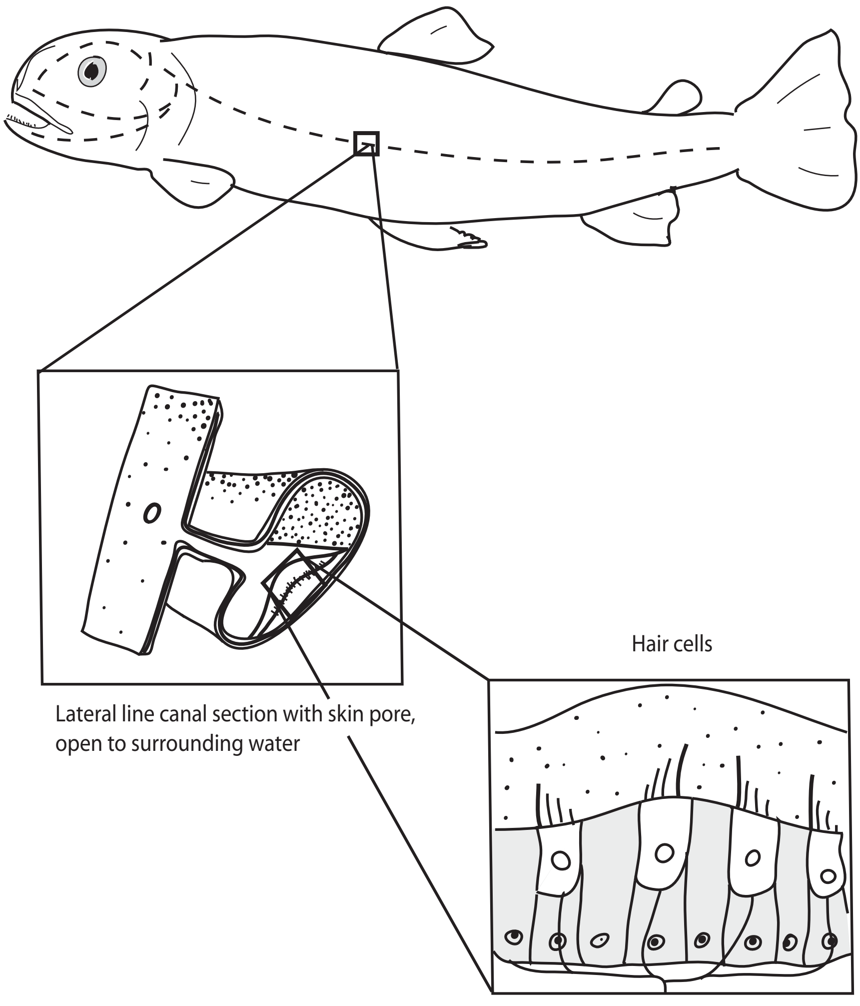

Figure 16-2. Fish lateral line system; schematic layout on the body; lateral line canal section with skin pore, open to surrounding water; and hair cells embedded in the lateral line.

may have poor vision in general or are even blind, whereas shallow water fish can differentiate colors; thus the size of the brain areas associated with color vision and vision in general are differentially enhanced.

Vertebrate Hearing

Hearing is the specialized reception and interpretation of another kind of vibration—sound waves, the mechanical displacement of the medium in which an animal lives, whether water or air. As with all sensory processes, the energy of the vibration must be transduced into an electrical signal. In the auditory system, this is done by the hair cells of the inner ear, as described in Chapter 12. As the term is generally used, "sound" refers to more than just general strength of vibratory signal—it also refers to the details and complexities of its frequency and "hearing" refers more specifically to its interpretation. The obvious reason is that this provides much more information about the source than just its presence.

The hair cells in the cochlea synapse with neurons whose cell bodies lie in the cochlear or spiral ganglion at the base of the *vestibulocochlear* or *auditory* nerve from each ear, which carry the signal to the *cochlear nucleus* in the brain stem. The vestibulocochlear nerves have two branches that project to the brain—one from the cochlea and the other from the vestibular system. The cochlear nerve contains about 30,000 afferent axons, most of them from inner hair cells in the cochlea, but 5–10 percent from the outer hair cells, which seem to play a role in tuning sound as a cochlear amplifier, among other functions.

The axon of each neuron in the cochlear nerve connects to an area of the cochlea that is most responsive to a particular *characteristic frequency*, and sound signals are carried to the brain in an orderly way that represents the spatial frequency tuning along the cochlea—the apical end receiving low frequencies and the basal end high frequencies. Moving from the beginning to the smaller inside end of the coiled cochlea, sounds of increasing frequency are detected.

The means by which each region along the cochlea responds most to a given frequency range is a complex matter of acoustical physics and hydrodynamics, and is not yet completely understood. Generally, the length and shape of the cochlear canal, and the differential stiffness of the basilar membrane along it, determine the vibration characteristics of the fluid in the cochlea. There are also differences in hair cell structure and firing behavior that vary along the cochlea. Sound energy applied to the basilar end sets up waves that travel up the cochlea through this fluid. The changing shape and stiffness of the basilar membrane along the canal dampen the passing waves, which as a result end up establishing standing peaks centered at a frequency-dependent distance from the basilar end.

In this way, a given sound frequency triggers hair cells in a specific region along the cochlea. Because this is a replicable property of fluid dynamics, the brain can recognize each occurrence of the same sound frequency by the location triggered. Further, while the relationship between frequency and position varies among species, and within individuals, and is not perfectly linear along the cochlea (it may be log-linear), the *relative* positions of activation can be used by the brain to work out complex sound characteristics, such as (at least for humans) harmonies, octaves, consonance, and the like. Thus, in a general way, successive octave frequencies activate regularly spaced locations.

Variation in gene expression leading to allometric variation in growth dynamics of the size, shape, length, and stiffness of the cochlea will produce a chamber with particular response characteristics, and these differ by species, as is shown in Figure 12-2. As a result, organisms are suited to particular hearing behavior. There is a danger that this will invoke an adaptive illusion, in this case of fine-tuning by selection. Such selection could in principle be what happened, but there are alternative potential explanations.

The basic mechanisms of mechanoreception and ion channel-based signal transmission were available early in evolution, and were cobbled together into the wide range of hearing mechanisms we see today. Rather than the sensitive touch of selection this could be explained as sloppy jerry-rigging of the pieces that evolution had to work with at the time, yielding hearing mechanisms that fell fairly randomly all over the map with regard to what organisms could hear. Organisms then made do with what they had, "tuning" themselves by organismal selection, regardless of the fact that the ability to hear a wider range of frequencies, or softer sounds, or from further afield would serve them even better. Evolution has built a basic vibration-response mechanism with wide interspecific, and even intraspecific, variation in how it works, with the only measure of importance being that it does work.

In fact, as with other senses, the sound wave peaks are spread over an area within the cochlea, so that there is overlapping sensitivity among hair cells. This and other aspects of interaction among cells along the cochlea give the brain the wherewithal to integrate the information and compensate for imprecision.

Relative position along the cochlea is conserved by relative position of fibers of the auditory nerve, with apical afferent fibers in the center and basilar ones around the outside. They thus arrive in the brain with locational information intact. This *tonotopic map* is analogous to the retinotopic map of the visual system in that an orderly spatial representation of the cochlea is sent to the brain but differs fundamentally in that the tonotopic map is totally unrelated to any spatial aspect of sound. The tonotopic map translates frequency into cochlear position, whereas the retinotopic map conserves the relative spatial positions of incoming photons. Perhaps this is related to the reason that, although sound is topographically represented in the brain, we don't perceive it as an image the way we do light (there is no reason, in principle, why the brain could not present sound to us that way; in fact, we make oscilloscopes to do just that). But the orderly map is in a sense a de facto result of the structural means by which sound is parsed so that individual frequencies can be detected. That structure takes advantage of the relative propagation properties of vibration of different frequencies in an enclosed fluid.

Auditory neurons are specific to different aspects of sound; some, for example, are responsive to the onset of a sound, transmitting information about the initiation of the sound but quickly dampening, whereas others don't begin to fire until the sound has been sustained for some time, thereby transmitting information about the sound's intensity and duration.

Neurons in the cochlear nucleus project to other auditory areas in the brain via three different pathways: the *dorsal acoustic stria*, the *intermediate acoustic stria*, and the *trapezoid body*. As with vision, sound is processed in parallel pathways that finally converge as complex acoustical information about sound source, intensity, frequency, and duration. Signals from both ears join in the *superior olivary nucleus*, which receives input from the trapezoid body. Postsynaptic axons from the superior olivary nucleus and axons from the cochlear nuclei project to the *inferior colliculus* in the midbrain via the *lateral lemniscus*, both of which, again, receive binaural input. Efferents from the superior olive project back to the cochlea to control sensitivity. The primary function of the superior olive is sound localization. Cells in the colliculus project to the *medial geniculate nucleus* of the thalamus, and axons from the geniculate nucleus terminate in the *primary auditory cortex* in the temporal lobe and in the *superior temporal gyrus*. See Figure 16-3 for a diagram of the central auditory pathways.

The primary auditory cortex is segmented in several ways, as is most of the neocortex. In particular, its segmentation is similar to that of the visual cortex in being laminar, with each layer receiving input from neurons from different places. It is also organized into functional columns, as is the visual cortex, with all neurons in a given column responding optimally to sounds with the same frequency. The dorsal portion of the auditory cortex responds to lower frequencies and the anterior portion to higher frequencies, with a gradient of frequency responses in the columns

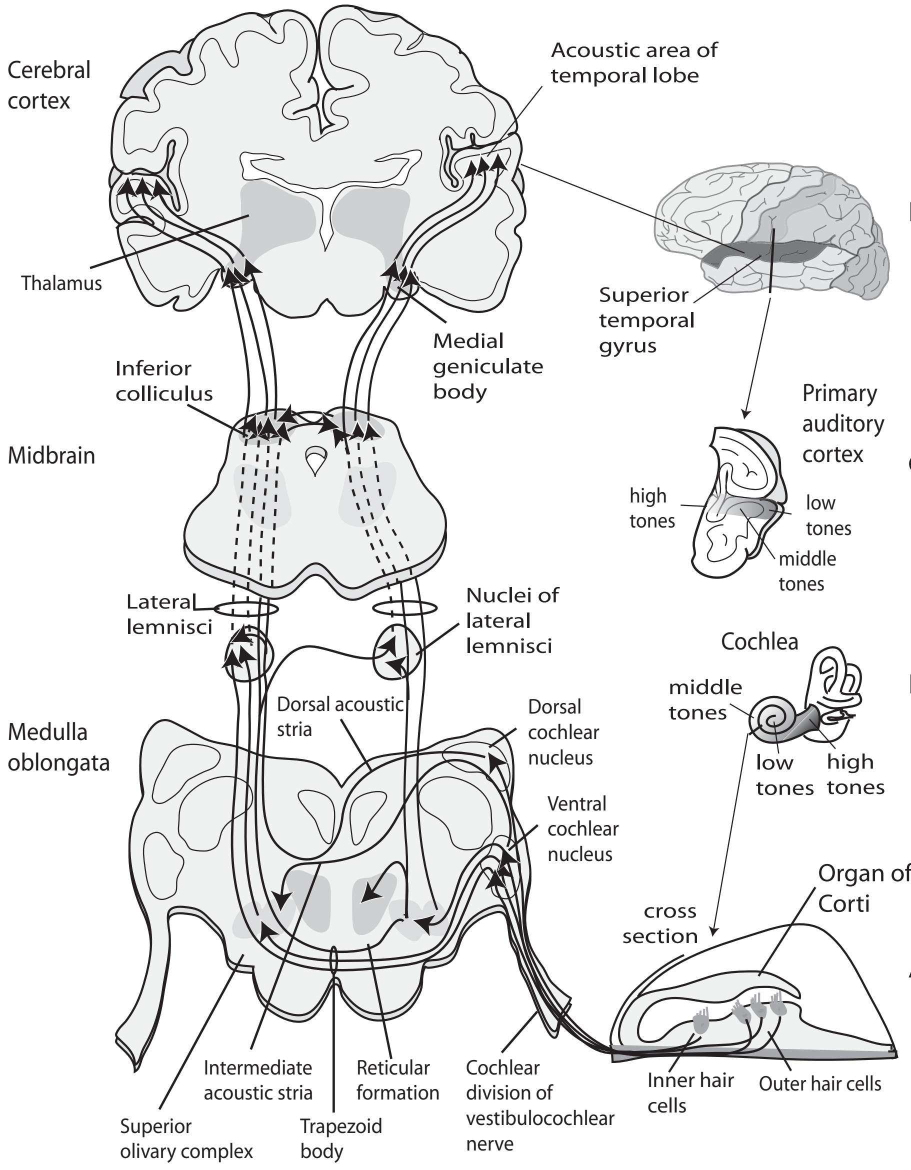

Figure 16-3. Central auditory pathways. Sound is transmitted from **A**, the organ of Corti in **B**, the cochlea, where different frequencies stimulate the hair cells in stereotypical areas; signal is transmitted from the cochlea through the brainstem to higher auditory areas, with the tonotopic map maintained, and ultimately to **C**, the primary auditory cortex, where signal is perceived in the acoustic area of the temporal lobe, **D**, still with the tonotopic map intact.

in between. The auditory cortex is also divided into two types of alternating "zones"—*summation columns* and *suppression columns.* The summation columns are composed of neurons that are excited by stimulation from either ear, and the suppression columns are composed of neurons that are stimulated by input from only one ear and inhibited by stimulation from the opposite ear. The spatial organization of these columns relative to the axis of tonotopic mapping enables the primary auditory cortex to respond to every audible frequency and interaural interaction (Kandel, Schwartz et al. 2000).

In mammals, sound is generally sent for further processing to a number of other areas beyond the primary auditory cortex, in a parallel and hierarchical way similar to the visual system, but the process is not nearly as well understood. In marsupials, sound processing seems to be confined to the primary auditory cortex; however, in eutherian mammals, there are at least nine regions beyond the primary auditory cortex, although the number seems to vary by species and/or depending on the method of data collection. In all cases, sound continues to be organized according to the same tonotopic map received and processed by the primary auditory cortex. The number of areas in insectivores seems to be three or four, in rodents four to seven, and in carnivores and primates six to eight or nine (Ehret 1997).

Again like vision, bilateral symmetry makes it possible to locate a sound in a more three-dimensional way when integrated in a stereophonic rather than simple left-right way. Unlike light, sound reaches both ears no matter which direction its source. This means that it is possible to detect the slight delay that results from signal coming from one side or the other and even to assess distance to some extent.

Radios rely on devices to filter out signals at all frequencies other than the one to which they are tuned. In that way, a signal is detected out of a background that can contain energy of all frequencies. Auditory systems in the brain also tease out single sounds or sound frequencies from the generally broad panoply of incoming sounds, so, for example, a person can hear the melody of the violins in an orchestra or attend to a single conversation in a crowded room. How we do this *auditory scene analysis* is an important but still not well-understood phenomenon that probably involves learning from experience combined with accurate representation and localization of sound by the ear and brain.

Sound interpretation is more than just the detection of characteristic vibration signatures, the way a mass spectrophotometer detects the characteristic spectrum of each type of molecule. Signature detection is certainly part of this, but, at least in higher vertebrates, many more subtleties and complexities are resolved. Also as occurs in vision, this is basically done in the brain, after the information has left the detector itself.

Human speech processing is an important higher area of sound processing in humans, but understanding of the neural pathways and processes involved is still fairly rudimentary, in part because there are no laboratory animals in which it can be studied.

Invertebrate Hearing

It appears that only a minority of insects have the ability to hear, but insect hearing may have evolved at least 19 times (Yager 1999). Like vision, we will probably learn that rudimentary elements of the system, perhaps shared with vertebrates, were present in stem animals (with whom animals on two divergent branches share ancestry) with morphological details independently intercalated in between an early induction signal and later receptor-system differentiation. Cytoarchitectural processes recruited for mechanoreception are likely to be its basis.

Paired hearing organs are generally peripherally located on almost any part of the body, including various abdominal or thoracic body parts, legs, wings, or mouth.

Almost all insect ears share a tympanum covering a tracheal sac and tympanal organ. The somites and dendrites of insect auditory receptors are in the hearing organs themselves; sound frequency is first analyzed there. The axons of the auditory receptors enter the nearby segmental ganglion or ganglia and carry auditory input in a tonotopically organized way, as in the vertebrate hearing system.

Thus, insects also interpret a signal that has been given an orderly translation from a frequency pattern into a neurological space-based "map." The signal is then sent to the brain for decoding of the sound's direction, pattern, frequency, and subsequent localization (Pollack 1998; Stumpner and von Helversen 2001). The specifics of the neuronal pathways differ by species; some, for example, use parallel independent processing to recognize and localize signals and some use a process in which localization depends on recognition (Pollack 1998).

As in vertebrates, sound direction is determined by comparison of the auditory input from each of the two ears. The vertebrate brain uses interaural arrival time difference and intensity difference to make that determination. The insect brain uses intensity difference as well but whether there is enough difference in time of arrival of a sound at two ears located so close to each other to be useful in determining sound directionality is still open to question (Pollack 1998). (Again, we must be careful not to view the world from our human frame of reference: some mammals that can detect directionality of sound, like voles and bats, are certainly also very small.) Inhibiting or destroying sound reception of one ear does result in an insect no longer being able to distinguish the location of a sound source, such as the call of an echolocating bat.

SOMATIC SENSATION

Plant Thigmotropy

We referred earlier to thigmonasty or vibration sense in plants. Plants also have *thigmotropy* or a sense of mechanical disturbance. The leaves of some plants, such as some mimosa or the venus fly trap, close when touched, in the same way that they close when sensing vibration. Roots often exhibit *negative thigmotropy* by turning away from objects in the soil that they touch as they grow. Perhaps the best example of thigmotropy in plants is the manner in which tendrils curl around objects. When they come into sustained contact with an object, the opposite side of the stem begins to grow more rapidly, thus elongating relative to the contact side. At the same time, the contact cells lose their turgor pressure and become flaccid and smaller. Sustained differential growth and turgid pressure cause the tendril to curl around the object it is touching. How the touch stimulus is communicated throughout the plant is not known and may work differently in different plants.

Biochemically, mechanical stimulation has been found to elicit the release of *alamethicin*, *jasmonic acid*, and *12-oxo-phytodienoic acid*, which induce coiling in some plants. Alamethicin is a peptide that forms voltage-gated ion channels and elicits the synthesis of a number of volatile compounds. 12-oxo-phytodienoic acid is a powerful inhibitor of growth and is differentially found on the contact side of a tendril (Engelberth et al. 2000); if the contact side slows its growth but the outer side continues to grow, the tendril will curl around the contacted object.

Plants growing in conditions of sustained exposure to touch, wind, water spray, and other similar physical stimuli are shorter and stockier than plants grown without exposure to these stimuli. This is called *thigmomorphogenesis.* At least four touch-induced (TCH) genes are known in *Arabidopsis thaliana*; they belong to a family related to *Calmodulin* genes, which code for calcium-binding proteins. TCH mRNAs accumulate rapidly on touch, within 10 minutes (Johnson et al. 1998), but the pathway by which these genes induce thigmomorphogenesis is not understood.

Vertebrate Somatosensation

As discussed in Chapter 12, many sensory receptors are located in the skin: mechanoreceptors that sense touch, pressure, vibration, and hair displacement; thermoreceptors that sense temperature; and nociceptors for damage and pain. These receptors are not uniformly distributed throughout the skin but instead are concentrated in some areas more than others. Proprioceptors relay information about joint and limb position.

The cell bodies for these receptors cluster to form the dorsal root ganglia along the spinal cord. The location of the ganglia provides a relative positional map of the location of the stimulating signal on the body. The axons of these neurons ascend to the brain in two separate pathways in the spinal cord—one that transmits information about highly localized sensations (fine touch and the proprioceptive system) in the *dorsal column medial-lemniscal* system and the other system, the interneurons in the spinal cord that synapse with receptors of poorly localized sensations, pain and temperature, in the lateral sensory tract of the spinal cord (the *anterolateral system*).

Each of these pathways projects to different areas of the brain. The fast-acting interneurons in the anterolateral system project directly into the thalamus without synapsing first in the brain stem. This is the pathway that is responsible for immediate pain perception after injury. The nerve fibers in this system are myelinated and large to transmit signal rapidly (diameter of the neuron affects the speed of conductance). The prolonged pain of injury is the product of the slower-acting dorsal column medial-lemniscal pathway. This system for perception of touch and proprioception ascends the spinal cord ipsilaterally (on the side of the spinal cord that corresponds to the side of the body that was touched), and axons synapse in the *dorsal column nuclei* of the medulla in the brain stem before moving on to the thalamus. The receptors in this pathway are unmyelinated small fibers.

In the dorsal column medial-lemniscal pathway, the first axons to enter the spinal cord, from the sacral (lower back) region, are centrally located in the dorsal column; axons that enter from successively higher areas of the body are progressively more lateral. At the upper region of the spinal cord, the dorsal columns divide into two bundles of axons—the *gracile fascicle* and the *cuneate fascicle.* The nerve fibers in these bundles are organized somatotopically—those in the gracile fascicle from the ipsilateral sacral, lumbar, and lower thoracic segments and those in the cuneate fascicle from the upper thoracic and cervical segments.

As in the visual and auditory systems, the somatotopic organization of the ascending axons is maintained as a map at each step of the processing of somatosensory information in the brain (Kandel, Schwartz et al. 2000). Touch and pain, however, require rapid reaction capabilities and in general (in mammals, or at least

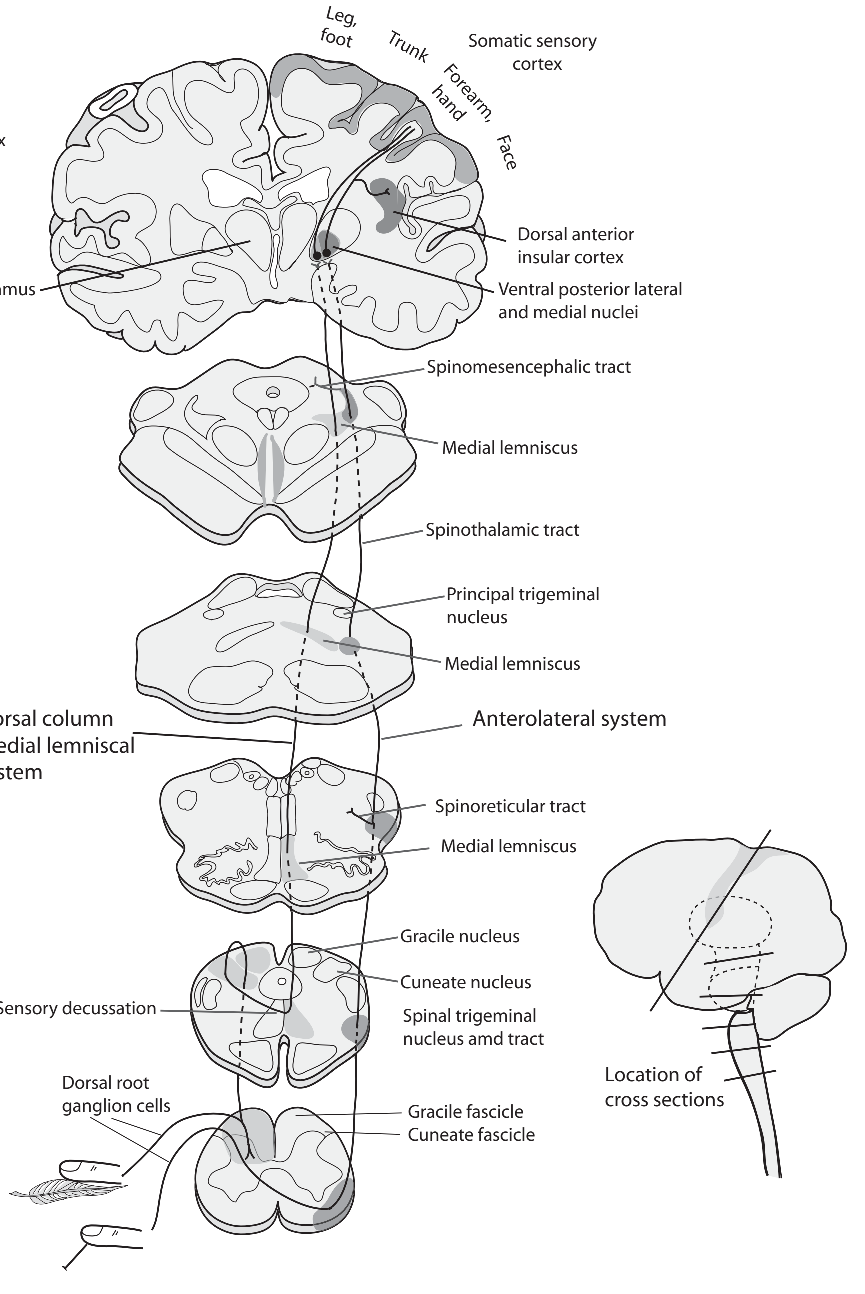

Figure 16-4. Anterolateral and dorsal column medial-lemniscal pathways: touch. Redrawn from (Kandel, Schwartz et al. 2000).

humans) less detailed understanding of the source of the sensation. We need to know the location of a cut or contact with flame, not the shape of the thorn or fire.

Because the brain stem was the first cerebral ganglion to form in the evolution of the cephalized CNS of vertebrates, most (but not all) sensory axons ascending the spinal cord, even in vertebrates with an extensive forebrain, still make synapses in the brain stem before traveling to higher parts of the brain. Axons from the gracile and cuneate bundles terminate in the gracile nucleus and the cuneate nucleus in the lower medulla. Somatosensory signals from the face and scalp are sent to the *principal trigeminal nucleus*, which is also in the brain stem. Axons that originate in the dorsal column nuclei decussate, or cross, the midline of the spinal cord and ascend in the medial lemniscus to the *ventral posterior thalamic nucleus* in the thalamus. As these fibers decussate, the body map becomes its mirror image, and the sacral segments are now located most laterally and the higher segments are located medially. Signals from the face, once they reach the principal trigeminal nucleus, are transmitted in the *trigeminal lemniscus*, joining axons from the arm and back of the head in the medial lemniscus, to the ventral posterior medial nucleus of the thalamus.

The information reaching the thalamus in the right side of the brain comes from the left side of the body and vice versa. In the anterolateral system, the crossover takes place in the spinal cord, whereas in the dorsal column medial-lemniscal system it takes place in the medulla. The decussation always occurs with the first interneurons to synapse with the primary sensory neurons (Matthews 2001).

From the ventral posterior thalamus, signal is relayed to the cortex. Most neurons from this area project to the *primary somatosensory cortex* (*SI*), with a minority projecting to the *secondary somatosensory cortex* (*SII*). Again, the body surface is represented in an orderly way. Some areas of the body are overrepresented in the brain with respect to relative body size, however, so that more brain area, for example, is devoted to afferent connections from a human's fingertips or lips than the legs or toes. More cortical area, in fact, is devoted to the palm and fingers than to the leg, trunk, and arm combined. These areas themselves also have a higher density of sensory nerve endings. The somatotopic map of the correspondence between skin surface and cortical area was first depicted as a *homunculus*, a human figure distorted to represent the amount of cortical area devoted to specific body parts, by neurosurgeon Wilder Penfield in the 1950s (see Figure 16-5). He determined, from testing individuals with epilepsy during brain surgery, the areas in the cortex in which afferent connections from different parts of the body terminated. The growth of individual nerves to the periphery of the body innervates the skin in a modular, segmented way, establishing *dermatomes* or areas on the skin served by branches of a given nerve. Since the branching is developmentally hierarchical, the brain is enabled to recognize regions and subregions in an orderly way.

This is something of a different kind of topographic map. Here, relative position in the body is represented, but there is differential distortion based on use or importance. Vision and hearing are not so distorted in this way, although retinal images have different strengths and sensitivities in the center and peripheral regions of the retina. There are also species-specific differences that, at least to some extent, correspond to use and need. In these senses, relative function *is* preprogrammed into even the brain (and innervation patterns of the body).

Like other cortical areas that we have discussed, the somatosensory cortex is laminar. Again, axons of afferents coming from the ventral posterior thalamus synapse with the stellate interneuron cells of layer IV, which in turn connect with the pyramidal neurons of the other layers of the cortex, including motor regions, and back to the thalamus in a feedback loop that supplies sensory information to the cortex. The motor regions of the brain can then act on proprioceptive information or cues about the body's contact with the environment.

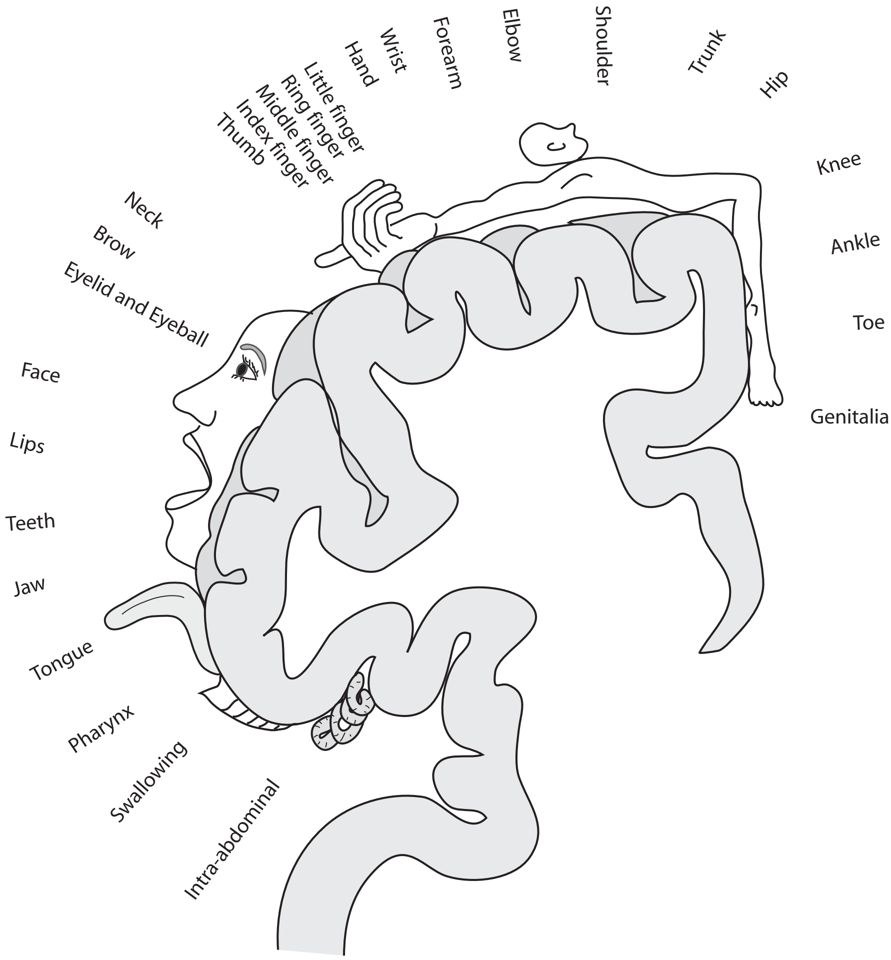

Figure 16-5. Homunculus, representing the topographic organization of the human sensory cortex, and relative size of brain areas dedicated to perception by various body surfaces.

Sensory modalities are separated in SI into four distinct functional areas, each of which contains its own representation of the whole body. Two of these areas respond primarily to deep pressure and movement, and the other two respond primarily to surface touch. Some neurons from these areas project to the secondary somatosensory cortex. Both the primary and secondary somatosensory cortical areas are organized in columns as well as layers, just as are the visual and auditory areas of the cortex. Neurons in each column respond to stimulation from a specific region of the body, and there is extensive interconnection between the four areas. This manner of morphological organization allows the somatosensory system to function via both parallel and hierarchical processing, again as with other areas of the brain.

The somatosensory cortex not only enables perception of touch and temperature but also is responsible for localization of the source and intensity of a stimulus, as well as further dissection of somatosensory signals. The secondary somatosensory cortex is innervated by neurons from SI. SII requires projections from SI to function. Neurons in SII project to the *insular cortex,* which in turn relays signal to portions of the temporal lobe thought to be involved in creating and storing tactile memory. Areas in the *posterior parietal cortex* receive input from SI and other areas and are involved in integrating tactile signals with proprioceptive signals and with integrating information from the two hands. Other areas integrate visual with somatosensory information and initiate sensing and movement as a response to tactile and proprioceptive information (Kandel, Schwartz et al. 2000).

Rodent Vibrissae

A much-studied example of topographic mapping of a somatosensory system is that of *vibrissae,* or whiskers, in rodents. The vibrissae are the principle tactile receptors in these animals. They form along a line by a repetitive periodic patterning process that generates them one by one, roman-candle fashion along the line. The number is generally invariant in the mouse (Dun and Fraser 1959), although artificial selection experiments and some naturally occurring mutation (in the *Ta*, or *Tabby* gene) can expose underlying variation. This is thought to be a cell-surface ligand gene product whose extracellular domain is a ligand for a TNF-class cytokine receptor (related to IL1 and NFκB pathways we have seen earlier), and is used in epithelial-mesenchymal patterning of structures like hair and teeth.

As the developing face projects outward, presumably each new vibrissal primordium produces its sensory and transmitting neurological elements when the vibrissa is initiated. As it develops, each whisker maps to a specific site in layer IV of the rodent primary somatosensory cortex. The somatotopic map represents precisely the position of each whisker on the rodent snout. A single whisker is innervated by about 100 neurons, which convey the directional movements of the whiskers to the brain. The afferents project from the whiskers to the SI through the thalamus.

In the SI, the neurons are arranged in functional units called *barrels*, one per whisker, corresponding to the array of the whiskers on the snout (see Figure 16-6). (The nomenclature is due to the shape of the cell bodies of these neurons because they appear to form barrel-shaped arrays when the cortex is cut parallel to the cortical surface.) As in other synaptic networks in the brain, there is a critical developmental period for the growth of the optimal network of connections among barrels in the cortex of the rat. When vibrissae are removed from a young rat at an early age, the cortical connections to the different cortical layers are much less dense than in normal rats, suggesting that experience "trains" the synapses (Fox 1994).

Interestingly, not all species that have whiskers have corresponding cortical barrels. Barrels can be found in, for example, the mouse, rat, squirrel, porcupine, and walrus but not in the dog, cat, or raccoon. In addition, barrels are found in some animals that have minimal use for their whiskers, such as the guinea pig and chinchilla (Nieuwenhuys et al. 1998). However our understanding of the brain is still incomplete if not perhaps only rudimentary, so caution is needed in generalizations about brain structure and function.

Star-Nosed Mole

An unusual example of an organ for touch that may give this animal the keenest sense of touch known in the animal kingdom is the star on the nose of the star-

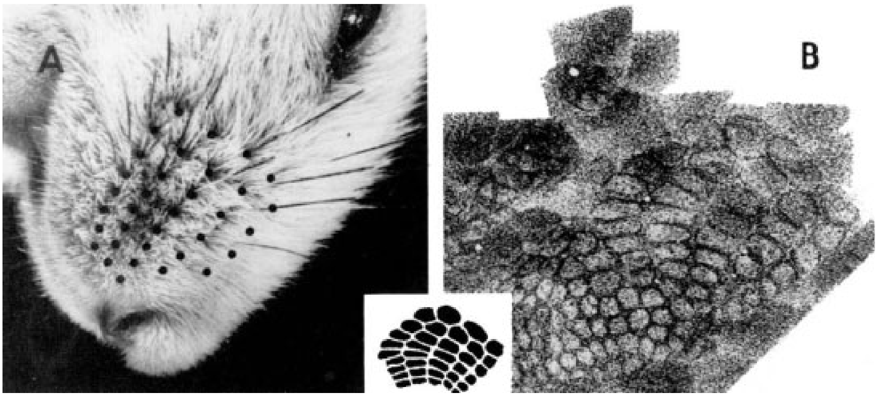

Figure 16-6. Topographic map of mouse vibrissae. (A) Vibrissae; (B) corresponding spatial organization of barrels in the cerebral cortex. Reprinted with kind permission of T. A. Woolsey, originally published in (Woolsey and Van der Loos 1970).

nosed mole. The snout of this animal extends in a splayed protrusion of 22 finger-like appendages that are used to explore the animal's immediate surroundings. This is shown in Figure 16-7. Despite its location, the organ is not olfactory, nor used for manipulating food or other objects, but instead for touch. The appendages are covered with tiny *Eimer's organs*, hairlike touch receptors, 1,000 or so on each of the 22 rays of the star. More than 100,000 nerves project from the nose to the brain, compared with the 17,000 or so from the human hand (Catania 1999). When the nose comes into contact with an object, it engorges with blood, and this pushes the receptors closer to the surface of the nose, thereby increasing its sensitivity. This provides a kind of light-less spatial representation of the environment in front of this burrowing animal, a kind of tactile retina, whose spatial aspect is vital to interpretation. Does the star-nose mole "see" its environment or just "feel" it?

Given what we know about the organization of the somatosensory areas of the cortex vis-à-vis touch sensitivity, it is not surprising that the star-nose is highly over-represented in this animal's brain. The star is represented three times by a stripe of tissue in the cortex of each hemisphere, once in the SI, again in the SII, and yet again as a smaller representation in an area termed SIII (Catania and Kaas 2001). This is unusual; the somatotopic map of vibrissae, for example, is represented only once, in SI. The amount of cortical surface devoted to the star-nose reflects the importance of this tactile organ to the mole. Furthermore, it is possible that distributing the cortical connections among three small areas rather than one large one may speed interneural connections, and this could be important for an organ upon which an animal relies so heavily.

TASTE

As we saw in Chapter 13, the receptors for the sense of taste in vertebrates are grouped together in taste buds, each of which contains about 100 *taste receptor cells*. Taste buds in mammals are located primarily on the tongue, but also on the pharynx, the laryngeal epiglottis, and at the entrance of the esophagus. Some fish have taste buds over the entire surface of the body (Butler and Hodos 1996).

Figure 16-7. The star-nosed mole. The splayed nose provides a spatial tactile sense of a light-less environment. Source: http://www.moleplace.com/photo_page.htm.

The surface of each taste cell is covered with microvilli, which protrude through a pore in the cell (the taste pore) to be exposed to the oral cavity. They are specialized to detect chemicals dissolved in the saliva on the tongue. The microvilli in turn are clustered into papillae that are embedded in the epithelial layer of the tongue. Afferent nerve axons enter the taste bud at the base and each one synapses with multiple receptor cells in the taste bud.

The taste cells can detect five basic stimuli: bitter, salty, sour, sweet, and umami. The ability to taste amino acids seems to be particularly acute in fishes, but it is not obvious why this particular sense developed. Combinations of the five basic tastant stimuli lead to perceptions of complex tastes. Each stimulus is transduced by a different mechanism, and the same taste may be elicited by different stimuli and different mechanisms. In addition, the mechanism used to sense the same stimulus may differ between vertebrate species.

Sour and salty tastants do not have a receptor molecule in the receptor cell in the usual sense; these tastes are due to the hydrogen or sodium ions in the substance. They are not detected with the usual receptors but by changing the membrane potential of ion channels in the taste receptor cell. Sweet, bitter, and umami reception are more complex, and transduction of these tastes is more similar to phototransduction and olfactory transduction: the tastant is bound by a G protein membrane receptor, which in turn activates intracellular signals that affect the ionic permeability of the taste receptor cell. Because many molecules can be perceived as sweet or bitter, receptors for these tastants are many and varied.

Two nerves carry input from the tongue to the brain: the facial nerve (cranial nerve VII) conveys information from the front of the tongue and the glossopharyngeal nerve (cranial nerve IX) from the back. These two nerves also contain the sensory fibers for transduction of touch, pressure, and temperature. Axons from these nerve bundles enter the brain stem in the medulla and synapse in a thin line of cells called the *nucleus* of the *solitary tract.* As with other sensory systems, the somatotopic organization of taste is maintained in the solitary, or gustatory, nucleus, such that gustatory nerves from different regions of the head, and skin in animals with taste buds on the body, enter this area in the order in which they are located on the body. From there, depending on the extent of development of the animal's thalamocortical system, input is conveyed to the gustatory region of the somatosensory cortex, the primary gustatory cortex, where the perception of taste is formed, and, in animals with more fully developed forebrains, to the hypothalamus, amygdala, and insula, the limbic area, where behavioral responses to the perception of taste—aversion, salivation, gastric secretion, pleasure, etc.—are triggered.

Fish have a very well developed sense of taste in general, with taste receptors on their lips and in their mouth and pharynx, but several suborders have evolved taste receptors over the entire surface of their bodies. In these fish, the gustatory nervous system is very complex and supplies them with a detailed topographical taste map of their surroundings. These *silurids* (catfishes, of which there are more than 1,000 species) and *cyprinids* (carps, minnows, chubs, and goldfish) are bottom feeders and have evolved an elaborate system for separating inedible particles from food as they pick up mouthfuls of sediment from the bottom of the stream or ocean (Butler and Hodos 1996).

Insect Taste

In insects, detection of soluble chemicals by gustatory neurons can elicit feeding behavior, but mating behavior as well (Scott et al. 2001). *Drosophila*, for example, have chemosensory hairs on their legs and proboscis that activate proboscis extension and feeding when they detect sweet compounds. Female *Drosophila* have bristles on their genitalia that elicit ovipositing upon detection of nutrients, which probably maximizes the probability that eggs are laid in an environment in which the newly hatched can feed.

Taste receptors in *Drosophila* are found in sensory sensilla located on the fly wing, legs, proboscis, and genitalia. Two types of sensilla have been characterized: *taste bristles*, which are located on the legs, wings, ovipositor, and mouthparts, and *taste pegs*, on the oral surface and in the pharynx. Bristles are hollow hairlike lymph-filled structures with a pore at the terminal end, through which nutrients enter to dissolve in the lymph. Each bristle contains neurons that are specific for sugar, salt, or water, as well as a single mechanoreceptor neuron (Shanbhag et al. 2001). Taste pegs, in contrast, do not have obvious pores in the termini or side-walls, but their role in gustation has been shown behaviorally in blowflies, and electrophysiological assays suggest that they have sugar and salt receptors, as well as a mechanoreceptor. Stimulation of these cells induces feeding (Shanbhag et al. 2001).

Adult *Drosophila* have about 2,000 chemosensory neurons in the sensilla. Up to four neurons innervate each sensillum. Taste pegs are innervated by two receptor cells: a chemoreceptor and a mechanoreceptor (Shanbhag et al. 2001). Sensory neurons from the proboscis extend to the subesophogeal ganglion (SOG) in the brain and then are relayed to other gustatory areas for further processing. Gustatory neurons from other parts of the body project locally to peripheral ganglia. How taste is represented in the *Drosophila* brain is not yet known.

Differential taste responses could arise from different locations on the body or tongue as a consequence of the developmental patterning mechanism that generates the taste buds or receptors. That there is a topographic map of taste receptors in terms of their location of the body is no surprise; it provides a sense of where a given tastant is being detected, but this need not have any *particular* evolutionary function and instead may just be a useful result of developmental patterning. But how different and variable locations on insect bodies are all interpreted as *taste*, or whether something "tastes" different depending on which sensors it activates, are more intriguing questions.

Odorant detection is odorant specific at the level of the cell in many animal species. This has to do with the developmental allelic and gene exclusion in generating the olfactory neuron. Taste is generally a different kind of detection, more general, and it is easy to understand why each taste bud might be sensitive to multiple tastant characters: it could enable the organism to determine the generic "flavor" of the stimulating substance, rather than its specifics. At the same time, it could also simply be the consequence of the way taste-sensing units develop.

OLFACTION

Odorants traveling in the air do not maintain a precise relative location and may mix and mingle. Most interesting objects have complex odorants that have no spatial relevance to a detecting organism. Except for strength and general direction, olfaction neither has nor requires a spatial map that faithfully represents details of the emitting source.

For these reasons there would have been little selective pressure to detect odors with a spatial orientation. However, olfactory signals *do* map spatially in terms of the relationship between the position of the detecting cells in the olfactory epithelia and where the associated neurons synapse in the brain. This relationship is probably used by the brain to keep track of what is coming in, because a given receptor will be triggered by the same odor whenever it occurs. The regionalization of the olfactory epithelium and the sets of related olfactory receptor genes expressed in each region, as described in Chapter 13, are easy to account for in principle by the generally modular nature of developmental patterning processes.

VERTEBRATE OLFACTION

The olfactory sensory neurons in vertebrates are in the nasal cavity, in the olfactory epithelium, an area of about $5\,cm^2$ in humans. Several million olfactory neurons are embedded in this small region, interspersed among supporting cells. A single dendrite extends from the olfactory sensory neuron to the epithelial surface of the nasal cavity, where it swells into a knob. Five to 20 cilia protrude from this knob into the mucus-covered epithelium. These cilia have receptors capable of binding specific chemical characteristics and hence able to bind specific odorants, which initiates the steps that transduce the olfactory signal to the sensory neuron and on to the brain. In Chapter 13, we described the genetics of odorant detection itself and the olfactory receptor gene families.

Each vertebrate olfactory neuron seems to express only one odorant receptor gene and thus transmits information about only one odorant to the brain, although one odorant may be detected by several different receptors and a receptor may respond to several different odorants. Animals can detect many more odors than they have odorant receptor genes because of this combinatorial approach to the open-ended nature of chemical detection. The perception is related to the combination a given odorant triggers.

The olfactory epithelium (Figure 16-8) is organized into four different zones. Groups of neurons with the same receptors cluster together in these zones, and axons from the different zones project to distinct regions of the olfactory bulb, thereby preserving the topographical organization of the receptors in the olfactory epithelium. The spatial map of these zones is nearly identical in both hemispheres of the brain and between individuals (Zou et al. 2001).

From the basal end of the olfactory neuron, a single axon projects to the olfactory bulb in the brain, above the nasal cavity, to synapse with olfactory bulb neurons. These neurons are organized into small glomeruli, of which there are about 2,000 per bulb in mice (Kandel, Schwartz et al. 2000). Several thousand sensory neurons project to each glomerulus, each of which apparently receives input from only one type of receptor (that is, one class within the odorant receptor (*OR*) gene family). Glomeruli that receive input from specific types of receptors are located in the same place in the olfactory bulb in different individuals. Each sensory neuron synapses in only one glomerulus. In each glomerulus, there are 20–50 different relay neurons, and axons synapse with three different types: the *mitral* and *tufted relay* neurons that send the signal to the olfactory cortex and the *periglomerular interneurons* that encircle the glomerulus and make inhibitory synapses with mitral cell dendrites.

Olfactory signals seem to be extensively processed before they are relayed on. The inhibitory connections made by the periglomerular interneurons may be part of this preprocessing, as are feedback connections from the olfactory cortex and parts of the forebrain back to the olfactory bulb. An odorant's effect on an animal's behavior may depend on the animal's physiological state; an odor may heighten an animal's hunger, for example (Kandel, Schwartz et al. 2000), or elicit sexual behavior only during receptive periods.

The mitral and tufted relay neurons project directly to the cerebral cortex for further processing and to the limbic system, which is involved in emotions and mediates emotional responses to smells. Olfaction is a primitive or early-evolving sense, and this may be why the olfactory nerve is the only cranial nerve that sends signals directly to the cerebrum, bypassing the thalamus. This probably reflects the importance of chemodetection in early vertebrate evolution, which would not be surprising given the aquatic environment in which chemicals more than light or sound might have been the primary environmental stimulant. The primary olfactory cortex is actually in the paleocortex, a part of the brain evolutionarily older than the neocortex with somewhat different laminar structure.

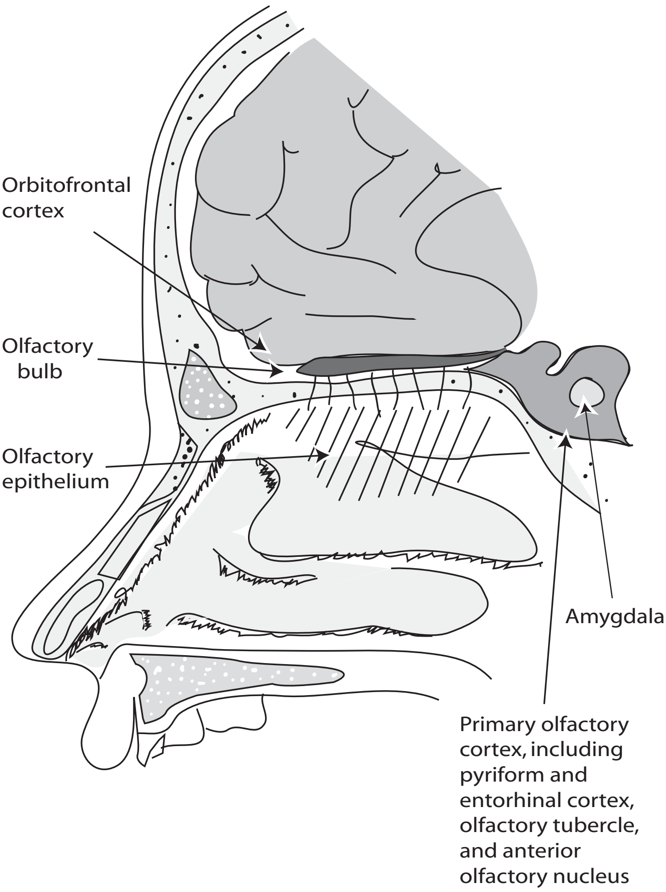

Figure 16-8. Diagram of the olfactory pathway to the human CNS.

The olfactory cortex is divided into five regions: (1) the *piriform cortex* (the largest olfactory area), (2) *olfactory tubercle*, (3) *anterior olfactory nucleus*, and parts of the (4) *amygdala* and (5) *entorhinal cortex*. Mitral cells project to all parts of the olfactory paleocortex, whereas tufted cells project only to the most anterior regions (Zou, Horowitz et al. 2001). From the primary olfactory cortex, signal is relayed to secondary and tertiary olfactory regions, including the hippocampus, ventral striatum and pallidum, hypothalamus, thalamus, orbitofrontal cortex, agranular insular cortex, and cingulate gyrus (Kandel, Schwartz et al. 2000; Weismann et al. 2001). Studies of people with brain lesions suggest that the pathway to the orbitofrontal cortex from the thalamus regulates perception and discrimination of odors, whereas the amygdala regulates emotional responses to odor.

Olfactory input seems to be relayed to the same brain areas across individuals, suggesting the existence of a stereotyped map of axonal connections to the olfactory cortex (Zou, Horowitz et al. 2001). This is what would be expected of hierarchical or regional patterning mechanisms, as seen in the other systems we have described. However, a stereotyped map of connections does not mean that this replicable organization is related to anything inherent in the objects emitting odors or the odors themselves that are being detected. Instead, today it mainly reflects the evolutionary history of gene duplication in the *OR* clusters, the developmental patterning process, and perhaps the nature and timing of the process by which each cell "chooses" which *OR* to express. As we have seen, this may affect the localized migratory pattern of that neuron into the brain.

Still, consider that to at least some extent, odorants with somewhat similar characteristics are today detected by receptors in similar *OR* classes, that are expressed in generally similar regions of the olfactory epithelium (Liu et al. 2003). Given the activation-inhibition nature of repetitive patterning mechanisms, it is likely that when there were but few *OR* genes just accumulating binding differences (and hence, the class differences we find today), the cell-patterning mechanism would leave them regionalized with a de facto function-location correlation. Whether that was important to the *nature* of the odorant classes that were detected by the simple early system for any particular functional reason, is a separate question.

As in other sensory systems, olfactory input is processed hierarchically as well as in parallel in different regions of the brain, when, as with other senses, olfactory input is sent to more than one area at a time. Unlike other senses, the olfactory topographic map in the brain does not maintain spatial information about odors, or retain the bulb map, but instead it may encode the quality of an odor (Mombaerts et al. 1996; Wong et al. 2002; Zou, Horowitz et al. 2001), with organized input coming from different glomeruli.

Information from different odorant receptors is segregated until it reaches the olfactory cortex, where input from many glomeruli clusters into overlapping neuronal groupings and the olfactory cortex integrates these signals to produce the perception of many different and complex odors in a way that is not yet well understood. Some olfactory signals are then relayed to the limbic system, where they may affect emotional states or instinctive behaviors, and to the neocortex, where they are further processed.

Insect Olfaction

We described insect olfactory reception in Chapter 13. Odor receptors in insects are found in sensilla, usually sensory hairs, which project from the cuticle. The sensilla have tiny pores on their surfaces through which odorants pass, and they stimulate the dendrites of the odorant receptors inside, which are covered in lymph. Most sensilla in most insects are found on the antennae.

Briefly, adult *Drosophila* have about 1,300 odorant receptors (Vosshall 2001), compartmentalized in sensory hairs on the surface of the third antennal segment and on the maxillary palp, an olfactory organ on the proboscis. This compartmentalization is a major difference between vertebrate and invertebrate olfactory systems. Each antenna holds about 600 sensilla of three different morphological types, whereas each maxillary palp has 60 of a single type. The *OR*s code for G protein-coupled receptors (GPCRs), of which there are three classes (Figure 13-2B), and each of the sensilla in turn contains two neurons, arranged in stereotyped pairs. Thus, there are six types of neurons, each with its own particular response. If each gene is expressed with the same probability, the relative frequency of the types will depend on the relative numbers of genes in each class. Whether this has functional or evolutionary relevance is not known and would probably require categorizing these genes in multiple species.

The sensilla are situated on the palp in a fixed configuration, and each exhibits a different specific response. A certain odor may excite one OR and inhibit another, and a single OR may be excited by one odor but inhibited by another. The ORs all behaving in this fixed way generate an olfactory code. There appears to be no sexual dimorphism in the kinds of sensilla present in male or female flies, in contrast to a marked sexual dimorphism found in the sensilla and olfactory glomeruli of some moths (e.g., de Bruyne et al. 1999).

As noted earlier, the *OR* gene class in *Drosophila* is quite divergent from the corresponding genes in vertebrates (Scott, Brady et al. 2001; Vosshall 2001). Each olfactory neuron expresses only one receptor gene, as in mammals (with the exception noted earlier of one *OR*, *Or83b,* that seems to be expressed in every olfactory neuron) (Scott, Brady et al. 2001; Vosshall 2001).

In arthropods, as in vertebrates, afferents from ONs project to glomeruli in the CNS, maintaining the topographic organization of the expressed ORs in the sensory projections (Wong, Wang et al. 2002). There are 43 olfactory glomeruli in the *Drosophila* antennal lobe, the fly equivalent of vertebrate olfactory bulbs. As in vertebrates, functional imaging of brain activity in insects shows that different odors elicit activity in different glomeruli (Wong, Wang et al. 2002). Projection neurons then relay input from the antennal lobe to the mushroom body and the lateral horn of the fly protocerebrum in a stereotyped set of axon branching patterns (Marin et al. 2002), for higher processing. If, as in vertebrates, a combinatorial code defines an odor, the specifics of how input from different ORs is integrated in the higher areas of the insect brain remain to be elucidated. It is known that projection neurons connect to a stereotyped set of third-order neurons, and it is at the level of these third-order neurons that the integration appears to take place.

MULTISENSORY PERCEPTION

We have largely discussed sensory perception in a Cartesian way, as though each of the senses were an independent function. Although we see orderly ways in which these things are organized, and we can relate that to the known kinds of developmental processes, the emergent nature of perception in each type of sense is still rather elusive. In fact, although many of these sensory systems do function without input from others, higher sensory processing often involves their interaction.

Morphological, visual, olfactory, auditory information is routinely integrated in evaluating threat postures, prey or predator behavior, and the like. Perception of smell and taste interact to produce a more complex sensation, and spoken language comprehension can be more precise when complemented with visual processing of lip movements. Significant interaction takes place between visual and auditory input in vertebrate development, and visual input can influence tonotopic mapping in the auditory cortex (King 1999). Indeed, it is this *intersystem* integration that constitutes behavioral responses to the environment, a fact reflected at the neural level as well. These statements are true of invertebrates as well as vertebrates and indeed, in their way, of plants and single-celled organisms, too.

In many vertebrates, some processing actually takes place in multisensory neurons. Cortical and subcortical afferents from a variety of different modalities converge in these neurons and allow enhanced (or sometimes depressed) perception of an event. That is, neurons that respond to visual cues may also receive auditory and touch cues and then issue multiple motor commands, controlling the orientation of the head, eyes, ears, and so forth, in response.

Located in different regions of the neuraxis, the most well-understood being the superior colliculus of the midbrain, where sensory cues are received and motor commands initiated, these neurons have overlapping receptive fields for each separate modality from which they receive signal (e.g., Kadunce et al. 2001; Meredith and Stein 1986; Stein et al. 2001). Interestingly, although they already have multiple receptive fields and act as fully developed multisensory neurons in their signal reception and sending capacities, these neurons are not able to process multiple modalities at birth. Their multisensory processing capabilities only develop with increasing sensory experience (Wallace and Stein 2001).

This fact is not a surprise, however, as it is known that if one sensory modality is lost, the parts of the brain that perceived and controlled responses to that sense can be recruited by other sensory modalities. This rewiring or remapping does not need to happen only during development; the brain is plastic throughout life. The visual cortex of someone who loses vision as an adult, for example, can be partially recruited for touch or hearing. As with many aspects of perception, sensory and multisensory processing in the brain is the product of hard-wiring *and* experience.

THINKING: WHAT PERCEPTION "FEELS" LIKE AND ORGANISMAL RESPONSES

It is difficult for us to assess the relationship between detection and perception without involving our own experience and notions that connect awareness and consciousness. Indeed, that experience places limits on our attempts to understand perception generally, especially in other species. Thus, a discussion of consciousness itself, speculative though it must necessarily be, is in order.

Nerve nets and ganglia allow some forms of coordinated or integrated responses. Plants, however, do not have any kind of organized nervous system, and, as we have seen, they manage to have coordinated and integrated responses to environmental signals nonetheless; therefore, a centralized perception of the environment or a higher-level organization of response is not necessary for organized multicellular life. With respect to plants and the simplest nerve nets, we may say that "all" this is is reflex triggering of neural impulses by input impulses—a purely mechanical process. In the case of ganglia and more complex systems, we might say that this is "simply" a more highly organized form of the same thing. Cognition—the *real thing*?—involves similar chemical reactions. What is the difference, and, especially, what is the difference when it comes to consciousness?

One standard if nearly metaphysical answer is that these higher-order processes are analogous to electromagnetic fields in physics: when many neurons fire at once, a higher-order emergent phenomenon occurs, say a "field" of "electro-chemomagnetic" energy, and if enough neurons fire at once, that becomes the phenomenon that we, at least, experience as *consciousness*. As humans, we tend to equate this with the organized unifying phenomenon we refer to as perception.

How this would work or what it means in scientific rather than metaphysical terms are difficult questions. But let us take the ideas as generically correct; if perception *is* an emergent phenomenon, can a simple organism with a neural net, or even a ganglion itself, also generate some form of emergent "aura" of awareness that we would recognize relevant to our own human experiences? For that matter, because plants also organize reactions in a complex way, is there any sense in which we could refer to this as plant "perception"?

We have several times referred to the complex social organization and problem solving abilities of ants and other social insects, who have the tiniest of brains. We can list neural wiring patterns, name some lumps and bumps and connection zones in the brain, and identify neurotransmitters, ion channel genes, and adhesion and signaling factors and receptors that are expressed in brain development. But do we understand what this means any better now than Aristotle did 2,400 years ago?

Natura non Facit Saltum: Perception is Not an All-or-Nothing Phenomenon

Humans with severe forms of epilepsy have sometimes been treated by severing the *corpus callosum*, one of the major neural throughways that connect the left and right sides of the brain. In these "split-brain" individuals, who behave and report still feeling their normal identity, the evidence suggests that only one hemisphere of the brain (usually, the left) is the seat of consciousness. The other hemisphere is a fully functional, problem–solving entity that can even be shown to be self-aware. But it may not have consciousness (an important part of this may be aspects of language and verbal expression, usually controlled by the left side). Perception and sophisticated problem solving are possible without explicit consciousness.

It is clear that there is no one such thing as consciousness or the experience of awareness. We sleep, dream, and can drive cars without "thinking" about it. Each person experiences things in different ways at different times and differently from other people. Some people have a "feel" for science, music, personal interactions, or the flow of a hockey game. Some smell or see things others completely miss, facts that in this case we know directly, from studies of opsin spectral analysis or odorant-specific sniff-testing. Some people even claim to have direct contact with the immaterial world, whereas others who lack that experience strongly declare that to be delusional. Fanciful as such contact experience may seem, areas of the brain that are responsible (for the experience, whether or not the reality) are identified by studies of brain activity scans and the association of some such activities with epileptic episodes and so on.

The most central of all observations of evolution is that systems in related organisms are similar because of shared descent with modification. There is considerable freedom in evolution, but not complete freedom; thus, this kind of relationship has an organic reality. There is no reason whatever not to extend the same principles to perception and consciousness itself.

Darwin was committed to the notion *natura non facit saltum*—nature does not take leaps. To him, this principle gave evolution by natural selection its plausibility, and he used this principle in defending many attacks on his theory (especially in expanded discussion in the sixth edition of *Origin of Species*). His inference was that evolution and selection worked gradually over time. Traits did not emerge de novo. This in a sense is the basis of the modern view by which the step-by-step evolution of complex traits occurs.

Evolutionary biologists have come, perhaps somewhat reluctantly, to accept "punctuated" events in evolution in two senses. First, it is generally accepted that there can be times of acceleration in the rate of change relative to longer times of slower change. This can be brought about by things like the invasion by a species of a new territory or by rapid climate change (or by intense use of antibiotics). Secondly, homeotic change is recognized as a mode of evolutionary "jumps" in which the number of segments like digits or vertebrae can change as a result of changes in meristic patterning processes. In a sense, the latter is just a quantitative change, requiring no sudden novel mechanisms. But beyond that, we have no good examples to persuade us that a new trait can suddenly appear.

It is thus interesting that a darwinian (gradualistic) view of consciousness should be controversial, but it seems to be so. Many scientists are reluctant even to acknowledge that chimpanzees share the human experience of consciousness in more than a rudimentary way. In what can be characterized as at least a bit anthropocentric, since humans and chimpanzees have evolved numerous important differences in other traits, efforts to explain our evolutionary difference from apes has often been focused on the brain. The classic example is the famous but vain struggle by Richard Owen in the 1800s to find a trait unique to the human brain. He thought he had done so with the hippocampus major but was famously embarrassed in that notion by the ever-combative Thomas Huxley. Recent invocations of the brain-centered bias have demonstrated genetic differences between humans and chimpanzees that involve language or differential gene expression in the brain (Enard et al. 2002a; Enard et al. 2002b).

Any application of human experience to "lesser" species—even dogs and their emotions, not to mention the social behavior of ants—is typically denigrated as anthropomorphizing (or worse). However, some authors have tried to justify the position of humans as part of the natural world (rather than above it, by virtue of our unique traits, like consciousness) by insisting that consciousness has arisen out of a phenomenon that somehow pertains to *everything*. These views have typically come out of philosophical or religious perspectives under notions that might generally be referred to as *animism*. Animism asserts some type of universal internal awareness and drive, of which the human mind is the logical, inevitable, or creational acme. Famous biologists and writers on biology including Lamarck, Henri Bergson, and Teilhard de Chardin (to name some of the most prominent) have held such views, often including rudiments in atoms themselves to make the system universal (e.g., Chardin's *Phenomenon of Man*). Even one of the cofounders of population genetics theory, J. B. S. Haldane, made this kind of point (Haldane 1932).

We can look at this subject from an entirely rationalistic point of view, with no mysticism attached. All we need do is apply the same evolutionary notions that we apply to other traits, asserting the gradual or quasi-gradual origins of new traits (no saltations). *Natura non facit saltum.* It follows almost automatically that other species would at the very least have identifiable run-ups to what we experience as consciousness, and these should involve the same sensory neural-integrating processes in other species that they do in humans. We may not know what context-dependent, intermittently flickering, or partial consciousness feels like. Nor are there unambiguous criteria for speculating how far into the range of species this phenomenon may reach. But by any consistent standard of evolution, this important trait must have evolved over time and rudiments of some sort may still characterize much of animal life.

It is hard to accept that even insects might have complex perceptions to go with the behavior that has impressed so many. Henry W. Bates, one-time exploring companion with Wallace, remarked of Amazonian sand wasps: "The action of the wasp [in building a nest for its young and stocking it with paralyzed insects] would be said to be instinctive; but it seems plain that the instinct is no mysterious and unintelligible agent, but a mental process in each individual, differing from the same in man only by its unerring certainty" (Bates 1863). Anyone watching social insects can easily see that their behavior is to some extent open-ended in the sense we have used the term in this book: they may or may not have a fixed or limited repertoire of basic interactions, but they use them in nontrivial, context-dependent ways that we should not diminish by a facile assumption that what appears so complex and well organized must be "just" instinct instead. One day we may develop some way to understand what the experience is like to be an ant or wasp.

The dramatic difference seen today between humans and others seems to rest clearly on language and/or whatever symboling abilities that entails and entailed evolutionarily. If we compare electrical engineers or Shakespeare, with chimpanzees, the difference of course seems qualitative. However, several species of human-like ancestors have existed in the past, and we can trace through the fossils the progressive increase in cortical brain size. It seems hard to envision that this was not accompanied by progressive increase in "consciousness." Not even *Homo erectus* suddenly emerged with consciousness.

To assert that consciousness is a totally new phenomenon or even represented solely by ourselves, would verge uncomfortably on an essentially unprecedented view of the sudden appearance of what Richard Goldschmidt, in opposition to the Darwinian assertion of graduation, famously termed a "hopeful monster" (a baby who could speak language to parents who couldn't?). It would be like invoking spontaneous generation or a form of special creationism—the very last thing most biologists would want to be accused of.

WHAT IS PERCEPTION ANYWAY?

It is easy to say that perception is an organism's way to develop an internal map of the external world, but, given that different species thrive with the external world internalized in such diverse ways and that even individuals within a species do not completely share internal maps, no specific internal map can be said to be a "true" representation of the outside world, but only one that suffices for the particular individual or species. In that sense, the subset of cues an animal takes from the environment are what its ancestors required to live and reproduce; however, in a sort of darwinian feedback mechanism, these cues also drive the evolution of the pathways that allow the animal to collect the information that it needs. And, in an echo of Kant, an organism's world only *is* those aspects it can perceive.

There are many other fundamental questions about sensory perception that we don't yet know how to answer. For example, given that there are critical periods for synapse formation, and if external stimuli and experience are crucial to their formation, how can stereotyped patterns arise? How much of what appears to be pattern merely reflects the shared experience (e.g., uterine environment) of all members of a species? If there are stereotyped patterns, brain areas "for" different senses, how can these areas also be so plastic, so readily recruited by other senses, in the event of, say, a brain lesion or the interruption of a neural pathway or its development or the loss of the relevant neural input? Is brain plasticity a generic phenomenon, so that the synaptic reorganization involved in acquisition of knowledge or new memories is the same as that involved in remapping of brain areas? How does the brain reintegrate sensory input into a single image or sound or smell after it is processed by a number of different areas? How does the brain integrate perceptions from separate modalities into one? How does a smell evoke emotion?

It is in this area, more than perhaps any other area in biology, that differences between molecular and organismal biologists' interests most diverge. There has been incredibly rapid progress in identifying genetic and molecular aspects of neural development and arrangement and of the genetic and neural wiring of various senses that must integrate information from the environment. But lists of transmitters, receptors, and gene expression cascades do not bring us much closer to understanding the phenomenon, or perhaps better the *experience* of perception, than philosophers do.

CONCLUSIONS

We referred earlier to Aristotle's defining of the classical separate senses. Here we have noted that not only is each sense a complex problem of perception of detection, but the senses are not really entirely independent in that their perception is often integrative. Indeed, the poet Dante Alighieri even suggested that shades (souls) of the departed were equipped with organs for each of the senses (Alighieri 1314). How else could he have spoken with them on his tour through the postmortem worlds of Heaven, Purgatory, and Hell?

We need not delve into that particular issue, but as this and the previous chapter suggest, our current repertoire of analytical techniques is good at uncovering neural pathways and visualizing brain activity given different sensory exposures. We have shown in a physical sense what topographic sensory maps are like, that is, how physical relationships among receptors are maintained by physical relationships in the brain. We have not discussed many specific genes in Chapter 16. Genes such as for olfactory receptors have made it possible to trace individual neurons, and transcription factor families have made it possible to characterize segmental development. Genes associated with apoptosis have been informative about the remodeling processes in brain development. And genetic disease and experimental mutations, along with informatic methods and expression profiling that allow unknown genes to be found that are related to genes of known function, have all aided in the understanding of the wiring and developmental processes.

The processes by which the brain develops and by which connections from the periphery to the CNS are built involve the same kinds of processes we have seen again and again in surveying the way organisms get through life. We have learned much in recent years about neural mapping, especially of olfaction and other senses. In brain development and sensory mapping, division, segmentation, differentiation, cell adhesion, signaling, receptor-ligand information transfer, and apoptotic signals are involved. We have seen these processes used numerous times in the development of plants and animals and have seen that they also involve many of the same genes.

In a way this reveals how the somatotopic mapping of inputs to the CNS is achieved, almost as the requisite spin-off of standard, relatively simple and straightforward patterning and developmental processes. No specific natural selection for topographic maps per se is needed. The organization of sensation of open-ended systems into topographic maps from sensor to brain is sufficiently accounted for by the directness of the processes by which segmentally or regionally organized tissues are laid down—progressively during development, and ramifying in nested, modular form from simpler beginnings over evolutionary time. Development of the tissues at both the detecting and the receiving end will have these modular characteristics, as they are the general characteristics of tissue development. Once the detection-reception connection was made in the rudimentary system, that it proliferates and retains spatial relationships as the system evolves could basically be predicted.

For visual imaging and some aspects of touch, the topographic map was of particular importance because it represented the actual spatial arrangement of the outside world providing the stimulus. For sound, the cochlear tonotopic map is a byproduct of the developmental process generating the physical structure by which wavelength detection occurs. For odor detection, the map is a useful matter of bookkeeping. In all these cases the nature of the developmental process involved in making the isomorphic maps did not require fundamental new mechanisms.

Generally, we have done a lot of hand-waving in this chapter. The genetic basis of these neural wiring patterns is only partially understood. More importantly, the perceptive side, the side that we may perhaps be most interested in as human beings and as scientists, remains largely a black box. Why is it that we have maps of sound, light, and touch in the brain, but these perceptions *feel* different? Why do we see an object as a kind of information panel but not hear sound as a straight line as on an oscilloscope? Understanding these aspects of perception is one of the most fascinating questions in biology.

An important fact is that the neurochemistry and cell architecture are essentially the same in the visual and auditory parts of brains, and indeed one area can take on functions of the other, as we have noted. This helps provide an answer to the question as to how such diverse and complex systems can possibly evolve and develop. Each sensory system uses its own means to detect signal. But *all* the systems are translated into the *same* lingua franca of perception: the *interaction* of signals in networks of neurons. Perception in this sense has nothing to do with the nature of neurons, any more than **AGA** has to do with the functional characteristics of arginine for which it codes, or Fgf8 has to do with the morphological structure of teeth. Unless a truly new principle is discovered, this fact would have to stand as an additional profound way in which the traits of life are achieved by functionally arbitrary means.

PART V

Finale

Evolutionary Order and Disorder between Phenotypes and Genotypes

We have reviewed many aspects of the lives and times of complex organisms. It can be seen both in terms of traits and their associated genes that life is a universally connected phenomenon, in many simple but elegant ways. Genes are complicated, but the process by which they have been strung together and used to make the Great Chain of Beings that live today, and that have lived, is a mesh of chemical interactions that follow some simple general processes that lead us to a unified view of life.

Indeed, even the traditional definition of what constitutes a "being" relative to the systems of life, has been somewhat arbitrary and restrictive. The relationships of genes within cells, of cells within organisms, of organisms within species, and of species within ecosystems have remarkably similar features. In many ways, it is the emergent properties of sometimes arbitrary interactions, not the chemicals themselves, which have the most profound meaning in life.

Genetics and the Logic of Evolution, by Kenneth M. Weiss and Anne V. Buchanan.
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Chapter 17
A Great Chain of Beings

It doesn't take a brilliant mind to discern that there are extensive and orderly relationships among the creatures on this Earth. But it has taken a number of brilliant minds to try to explain *why* that is. Charles Darwin and Alfred Wallace were the first to articulate clearly the basics of what is still the prevailing view, that natural selection is responsible for the diversity and connectedness of life as we know it, giving a new kind of meaning to the order of life classically developed by Aristotle. We use these thinkers as representative of many others, whose work also contributed to our evolving modern understanding. However, their explanation didn't satisfy everyone—many naturalists thought it was incomplete and it challenged the fundamental world-view (not to mention long-held pronouncements) of many others. Beginning almost immediately after the publication of the first edition of *Origin of Species*, there has been a steady stream of resistance to the idea of natural selection, and many of the same arguments are still raised by skeptics nearly 150 years later. There are still religious and political agendas that lead to some opposition, but some unease continues within professional biology as well, and the basic problem has not really been resolved.

To biologists satisfied with the classical darwinian explanations, a blanket invocation of evolution by a persistent, consistent directive *force* of selection seems to suffice, though the long geologic time scale means this must generally be surmised rather than observed directly. To those who are not satisfied with so generic an assumption, the major problem is to understand more specifically how a simple process like blind (not teleological) natural selection could bring the diversity and apparent high degree of specificity of complex adaptations about, especially in terms of the genetic mechanisms that are responsible.

Here, we will try to present an integrated summary and overview of the major points of this book, much of which deals with this basic question. A modest number of basically simple general principles show how evolutionary processes can indeed have achieved what we see. Some of the generalizations are the classics that go back to Darwin and before. Others are not usually thought of as general evolutionary principles, although we believe they deserve that status and have tremendous

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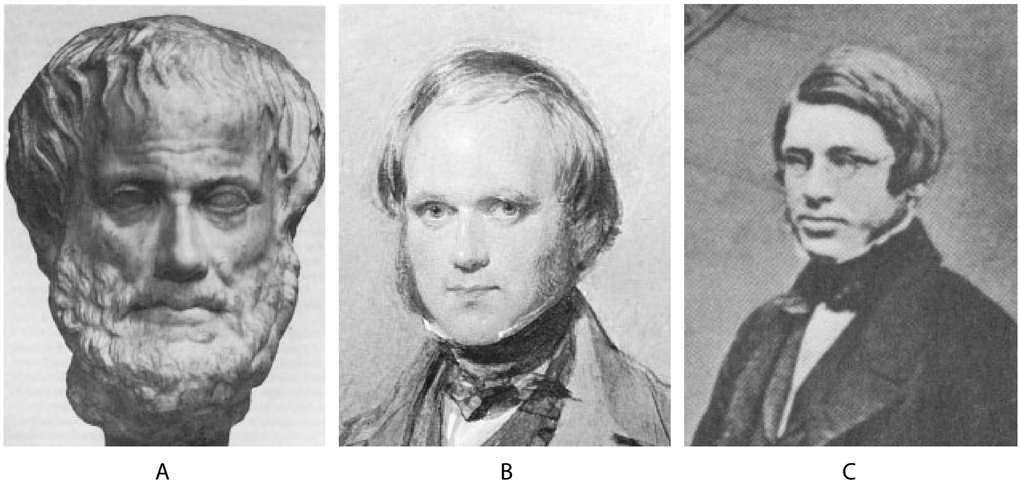


Figure 17-1. (A) Aristotle; (B) young Darwin; (C) young Wallace. (A) Statue in Vienna art museum, copied from (Bowder 1982), (B) 1840 painting by George Richmond.

explanatory power. Among our points is that one can accept Darwin and Wallace's deep insight, yet need not *overly* or uncritically invoke the single force of systematic natural selection as we try to understand the evolution of life.

Before there was much formalized history and essentially no paleontology—no sense of great time depth in the affairs of our planet—a static view of the nature of the relationships among the different forms of life on Earth was not just natural but probably the best supportable one (although there were dynamic cosmologies even among the ancient Greeks). In a static worldview, it was easy to group similar-looking animals, like different types of fish or birds or butterflies, and to infer that they have a relationship with each other.

Aristotle is widely credited with expressing the first systematized biology, which included the notion of a qualitative *scala naturae* (natural ladder, scale, or order of all life), the natural ranking of beings according to their relationships, and considering them in order. In a classic book, Arthur Lovejoy (Lovejoy 1936) traced the history by which this idea, which came to be called the Great Chain of Being, was established in Western culture; mainly, it was fitted into our biblically based cosmology with a point of origin at the Creation. In most Western versions, of course, this is a linear hierarchy with humans at the top (but under God) and the rest of the chain in service to us.

However, evidence was accumulating into the 19th century that the Earth was older and organic beings were not so static as had once been thought. Discoveries in geology, agricultural breeding, geographic distribution of forms found by world exploration, and a growing appreciation of the meaning of fossils all led to the burgeoning realization that species had changed over time. Darwin and Wallace provided a very general *process* that could account in principle for this dynamic history and connect the entire living world—plant and animal—into a single phenomenon. They stressed selection, but probably as important was that *time* and connectedness through *common ancestry* became key variables in biological explanation.

Biology has not abandoned a great chain of being, but we account for it in a new and less arbitrary, materially more testable way. And, taking organisms past and present, the chain has been rearranged. The evolutionary process leads to diversifying rather than linear connectedness, and does not imply that the tips of one branch are qualitatively better than those of another, only that their ancestors were successful *in their own times and relative to their local situations*. Despite this, however, an informal notion of progress, basically of a qualitative hierarchy in nature, is surprisingly persistent in our culture and in biology itself (Ruse 1996), even though it is manifestly true that simple organisms still thrive today. The notion of progress is more than a human conceit; it leads to a kind of tacit general inference of perfection and tightness of adaptation in the world.

The new evolutionary worldview was wondrously reaffirmed during the 20th century in a totally unpredicted way. Biologists adopted the molecular reductionism of the physical sciences and used it as the framework for looking inside organisms, to show that their parts, including their genes, had their own reality as biological entities, and were also connected in an historical way and that this was related to the general pattern of connectedness of whole organisms. Rather than a great chain of static being, what we see now is all organisms *and* their multilayered internal constituents, interdependent today and linked through a three- or four-billion-year-old continuum of interwoven connections: a Great Chain of Beings.

It is worth quoting Darwin's famous reflection on the grandeur of life by which he closed *Origin of Species* in 1859:

> It is interesting to contemplate an entangled bank, clothed with many plants of many kinds, with birds singing on the bushes, with various insects flitting about, and with worms crawling through the damp earth, and to reflect that these elaborately constructed forms, so different from each other, and dependent on each other in so complex a manner, have all been produced by laws acting around us. These laws, taken in the largest sense, being Growth with Reproduction; inheritance which is almost implied by reproduction; Variability from the indirect and direct action of the external conditions of life, and from use and disuse; a Ratio of Increase so high as to lead to a Struggle for Life, and as a consequence to Natural Selection, entailing Divergence of Character and the Extinction of less-improved forms. Thus, from the war of nature, from famine and death, the most exalted object which we are capable of conceiving, namely, the production of the higher animals, directly follows. There is grandeur in this view of life, with its several powers, having been originally breathed into a few forms or into one; and that, whilst this planet has gone cycling on according to the fixed law of gravity, from so simple a beginning endless forms most beautiful and most wonderful have been, and are being, evolved.

This "vision statement" is interesting in ways that are perhaps unappreciated. Much food for thought is here, including mistakes relative to the currently prevailing view. Darwin uses the image of an "entangled" bank, of life interdependent, brought about through common "laws" that we still see "acting around us." He states the fundamental core principles we outlined in Chapter 1. He keeps open the possibility that life may have derived from more than one founding form—and we have seen some evidence of a reticulated rather than simple Tree of Life, perhaps particularly early on before the highly structured cell that we know today stabilized as a basic form (Chapter 2). Darwin stresses the divergent nature of the process, but his view is somewhat lamarckian ("variability from . . . use and disuse"). His evolutionary theory was so powerful and represented so deep a truth that it withstood incorrect notions even about core elements such as the nature of heredity and the age of the Earth.

Darwin's is largely a metaphor of the world as a Hobbesian war of all against all, driven by overpopulation relative to resources. Historians have suggested that his view of life was colored by his being a wealthy member of the world's Imperial power. This predisposition is said to be reflected in Darwin's regular allusion to the struggle for survival as an essentially deterministic and gradual view of life driven by laws as ineluctable as those of physics. Systematically, the better and more powerful are destined to prevail; the law of the jungle is the law of nature. Whether or not Darwin's view is based on his wealth and privilege in the British Empire, that view is the prevailing one today.

When Mendel demonstrated the discrete nonblending nature of heredity—that heredity was a natural "atomic" process of some kind—he opened an important door to further the understanding of evolutionary mechanisms. A century of intense work identified the molecules involved and demonstrated that they were inherited and modified in ways that could be fitted to the general processes leading to accumulated biological diversity that Darwin and Wallace had invoked. The result has been a set of broad, general principles, formalized as population genetics, which are applied universally to the evolution of life. That is what makes the work of Darwin, Mendel, Wallace, and other founders of modern biology so powerful.

In this book, we have looked at how evolution and genetics apply to a selection of different kinds of biological processes that exemplify the basic aspects of biological complexity. These include:

Figure 17-2. An entangled bank. Central Pennsylvania, USA. Photograph by E. Weiss.

1. how the largely prespecified complex forms of a differentiated organism start from simple beginnings like a single cell;
2. how complex differentiated entities can reproduce;
3. how the components within an organism communicate to bring about a unity of coordinated functions;
4. how organisms detect a variety of external conditions that cannot be prespecified but that are important to their life ways; and
5. how the latter information is interpreted and translated into responses.

We have discussed how a set of simple principles stated by Darwin and Wallace, a few others that supplement their theory, and a modest toolkit of genetic mechanisms can account for the diverse world of biological complexity and achievement. Some of the principles are not typically included as formal premises of biological theory, but including them helps integrate a unified and more complete understanding of the great chain of beings in the world today.

Darwin's Basic Postulates

For most purposes, all living forms can be viewed as descendants of a single origin on Earth. A "single origin" for all life does not mean a single original molecule, species, trait or gene. All have their own individual, partially independent nested origins over time. Rather, biology posits a single set of starting conditions and that the essence of the system, once begun, did not receive meaningful extraterrestrial contributions (even if, say, amino acids continue to rain down on Earth) and did not keep originating (and in particular, required no spontaneous creation of complex organisms). In many ways including ancient or even occasional modern horizontal gene transfer, sexual reproduction, and recombination, life is as interconnected today by biological processes as it is by a unitary ancestry, and the former is ultimately because of the latter. If other forms of biological activity—for example, those not relying on DNA or RNA or protein coding—existed back at the beginning, they have become extinct without leaving a trace that we recognize today. They need not be considered to understand how biology works today, but it is worth noting that if such other orders did exist, our *assumption* that the coalescent structure of all life today reconstructs the *origin* of life, will blind us to them.

Biological information that is replicated across the generations of reproduction can accumulate a trace of its past, as we see in DNA. This is due to the modular, slowly mutating nature of DNA but is not a formal necessity of darwinian evolution per se. This is clearly so, as Darwin's ideas of genes were largely wrong. Many aspects of a cell are specific to each organism and are inherited but do not bear such a trace; examples are the particular mix of minerals, salts, pH, vitamins, minerals, and so on. These are inherited in the fertilized cell by which life begins but do not retain a permanent kind of information the way genes do. Still this illustrates a legitimate and important point about primary versus secondary causation in life: is it DNA or cytoplasm? Chickens or eggs? This is not a conundrum at all: the answer is *both*. Eggs are continuations of chickens, a chain of cell division going back billions of years.

If we do not force ourselves to be constrained by an overly rigid definition of inheritance as strictly applying to genes, similar statements are true of aspects of life that are inherited in other ways (Chapter 3). Indeed, some are inherited in a lamarckian fashion; for example, behaviors learned or traits acquired in life that are transmitted to offspring, like local microenvironments, birds' nests, and aspects of verbalization. Of course, cultural inheritance is vital to human survival, and its accumulated sophistication makes possible our *social* and *technical* complexity—that makes us *seem* qualitatively different and superior to all other organisms. But a human stripped naked and dumped alone deep in a wilderness would be a laughable King of Life. If John Milton (*Paradise Lost*) is any guide, Adam and Eve were created with a house, plantation, and cookware. Culture is inherent in what differentiates us from other animals, not a trivial add-on unrelated to our evolution, and something similar is true of most if not all animals. Of course, the inheritance of acquired characters we are referring to has no implication of inner drive toward some long-term goal, although human culture is probably more truly Lamarckian in that respect than almost anything else in the living world.

Malthusian population pressure was an important stimulus for Darwin and Wallace, and is important in life but not necessary to evolution. Phenotypic change could in principle occur in a population that did not suffer from overreproduction, for example, if all individuals had equal chances of reproducing (that is, change by *drift*). But *if* more individuals exist at any time than their source of nutrients or propagation can support, and *if* relative success depends on heritable information, and *if* that information relates to form, then whatever the cause, there will be change of form. Darwin and Wallace had the great insight that *if* there was a systematic favoring of a particular subset of competing forms, that could produce particular kinds of adaptation, and natural selection became a transforming concept. But population pressure does not guarantee by itself that there will be this kind of adaptation, and we know that differential reproduction is heavily affected by chance. We also have seen in Chapter 2 and elsewhere ways in which the fact or nature of adaptation, always viewed after the fact, can be illusory. *All* organisms are the descendents of an unbroken chain of nearly four billion years of successfully *adapted* ancestors—whether or not they had the highest fitness, as measured by population genetics, among their respective peers.

The basic darwinian principles not only provide a logically coherent explanation of how complex evolution can occur, but *predictions* from this reasoning are borne out regularly in new data that were not used to develop the theory. Examples are the statistical correlation of DNA sequence differences with times of separation among species, and the lower level of variation in DNA sequence affected by selection. Although the theory does not enable us to predict the specifics, the general kind of predictive power of evolutionary biology constitutes a convincing demonstration that the notions are compatible with a large body of data, and that is the most we can ask of any science.

Evolutionary theory allows fewer "bits" of information to account for at least some aspects of life than their complete description and enumeration. However, there is a danger that a convincing theory becomes a constraining ideology, with vested interests that resist dissent just as in any other ideology, and in some hands this has occurred in biology. Ideology fetters thinking, and this is important to avoid as we try to understand how life has evolved because there really are a few problems, of incompleteness if not also of stress, with the elements of the theory of Darwinian evolution.

Some Additional Principles: Modularity and Sequestration

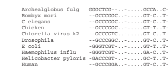
Darwin's notion of divergence of character is empirical rather than logically necessary. Competition by itself does not imply systematic or deterministic change of an adaptive nature. Even when it has a genetic basis, evolutionary theory does not claim to provide any guide as to *how* adaptation will occur, only *that* it will. Nor does the theory say anything about the mechanisms by which complex traits come about in organisms. A few additional principles help fill those holes.

The DNA/RNA coding system, based on complementary base pairing between nucleotides, and the nature of protein-protein interaction specificity are principles of evolution as it has happened on our planet. DNA base pairing provides a universal, modifiable system with historical memory that, once evolved, makes the rest of divergent evolution and adaptation relatively easy and plausible to understand. As has been said of cells, after they evolved the rest was easy—and is explicable in cellular terms. In many ways, after the RNA/DNA system, the rest can be explained in terms that relate to this system.

Everywhere in life the importance of *modularity* or segmentation is clear. From the modular nature of proteins and DNA, through cells, through perhaps the vast majority of the structures and systems of complex organisms, nature is composed of modules. This is immediately apparent as a fundamental characteristic of genes themselves. New genes could in principle arise by the incremental accretion of nucleotides to the ends of chromosomes until, by chance, they formed a valid coding unit that could be expressed, but they don't. The genome can be viewed as an assemblage of modular elements, including gene family members (each with internally replicated structures like exons, splice signals, and the like), replicated telomeres and centromeres, and regulatory elements.

This reflects billions of years of duplication events plus the evolution of short elements like enhancers by mutation in otherwise noncoding DNA. These processes result in genes connected together on chromosomes along with regulatory sequence, which allows for differentiation among cells, and which in turn allows for multicellular organisms. Indeed, life works the way it does fundamentally because genes are concatenated on chromosomes and thus fundamentally through modular organization that was made possible by this duplication history (something presciently seen, with relatively little data, by Ohno (Ohno 1970)). In retrospect, we can see the chemical ways in which mutation and erroneous replication lead to duplication of structures. Thus, duplication is a commitment made so early that it is now as fundamental as any other of the postulates of biological evolution as it happened on Earth.

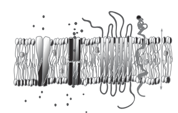
Modular physical structures occur from cells on up and across all of life. They include direct repeats, as in hair or leaves, or modified repeats as in regionally differentiated vertebrae, digits, or limb segments or butterfly color spots. Physiological systems are similarly organized. They involve the interaction and differentiated use of the products of duplicated, differentiated gene families. From lipid and oxygen transport to neurotransmission, transcription regulation, olfaction, immune response, selective sampling of frequencies in the light spectrum, hormonal signaling, ion channel formation, and most else, this is the molecular nature of cells and organisms. In fact, the core constituents of life themselves (amino acids, nucleic acids, lipids) are modular branches of the core biochemical energy-processing system of life (Morowitz et al. 2000).



Gene duplication provides some of the wherewithal for repetitive structures, but modular structures themselves consist of spatial or temporal repeats of patterns of similar gene expression rather than literal duplicates of a separate set of genes for each module in the system. Here, replication, sometimes with modification, occurs via interacting combinations of products of gene families and results in spaced replicated zones of structure activation surrounded by inhibition zones. In the sense of scientific understanding, but also in the sense of the phenomenon itself, repetitive patterning as a *process* involves fewer "bits" to make something complex.

None of this works without extensive *sequestration*. Darwin knew that there were problems with his blending theory of inheritance, but the fundamental importance of nonblending applies at all levels of life. Life may have begun in a fluid soup of some sort, in which simple chemical reactions took place in public. But at some point billions of years ago, this changed, which explains differentiated life as we have seen it from the earliest fossils to the present. From the nonblending nature of DNA sequences (including codons, enhancers, and so on) to the origin of cells as ways to protect an accumulating repertoire of vital internal reactions in controlled isolation from the external environment, sequestration of biological elements allows them to differentiate, specialize, and interact.

Sequestration makes possible the discrimination of immune, visual, auditory, and olfactory signals. To the extent that signals from different receptors activate the same neuron, the organism cannot make specific discrimination as to the source (e.g., in nematode olfaction each cell expresses multiple receptor genes and, in some insects, brightness and color signals may merge in their path to the brain). Diffusible signals in development are by definition not sequestered for the very reason that their relative strength across a tissue field is presumed to carry the developmental information for local cells within the field. But the reception of these signals *is* sequestered in that there are distinct, separate receptors or binding proteins for each signal to be received and specifically recognized. The receiving cells must be sequestered so that they can develop independently, which is how such a uniform tissue field becomes differentiated.



Actually, sequestration has led to life being a network of interactions among otherwise isolated elements. Divergence is important, but horizontal interaction (that is, among contemporary molecules, cells, organisms, and even species) is completely essential to organized life. Sequestration in many if not most instances is not or perhaps cannot be complete. Interactions are fundamental to development, homeostasis, reproduction, and even gene evolution (recombination, gene conversion, and gene regulation by *trans* factors acting on *cis* gene-regulatory elements). Complete sequestration would prevent that. And if hormones are diffusible signals within an organism, pheromones, flower color or odor, and ritual display or sound are examples of diffusible signals among organisms and species. In each case, the recipient has to be enabled to receive the signal, and we have seen many ways in which this happens; much of life depends on ligand-receptor binding to transfer information of diverse kinds in diverse ways.

Related to the notion of sequestration is the fact that most biological processes are *contingent*. This is implied by modularity, sequestration, and nested hierarchical organization because each new step in a process or stage of development is based on conditions at the time. However, it is the *partial* nature of sequestration that enables divergence to occur but not to isolate components completely. This is an important way in which the notion of cooperation in life is extremely important, to keep systems, organisms, species, and ecosystems integrated to varying degrees.

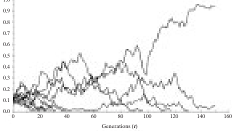
An Expanded View of the Role of Chance

Because we can always observe the current adaptation of a species, nothing prevents us from expressing (and, if we wish, believing) the idea that the adaptation is "remarkable" and "had" to have been molded by natural selection. But the after-the-fact nature of our observations means that this is a human judgment, and we are necessarily blinded to the nature of time periods far longer than our direct experience; we have to model them with some sort of law-like theory, often using the mathematics of population genetics because that captures general ideas in somewhat tractable form—abstract, but digestible.

A modified view actually makes Darwin's central idea of divergence from shared ancestry even more plausible: present-day adaptation does not in principle require a systematic, steady, or prescriptive selective environment. Chance is a frustration treated as a source of measurement noise in a science seeking deterministic highly predictive laws of nature, but chance has much more of a direct influence on life itself than is generally credited.

The difference between chance and selection largely depends on the parameters of the process, including population size and structure, intensity of selection and the like. The question as to whether there has been "enough" time for chance to have brought about given adaptation really rests on the general property that change happens faster when directed than when meandering. In this sense, as the estimated age of life moves backward (getting older), the role of chance via phenotypic drift and phenotypic substitution becomes more tenable in principle.

But since what is here is here because it has worked, and no specific thing *had*, a priori, to be here, the time question is a somewhat moot point, as explained in Chapter 2. Has there been enough time for *R*-genes or olfactory genes to evolve sufficient diversity? Is this even a meaningful scientific question? Of course there has been enough time, unless our most fundamental idea about a terrestrial, unitary origin of life is wrong. The more important issue is what mix of factors may have been responsible.



A highly deterministic view that assumes that selection is intolerant of variation, and highly prescriptive in nature is not necessarily accurate, and we know it is often inaccurate. In most of the examples we have covered in this book—from the constituents of cells on up—it would be difficult to argue, other than post hoc, that what exists today is actually optimized. Would one be able to say that about sex-determining mechanisms? If so, which ones? What about vision or olfaction or immunity? If these have been optimized, in what sense, and why are they so different, both among species and between individuals? What about people who type with two fingers rather than all ten? Daily life all around us is manifestly inefficient. Perhaps a better question is *how* prescriptive can we expect selection to be?

The answer seems clearly to be contextual. Optimizing or any other kind of tightly determining selection can certainly occur in principle, in the laboratory, or under the appropriately strong limiting conditions (like nutritional stress or the imposition of antibiotics or pesticides, but breeding experiments show that there are limits to what even artificial selection can accomplish). Nonetheless, although conditions for strong selection *might* occur, chance is *always* a factor in change from one generation to another. Phenotypic drift is a legitimate and logically plausible means of change. Humans are builders, and it seems difficult to understand that chance can play a role in what from our perspective seems clearly designed for its present adaptation. But accepting a greater role of chance relieves some of the need to construct specific post hoc selective scenarios that, no matter how attractive, often show exceptions and complications that must then have to be accounted for by caveats and additional explanations.

For most traits that we have discussed in this book, there is a diversity of standing variation. This applies to perceiving light or chemicals, defending against microbial attack, and even the essential phenomenon of sexual reproduction. These traits seem to have been shared by common ancestors of animals and plants, so their diverse forms today are one kind of evidence that *non*specificity of mechanism is important, not just a curious observation, and is a widespread if not fundamental characteristic of evolution.

Also as noted in Chapter 3 and elsewhere, change in the genetic basis of phenotypes can occur even when classical natural selection is taking place. Selection screens on phenotypes and does not "care" or detect how those are brought about. It works only indirectly on genes. This is a powerful protective mechanism, in that it allows for redundancy and alternative genetic pathways that may make survival and persistence more likely. Some core metabolic processes and highly specific communication via pheromones seem to have been rather tightly controlled even at the gene level, and may place deep constraints on evolution. However, as frequently demonstrated by gene knockout experiments in mice that fail to show the expected phenotypic effect or show it only in some background strains, even basic developmental systems that produce rather invariant phenotypes have buffered or alternative mechanisms. And some important functions, like olfaction, vision, and immunity, have specifically *imprecise* mechanisms that increase the chance of success: their sensitivity does not depend on the presence of one specific allele or gene.

Tolerance of variation is itself a survival mechanism, and every redundancy and imprecision is an opportunity for drift. It is in this sense wrong to think of variation as "noise" around some true signal (often called the "wild type") specified by selection. It is true that rabbits breed rabbits rather than mice, and this is certainly the result of genes. But the appearance of specificity in comparing species is at least in part an artifact of reproductive isolation. After long time periods, the variances around trait modes among related species may no longer overlap.

But within species there is always variation, and gene mapping studies and studies of allelic effects on most complex traits that have been looked at carefully bear this out. This is true of human and mammalian disease mapping, studies of yeast and bacteria, and natural and experimental observations in fruit flies, agricultural breeding, and so on. Many genes contribute allelic variation, generally of low individual penetrance, and there is phenogenetic equivalence. One cannot accurately predict the phenotype from knowledge of the genotype (except under favorable or highly constrained circumstances, or in a probabilistic sense). Similarly, and for related reasons, we usually cannot infer the underlying genotype from the observed phenotype, and if we can't do that, natural selection—much blinder than we are because it only uses the one criterion of fitness—can't do it either.

An important lesson we have learned from experimental and observational studies of genotype-phenotype relationships is that they often depend on the population of inference; they are not universals (Lewontin 2000; Schlichting and Pigliucci 1998; Weiss and Buchanan 2003). Related to the presence of redundancy at any given time is the evolutionary consequence that selection can keep a trait or function around over evolutionary time while its underlying genetic basis or even its physical basis changes. We have described examples of phenogenetic as well as phenotypic drift. Phenotypes, by and large, are not inherent properties of genotypes, and this is even truer of fitness, and as we tried to stress early on, it can be a mistake to treat selective coefficients in that way.

One gets a rather different view of evolution from the perspective of the trait or organism, not the gene. However, this is more in line with what motivated Darwin and Wallace in the first place, and it has ironic implications. Genes may bear the information trace of the past, and organisms may typically develop from single cells and hence be genetically driven; however, if the trait rather than the gene is what selection maintains, the ephemeral trait may be more "real" or lasting in that sense than its underlying genetic basis.

This certainly does not imply that traits evolve without underlying genes. Nor does it imply that there is no conservation of mechanism. Indeed, we have seen throughout this book exceedingly deep conservation of at least some aspects of phenogenetic mechanisms. The role of *Pax6* and opsins in vision across the animal world and extending even to algae and of genes inducing dorsoventral patterning between vertebrates and invertebrates are examples. But phenogenetic drift does occur and provides a view of evolution that is less imprisoned than reasoning that demands more causal genetic precision by selection need be.

Adaptive Evolution: Selection and Drift are a Continuum of Effects

There has long been debate about the relative prevalence of selection *versus* drift in evolution. In fact, these constitute a continuum. We discussed the key elements in the early chapters of this book. Genetic drift occurs when selective coefficients, s, equal zero (that is, the genotypes being considered have equal fitness). But this parameter is fundamentally context-dependent (it is *defined* as pertaining to peers within a population of inference). Unfortunately, this leads to a practical problem, because of the notorious difficulty in detecting selection in nature.

Even some of the most classic examples of selection such as related to the beaks of Darwin's famous Galapagos finches, or protective coloration associated with industrial melanism in peppered moths over the past two centuries, are not as clear as had been thought (Grant 1999; Grant and Grant 2002; Weiss 2002a). These are cases where the supposed selection probably *is* at least one of the important factors involved, and is probably a relatively strong force.

Instead of direct observation, we usually have to detect the effects of selection in a general way by comparing genomic sequence in various ways, and inferring that selection is responsible for the more stable aspects of the data. Thus, we infer that selection is responsible for the systematically lower observed variation in coding than in intronic DNA, or in regions of a gene showing more sequence conservation in one species than in another. Equating function with reduced variation is at least a little bit circular, but it generally corresponds with what we know of DNA function. It is this kind of analysis that led to the genetic load problem we discussed in Chapter 4, because while selection is so difficult to document in regard to genes on the ground, the statistical evidence for it is pervasive across the genome. It is not easy to account for how daily life can support the amount of selective loss required to maintain so many variation-constrained regions across the genome.

The most persuasive resulting generalization from this kind of population genetic approach to sequence variation is known as the "nearly neutral" view of evolution. As mentioned in Chapter 3, it seems that most of the time most selective coefficients are so small that the future fate of an allele is as much or more affected by drift than by the effects of selection (Ohta 1992).

Population genetic data have been showing things like this for a long time, but it is nonetheless common, if not usual, to see evolutionary explanations that equate present-day function with selective history. This often assumes steady, gradual selection as the causative agent and accepts the *net* result of selection as an adequate way to account for the local, day-to-day events that were actually responsible over evolutionary time and space. That verges on a kind of determinism that implies that what is here was destined in advance, and is in part a product of the adaptive illusion.

Despite this tendency, biologists are in unanimous agreement that this is an incorrectly teleological view. To account for complex evolution in a nonteleological way, we take from Darwin himself the assumption that complex traits got here through a series of intermediate precursors (sometimes called "exaptations") that had their own evolutionary reason for being. The usual reconstruction is to attribute selection to each such stage. The fossil record or comparative biology sometimes shows us what these earlier stages were, and sometimes we guess at what they may have been. Evolution is contingent in that changes from one stage to the next depended on selection among variation and by conditions that existed at the time, having nothing directly to do with what might happen in the future, and in that sense the model is one of "chance" evolution, even if selection is responsible all along the way.

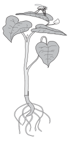
However, as we noted in Chapter 3, our everyday experience trying to demonstrate selection in action today, tells us that these local stages probably were, in their time, typically not under intense selection. Slow evolution with small selective coefficients at any given point in turn means stepwise nearly-neutral evolution. This in turn implies that drift can be important if not predominant at many or even most stages of the process. Of course, there is no way or reason to rule out occasional bursts of more stringent selection, nor that at all points on the way selection may have truncated phenotypes that were out of bounds for their local conditions. But small *s* means the bounds were broad, and tolerant of variation, for example, culling only at the extremes and shaping variation, but only weakly.

This scenario allows phenotypic drift to apply to the incremental changes, allowing a much greater role for chance than in the usual view. Indeed, pushed to its extreme, it allows what we judge retrospectively to be highly molded and focused to be due instead largely to chance, both in the contingency and the drift sense. This general model of complex trait evolution is consistent with the combined action of natural and organismal selection, various levels of drift, the extensive evidence for nearly neutral evolution inferred directly from patterns of DNA variation, phenogenetic equivalence due to many loci affecting complex traits, the existence of standing variation in populations today in essentially all traits (indeed, the fuel for future evolution), the imprecision and "imperfection" of biological traits, the widely varying forms of traits even among related species (that the examples in this book clearly show), and the correlation between separation times and phenotypic differences.

This view is completely consistent with the genetic theory of evolution, though it does not over-invoke classical, gradual, steady selection. Its elements are all familiar. The most difficult thing is to shake the anthropic illusion and to allow something that is functional to have evolved largely by chance in a much deeper sense than we usually accept. But because selection and drift are themselves in many ways different points on a continuum of effects, between this view and one invoking selection more strongly, the differences are matters of scale and perspective, as is so often the case in science.

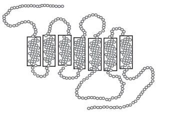
Some Additional Points

Watson and Crick characterized the basic nature of genes, which shows how the replication of chromosomes makes it possible for every cell in an organism to contain the entire inherited genome (except for somatic mutations). But this did not explain how cells with the same genome could produce a differentiated organism. We now know quite a lot about the way in which specific subsets of genes are activated in specific cellular contexts via *cis*-regulation using modular response elements in and around them. We have learned of other ways gene expression level is adjusted, including RNA interference, and quantitative effects of different numbers of copies of given enhancers, tolerance of variation in enhancer binding sequences, and the packaging or chemical modification of DNA near a gene that may affect transcription factor binding.



We have learned in recent decades of other sequence-based functions in DNA, including chromosome protection (telomeres), separation during cell division (centromeres), packaging (histone binding sites), and multiple RNA splicing to affect protein structure. These aspects of DNA sequence expanded the traditional meaning of the word "gene" from just protein coding to include these many additional functions. Others likely remain to be discovered. This changing understanding of the nature of the gene does not challenge the essential ideas in the genetic theory of evolution, which is based on the change in frequency of heritable variants, whatever their nature. But the diverse functions of DNA add richness to our understanding of evolution's mechanisms, and contribute to a more complete theory of evolution.

The function of many genes (perhaps most genes in complex organisms) involves regulating the expression of other genes, signaling, or other similar kinds of indirect function. Genes responsible for the final aspects of the traits—that is, the physical phenotypes of traditional evolutionary interest like morphology or behavior—are buffered from direct detection by selection by layers of causal interactions. Unlike specific proteins such as hemoglobin, most of these indirectly acting genes have multiple uses and interact with different genes in different circumstances. Epistasis and pleiotropy can constrain the freedom of action of selection because tinkering with these genes via selection on one of the traits they affect can have negative effects on the other traits.



However, the system that evolved uncoupled these pleiotropic regulatory proteins from what they regulate, because the protein can remain protected from mutational damage by selection, so it can do its work whenever needed, while mutational change can add, delete, or modify enhancer sequences the regulatory protein recognizes, allowing each *use* to evolve more independently. This allows considerable flexibility for selection while preserving a relatively stable toolkit, and is another way in which it has not been necessary for each new function to be built de novo by entirely new genes. The indirect, sequestered control of gene function by regulatory elements is another fundamental characteristic of life.

Some of the regulatory processes are subtle, including the way that mammalian X chromosomes are inactivated by being covered with *Xist* RNA; the competitive binding (or other) mechanism by which the members of *globin* or *Hox* gene clusters are sequentially activated; the means by which only one X-linked red or green *opsin* is expressed in a given retinal photoreceptor; the rearrangement of *immunoglobulin* or *T cell receptor* genes and the sequential use of the constant regions as response to infection proceeds; the allelic exclusion by inactivation of the other chromosome for these genes that leads to unique antibodies being produced by a given lymphocyte; and the near total exclusion of all but one *olfactory receptor* gene from expression in a given olfactory neuron. Posttranscriptional regulation by antisense RNA interference is an additional subtlety that we have mentioned.

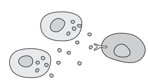
A consequence of this layered nature of causation is that a major fraction of biological activity arises from the action of genes that have nothing specific to do with the nature of the final trait to which they contribute. We have referred to this as logically necessary (the trait must be produced) but functionally arbitrary (it doesn't matter how it is produced). Signaling ties the living world together and is another of the general characteristics of life not specifically implied by the fundamental darwinian postulates.

Yet diffusible signals are a kind of arbitrary information-by-agreement. Nothing about an *Fgf* gene or its receptor need have anything at all to do with the nature of the final trait being made. The same is true with the many transcription factors, second messengers, and the like. Because they are arbitrary they, like codons and enhancer sequences, can form a general-purpose toolkit. However, this is not the same kind of "code," in that there is nothing in *Fgf* itself that is a stand–in for some trait the way a codon is for an amino acid.

In the last couple of chapters we have taken the notion of functional arbitrariness even further. If a generic toolkit of regulatory factors can in a sense account for all morphologies, it seems true that all of the physically diverse sensory inputs can be *perceived* by a single set of tools. The diverse functions of perception and central control are achieved by the interaction arrangements among a few types of neural cells. But the cellular properties of neural cells *per se* have little if anything to do with the nature of an image or smell, with whether one is looking at, touching, or tasting ice or ice cream.

The use of functionally arbitrary processes is deeply a part of life. Neural behavior most compellingly forces attention on the importance of learning how to understand "emergent" traits—that are fundamentally due to the *interaction* of elements rather than the sum or nature of those elements. How this will relate to or can be achieved by reductionist approaches, with their centuries' head start and long and successful record, or whether entirely new conceptual approaches will be developed, only the future will tell. But functional arbitrariness does seem to imply that we will have to understand many of the traits in life somewhere "above" the level of the gene.

As we saw in Chapters 4 and 5, a problem with genetic mechanisms being so utterly modular is that the genome is saturated with potential functional sequence. As scientists, we understand that function because of decades of experimental investigation carried on from without the organism. Organisms, however, have to figure this out for themselves, and from within. The DNA sequence of a new organism can be interpreted only because it arrives in an appropriate interpretive environment (e.g., mRNA, pH, and so forth in the cell). Again, this is why the distinction between a chicken and an egg is largely a false one: *life is continuity*. (Various proposed attempts to generate life spontaneously in a test–tube of ingredients may ultimately succeed but will not invalidate this last assertion, because the recipe that will be used will be derived from a knowledge of the nature of current life, that took billions of years to produce from the inside out).



There has been tremendous recent progress in identifying genes and processes involved in regulatory aspects of complex traits. But gene lists are not of themselves particularly more informative than was the classificational analysis of beetles in the 19th century, which is often denigrated from our modern perspective as having been "just" descriptive natural history. Dartboard-like today, regulatory pathway diagrams may ultimately become complete, via tools like expression profiling. Experts in the area are eager to deal with such data (e.g., Davidson 2001; Davidson et al. 2002a; Davidson et al. 2002b), but precedent and even the existing difficulties of understanding known pathway stereotypes suggest to us that how we will deal with the impending information inundation to move from even longer lists to better understanding, is by no means clear. For one thing, each new element adds another source of inter-species, inter-strain, quantitative, or stochastic variation. However one thing even incomplete gene lists *have* already shown is, again, the ubiquitous use of a few pathways, like *Wnt* and *Fgf* signaling and the like. In a sense, this shows how life is a combinatorial molecular phenomenon, another aspect of its fundamentally modular nature—and another challenge to understand emergence.

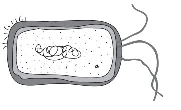
COMPLEX PHENOTYPES ARISE OUT OF A FEW BASIC PROCESSES

Even the most complex of phenotypes are produced to a great extent by a few basic developmental principles, a fact that makes it easier to understand the evolution of the diverse complexities of life. Not surprisingly, this directly reflects the underlying modular genetic toolkit and shares many of its organizational properties.

Complexity is an illusive term, but getting big is one kind of complexity. Size can be achieved simply by mitosis of adherent cells that function together as a physical entity. Spatially varied scaling (allometry) or temporally varying growth rates (heterochrony) among the parts of an organism are simple kinds of processes repeatedly used in different ways among organisms. But to be more than a structureless lump, such growth needs to be regionally differentiated, and temporal or spatial asymmetries that locate function are vital to many complex phenomena. We see this even in single-celled organisms, the classic example being the syncytium of a fly egg that establishes polarity via the interaction of factors with different concentration gradients within the cell.

Given the redundancy and diversity of function occurring even within a single cell, there would perhaps be too much chaotic cross-reaction had internal organization not already evolved early in the history of cells (and there is plenty of molecular chaos in cells as it is!). For example, cell membranes are loaded with ion channels and signal receptors responding to a host of different conditions, but they use similar sets of internal mechanisms such as protein kinase reactions. Intracellular membranes, lipid molecules, organelles, cytostructural molecules, transport and packaging proteins, nuclear membrane structures, and chromosome modification produce extensive intracellular sequestration.

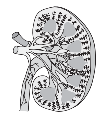
As noted earlier, morphological and physiological systems are typically characterized by repetition of modular units and/or segmentation. Rather than gene-for-unit specification, quantitative patterning *processes* bring repetitive structures about and as we have seen in several chapters, such processes may only require a modest number of interacting critical factors.



In some of the most well-understood examples, the factors are diffusible signals and their receptors that activate or inhibit selective gene expression. In principle, a two-component reaction-diffusion-like process can generate complex periodic patterning. We do not yet know how many factors are critical for the kinds of repetitive patterning we have described, from ommatidia in flies to hair to the arrangement of regions in vomeronasal or olfactory epithelia and the like. We know that there are some simple components, such as interactions between Bmps and Fgfs, or the *Delta-Notch* system, but we know that many if not most of the systems studied so far have redundancy or alternative pathways, that organisms can compensate for missing components, or, as revealed by many mouse knockout experiments, that the effects of a pathway are variable depending on naturally occurring variation even within a species.

Simple patterning processes can work because of the partial sequestration among the cells in a tissue field, which enables each cell to interpret and respond to the relative concentration of diffusing signaling factors in its particular extracellular environment. This allows repetitions of the same structure to occur, surrounded by inhibition zones. Making this even easier to understand is that these patterning processes are nested from the first stage of an organism. One process sets up fields, for example, basic polarity, anatomic axes, or organizing centers, which provide partial isolation or differentiation that can initiate subsequent more regionalized patterning, generating a hierarchy of ever more localized differentiation. This seems at least in part to be how limbs, vertebral columns, and invertebrate segmentation work. Because of the evolution of families of signaling and transcription factors, and their receptors and enhancers, initial pathways can diversify by specializing the various gene family members, another aspect of sequestration (that is often only partial because there can be cross-reactions).

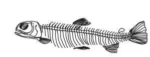
Morphological modularity is facilitated by, but also facilitates, the reusable nature of regulatory and signaling genes. Because these genes can be used in different combinations at different times or contexts in the same organism; prior stages in which a gene has been used lead to sequestered, differentiated descendant cells that then can respond to the same signaling factor in a different way. Some signaling pathways are used multiple times in different ways, even during the development of a single structure; an example is *Fgf* and *Hedgehog* signaling in dental patterning and tooth development (Jernvall and Thesleff 2000). It is the *cis*-regulation of gene expression that makes this combinatorial phenomenon possible, and these ubiquitous facts constitute another major fundamental commitment to a particular way of life that was made billions of years ago.



With the use of this set of mechanisms, complex traits develop via a few simple processes in addition to periodic patterning. Budding out or invaginating inward is brought about by local asymmetric cell growth, a signaled process. Branching is another simple process that is widespread in nature and can be repeated to generate a nested hierarchy of components. Among the widespread examples of complexity produced by such processes are the lung and bronchial trees of animals, the distribution of blood vessels, and, of course, *real* branches on plants. Plants avoid the need for layered repetitive patterning by retaining the potential for generating diversified structures that can be repetitively invoked in the sequestered environments produced by branching.

Differentiation by combinatorial expression of regulatory genes also applies to physiological traits and sometimes involves related or identical genes. This is the case, for example, for ion channel function, cell adhesion genes in neural development, the control of neuronal firing, osmotic function, lipid transport, the differentiation of cells from common precursors in the blood system, and many others.

We haven't covered all biological systems in this book by any means. We've concentrated on particular phenomena that are important to understanding how complex organisms work and how they got that way. The principles and often even the specific genetic phenomena are, however, similar for systems that we have not mentioned. For example, digestion involves the breakdown, absorption, and so on of proteins, fats, sugars, and other carbohydrates, using gene products just like the other systems (e.g., proteases, binding factors). Kidney filtration rests on ion channels and similar structures. Many of these are pure "chemical" processes; that is, they involve genes to the extent of synthesizing chemicals (e.g., HCl, pancreatic enzymes) and secreting them from cells but are mainly not "informational" in the sense of most systems and phenomena in which we have been interested here.



We can illustrate the kind of evolution that leads to diverse complex traits via a single set of mechanisms by an example from our own work. Among the most important characteristics of vertebrates are their mineralized tissues. As vertebrates evolved, the initial calcified tissue of external scales expanded to include teeth and bones (actually, it is not entirely clear which of these came first or whether the pattern was the same in all early vertebrate lineages). A class of secretory calcium-binding phosphoprotein (SCPP) genes, almost all still linked in a single chromosomal cluster (and that appear to have evolved through a series of gene duplication events in ray-finned and tetrapod lineages), is involved in the formation of different mineralized tissues in different species (Kawasaki and Weiss 2003). Some of these genes are expressed in bone development, others in forming the mineralized parts of teeth, and still others in lactation and salivary secretion (calcium binding in saliva can secure the mineral and probably has antibacterial function), tissues that arise from different parts of the embryo's developmental tree. Most or all of these tissues involve epithelial-mesenchymal interaction of the type described in Chapter 9.

These SCPP genes are physiological and not informational; they directly serve a final structural function rather than a developmental or patterning one. But the nature and arrangement of the gene family shows how step-by-step a diversity of complex traits can evolve through or taking advantage of gene duplication. The first SCPP gene was present before vertebrates, serving calcium-related metabolic function(s). In vertebrates, this trait could be used in the evolution of anterior feeding mechanisms (teeth), body protection (scales), and internal body support (bone). Subsequently, new functions took advantage of these genes and their epithelial expression pattern in the evolution of salivary and lactational functions that are basically unrelated to bones (except perhaps in the indirect sense like the use of casein to make calcium available to mammalian infants). Thus, metabolic function evolves through the same kind of gene duplication and subsequent divergence, resulting in the modular function "strategies" that we see in complex morphological structures.

Branching: A Common Metaphor with Differences

This is a point to note that some of the same metaphors apply at many different levels, as a reflection of correlated effects of the basic nature of evolution as revealed by the principles we have been describing. The idea of branching and nested hierarchy is a prime example. This is indicated in Figure 17-3. Karl Ernst von Baer was one of the 19th century founders of modern embryology. In pre-evolutionary times, he noted the way the embryos of collections of species, like vertebrates, begin life looking very similar. But as the embryos age, they diverge in form in the various species, ending up with adult variations on the theme of their shared overall body plan.

Figure 17-3A shows an attempt by Charles Darwin to understand von Baer's notions. Evolution in its original sense of developmental "unfolding" from a shared body plan might be due to a Creator's design, but Darwin saw that the embryological data were highly relevant to the problem of the evolution of species. He was among the first to use a similar branching metaphor for the divergence of species from a common ancestor (Figure 17-3B). Again there was a shared form, but that reflected the state of the ancestor. Development and evolution relate morphological similarities to very different time scales. There was both confusion and connection between the two, accounting for the famous Biogenic Law of Ernst Haeckel (shared to some extent by Darwin) that during embryogenesis (ontology) species sequentially recapitulate their ancestral forms (phylogeny).

Figure 17-3C schematically shows the somatic divergence of organ systems within the body of a vertebrate, starting from a single cell, the fertilized egg at the top, and ending up with the shedding of another single cell to form the next generation (sperm cells at the bottom). Finally, Figure 17-3D shows a fractal simulation of branching that, as we saw in Chapter 9, has been likened schematically to the structure *within* organs like the lung.

Metaphors are only so useful, and the phenomena in Figure 17-3 are different in important ways. But to a very real extent they are all the same, and for similar reasons: they all relate to the descent with modification, by duplication with variation, of partially sequestered lineages of genetically differentiated modules of cells. We have tried to show in this book how parallels like these are found throughout the living world, from genomes to species.

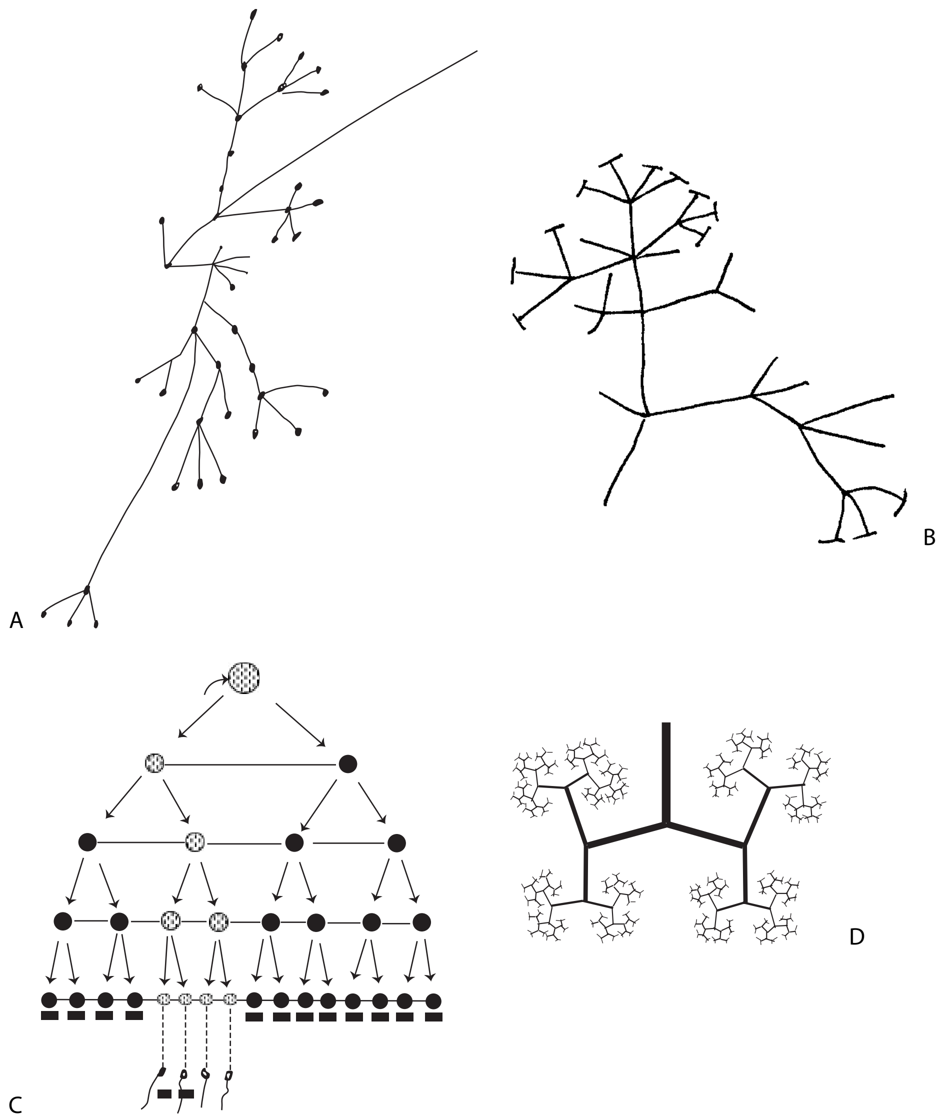


Figure 17-3. (A) Darwin's drawing of developmental divergence of basic body plan components from a common core body plan as proposed by von Baer; (B) Darwin's early sketch of the divergent nature of his emerging view of species evolution; (C) a somatic tree of development of organs within an organism; the sequestered germline cells are shown in gray; (D) a fractal pattern of branching as roughly approximates much of organ branching seen in veins, nerves, bronchial trees and real trees. (A,B,D) redrawn by the authors (Darwin's original figures can be seen in Richards 1992); (C) from Wilder (Wilder 1909).

HIGHER LEVELS OF EVOLUTIONARY ORDER

In Chapter 8, we alluded to various concepts of organization above the level of what is typically considered to be the organism. Such organizational structures have been referred to as "superorganisms," but for reasons discussed in that chapter, the definition of what constitutes an organism or "being" is more arbitrary than is often thought.

Bacterial biofilms and slime molds are among the most simply organized coalitions of individual organisms, although social insects are probably the most often-used example of clearcut organization above the level of what is traditionally considered an organism. Of course, human societies and hundreds of less elaborate animal social structures can be thought of in the same way.

From a darwinian point of view, of ever-present competition for reproductive gain, social organization should occur only if it confers a benefit that outweighs to the risks of group life (like potential exhaustion of food supplies). A major way this has been put is the need to explain the widespread occurrence of altruism, or how genes can evolve whose bearers sacrifice some or all of their own reproductive potential for that of others. This is an inflammatory dispute within biology, fought with ideological vigor. The argument is often engaged under the term "group selection." Strict genetic darwinism is individual-focused, strongly selectionistic, and will at most grudgingly accept sacrifice for the good of the group because of a lack of an obvious mechanism by which that can evolve. (Group selection was important in Wallace's view of evolution, however.)

The most common explanation for altruism is that its beneficiaries are close relatives who share similar genotypes with the individual making the sacrifice. Formally, in the general case, what an individual should be willing to sacrifice for another individual depends on the precise degree of relationship between them. As a general principle this seems reasonable, though it fails to account without contortions verging on implausibility for many of the social behaviors found in the world.

The individual cells in some bacterial biofilms are clones of each other, but many biofilms are aggregates of different kinds of bacteria. Why would different strains of bacteria be "willing" to enhance the survival of others? Is it that they are closely related *enough*? In *Dictyostelium* the slug forms from a collection of (former) individuals and it appears to be only a matter of chance which of them end up producing spores for the next generation. But to a considerable extent the issue is artificial. Most groups of organisms are collections of relatives to varying degrees, simply because individuals of most species do not disperse very widely from their place of birth or, if born into a horde or school, disperse *as* a reproductive group. Giving your all for the group usually means giving your all for your relatives in one way or another. Your peers are extensions of yourself. Precision in evaluating degrees of kinship before acting, or precise screening of individuals by selection in this regard, for the many reasons we have discussed, is neither required nor to be expected.

More interesting than the fine points of the altruism debate is a broader view of the nature of communities and the concept of community itself. An entire insect colony is produced by one or a few queens, and hence the entire progeny shares the half of the genome they inherit from her, with the other half coming from reproductive males. This means a peculiarly close kind of genetic kinship among members of the colony, as discussed in Chapter 8. This very close connection is why the term "superorganism" has been used to describe insect colonies.

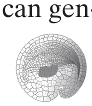
But we can extend the concept. Even a cell can be viewed as a superorganism, of individual genes, and an individual organism a superorganism of cells. In each case, there is internal connectedness, communication via signaling and response elements, with associated specialization of function and coordination of action. The immediate life history as well as evolutionary fate of one is dependent on that of others. There is active and sometimes externally instructed self-sacrifice, as seen in the widespread apoptosis among the cells in the life of a complex organism (or controlled mRNA degradation within a cell). Gene inactivation is to a cell, and apoptosis to an organism, what altruism is to a member of a society.

These internal characteristics of organisms—genealogical connections among genes, signaling systems that alter gene expression, and responses by one component to conditions of another—also apply to relationships between members of the same species and, indeed even between members of different species. Social insects represent but one way in which this occurs; in fact, it is a phenomenon found from biofilms to ecosystems.

Organisms communicate and coordinate by pheromones, visual, auditory, tactile, olfactory, and other means of message transfer, reception, interpretation, and response. Even under the most classical of darwinian scenarios, flowers evolve odors that insects can smell and vice versa. Signal and ligand, reception and response. The genomes are intimately connected and literally interdependent.

As noted in Chapter 8, one distinction used in the definition of what we refer to as an "organism" is that its cells are connected together to form a single body, and derive from a common cellular ancestor. But this is a somewhat false distinction. Cells and other elements within an organism are not always physically attached (for example, circulating blood cells), whereas in many species distinct organisms *are* physically connected when they reproduce—usually an essential part of reproduction. And members of a population are clearly connected by their unbroken physical chain of shared cellular ancestry.

Social and ecological interactions affect the collective genome(s) involved, which of course is thus of direct evolutionary import. Like any other set of interacting factors, the relative frequency and spatiotemporal relationships among the factors depends on the dynamics of their interactions. Just as in development, they can generate gradients of location in space (such as microenvironments), wavelike oscillations (such as predator-prey cycles), or relatively stable frequencies (such as population size). These are "emergent" properties of communities in the same way that similar patterned traits within individual organisms are properties of interactions among signaling factors.



At the embryological, organismal, and ecological levels, the interacting system involves the partial sequestration of its modular components. Indeed, though we think of organisms as entirely different kinds of "being," their interactions, as in pheromone signaling or sepual reproduction, involve exactly the same molecular processes—even genes and types of genes—as found in developmental signaling. If biofilms are any indicator, it may be that interactions among single-celled organisms were exaptations for the evolution of multicellular organisms in the first place.

In this sense, it is rather arbitrary what we should call an "organism." For appropriate purposes, even the entire biosphere can be viewed, truly, as composed of comparable entities interacting in complex, hierarchically nested and networked ways, by related genes and common molecular mechanisms. The distinctions between kinds of biological entities blur because they are all interconnected, in similar ways, now and by common ancestry back through the entirety of life. This is the Great Chain of Beings.

A BIG BLACK BOX STILL REMAINS

Discussion of the nature of emergent traits draws attention to at least one box that remains rather completely black. Inside that black box is *perception*. Perception is made possible by patterning processes, activation/inhibition, cell adhesion, ligand binding to alter gene expression, and so on—simple developmental processes that in natural and understandable ways provide spatiotemporal maps from sense to sensor. The developmental processes generating sensoritopic maps account for how individual incoming signal properties, like wavelength, sound frequency, or specific odorant molecules, can (in some species and for some senses) be distinct in an orderly and sufficiently replicable way. This is what we have seen for other organ systems.

Perhaps the most interesting aspect of sensory and cognitive systems, however, is not their molecular components but what these perceptive systems "feel" like to the organism, their emergent property. For example, how and why does hearing feel different from seeing when both involve structurally and chemically very similar neural systems (indeed, that can remap from one type of sensation to another)? This immediately leads to the issue of most interest to humans: the relationships between perception and consciousness.

As far as keeping track of incoming signals, one might liken the brain to a building guard sitting in the guardroom of a bank, watching the halls, teller windows, and so on through television monitors (it is in fact debated, but immaterial to our point here, whether the brain actually works by having a monitoring center—sometimes also referred to as a "homunculus"—or not). The guard does not see anything directly and must respond on the basis of video images provided by the monitors. This is inherently a limited amount of information compared with what is really going on out there in the bank. However, because television monitors are *designed* specifically to mimic human visual perceptive experience, the guard does have a rather natural, if one step removed, sense of at least that slice of external reality.

A somewhat more apt image might be to submerge the monitor more deeply, and think of a sailor in a submarine working with sonar and radio information. Sonar does not represent a natural human information gathering system. Why? Because undersea sounds are not the kind we are used to or evolved for interpreting. Undersea sound has to be translated into something interpretable, and we choose mainly to do that on a video screen, as blips identified as to distance, direction, and azimuth (or as audible beeps produced in earphones). These are all arbitrary ways to represent the information. Electronic detection (e.g., of radio signals) may be presented in some other computer-digested form. Indeed, sensors are made to detect things humans have no sensory means to detect (e.g., very low frequency radiation), and this information may be presented as spectral frequency pattern data, such as on an oscilloscope. In some cases information is presented in the even more abstract and arbitrary form as text on a screen, representing a totally black box analysis by a computer. Again, the data are translated into a form interpretable by humans, and the person has essentially only limited data, indirect contact, and must rely on integrated experience to interpret what the sensors provide. Every organism lives in a Kantian world, and can only know aspects of reality for which it has some form of receptors.

In this imagery, what counts is that some aspect of the signal, which we select for our own purposes such as to thwart burglars or

survive attack, is chosen for detection. We design our interpretive machinery from the outside, perhaps using cost or other considerations to decide what aspects of the signal to bother about. Evolution has had to do this by trial and error from the inside, and the only known criterion is reproductive success. Each natural sensory system is built today to detect a subset of environmental information in a way that was evolutionarily sufficient in the past. But the neurons themselves, like the electrons used in the submarine's systems, are not directly related to the nature of what is being detected by them.

It is not surprising that in many systems the topographic relationship between the signal and the interpretation is maintained. This helps the brain keep track of important aspects of the signal, although what counts as important varies. Above all, it seems to mean replication: the same signal will fire the same distinct set of neurons. In some systems like visual image perception and touch, the map directly reflects the structure of the outside world, while for others like olfaction it seems mainly to serve a bookkeeping function. Vertebrate hearing uses a physical trick of forming a fluid-filled tube to decompose a complex wavelike signal. None of these particular kinds of sensory maps are essential, however; some species ignore aspects of these information sources while detecting others.

We have not really considered other "higher" aspects of the integration of environmental information, like prey tracking, locomotion, mating, eating, and the like, which integrate environmental sensing, physical activity, proprioception, and so on. These are internal supersystems, integrating muscular, neural, sensory, and motor functions, each of which is an organized system in itself. Emergence upon emergence. The overall ways these complex integrative functions are orchestrated are not understood (although some of the wiring is known).

We have suggested that these functions, including consciousness itself, evolved gradually and may constitute traits of which we have not much more direct sense of the reality than a sonar operator does of things detected from the murky deep. But this intriguing subject has to remain for the future. Even attempts to specify what the phenomenon of consciousness actually is are confessedly highly speculative (Crick and Koch 2003). Many biologists would argue that, with present knowledge and tools, these questions are currently beyond the reach of science.

SOME CAUTIONS ABOUT EVOLUTIONARY AND GENETIC INTERPRETATIONS

One problem in reconstructing life is that adaptation—whatever its cause—is not perfect, and there is no way to define what "perfect" might mean. If the only criterion is that the fit of the organism to the environment be "good enough," and this depends on the changing landscape of local environments, competitors, colleagues, and the genotypes available, then we cannot expect definitive answers. Rather, the generalization seems to be that there is no one way and no need to be better than good enough under the circumstances—and lucky.

Members of some species are largely safe from predation (elephants, lions, humans, giant turtles, probably many viruses and bacteria, and so on). Some seem not just essentially safe but to have rather open life spans, such as venerable olives, the famous Tule tree in Oaxaca, Mexico, and some giant sequoias in California that are thousands of years old. Other species live but a fleeting hour or so before reproducing (e.g., bacteria, adult mayflies). Some produce young that are almost all immediately devoured. Every creature gets through life differently.

Interestingly, however, each must face the same kinds of challenges, of escaping from predators and microbial attack and finding food sources. The differences are mostly of scale and circumstance. There is nothing inherent in mammaldom that should make them vulnerable to a greater diversity of parasites, for example, than a tree that has to just stand there and take it. This means that there is little in the way of a priori criteria by which to predict how or what a given organism will do or be like. It is in this important sense that the organism-as-machine metaphor can be highly misleading; it is thinking of organisms in human terms, as being designed to solve problems laid before them by nature. An individual organism may try to get "there" from "here" (e.g., to catch that rabbit), but evolutionarily the only problem it is solving is to persist and reproduce (to catch a rabbit *today*). And this is why it can be misleading to think of there being a fixed number of senses, or ways to hunt, or a single kind of "immune" system, and so on.

Not only is chance a pervasive factor in the environments inside and outside of organisms, but imperfection, if we can use the term, is an essential factor that makes evolution possible and ongoing. Energetic *inefficiency* is what enables evolution and in that sense it is non-sense to ask whether a trait has evolved to be energetically efficient. The puffer fish makes do with a genome only about 1/8th the size of its vertebrate relatives who, nonetheless, have roughly the same genes. This shows that such baggage can in practice be off-loaded to save on the substantial metabolic demands it must make. Yet selection is often invoked to account even for what seem to be the fine points of nonfunctional DNA (for example, it is said that there can *be* no truly nonfunctional DNA or selection would have eliminated it). But if energetic considerations of that kind applied as often as they are invoked, most individuals in most species would be on the brink of starvation, and thus need to shed every ounce of needless base pairs. In energetic terms themselves, purging has clearly not been worth the cost. If selection were typically too stringent, evolution as we know it might not have been possible—nobody would have survived.

Again, it is necessary to think contingently and that means separately for each case, and one result is a substitution of description for scientific generality. But that itself seems to be one of the realities. For example, one can ask whether evolution has made some trait more efficient than it used to be for the same use, but the truth is only that the trait is as efficient as it has had to be and the uses are always changing. A classic example is the argument that mammalian legs are oriented to work more for-and-aft than the arc-sweeping of reptile limbs, and that this evolved because it produced more efficient locomotion in mammals. But think of alligators in motion. Whether they use more energy per foot traveled than mammals, many a mammal has paid the ultimate price despite having "more efficient" locomotion. Or are the winners among competing alligators those with more efficient inefficient locomotion?

This exemplifies the problem of making evolutionary reconstruction stories based on how any particular trait "must" have come about because the post hoc nature of evolutionary inference means that multiple explanations can have comparable plausibility. This is known as a *nonidentifiability* problem, and it is a ubiquitous fact of evolutionary biology.

This is relevant to the persistent division of opinion about the roles of competition and cooperation in evolution. If a genetic variant becomes more common over time, one can always *assume* or *define* its increased frequency as due to adaptational competition and can in principle then look back into the specific history of the gene(s) and infer

that what happened was deterministic. From this point of view, persons who see (or desire to see) cooperation as being as important in nature as competition can always be refuted: if social organisms, organelles, genes, or individuals cooperate and this leads to differential proliferation of associated genetic variation, then that cooperation can be expressed in a consistent way in terms of competition (some alleles do win, after all, if that's how we define winning).

But if the World According to Hobbes is a consistent one, the same outcome—what we observe today—can be accounted in a different way. Chance and organismal selection can lead to genetic change without the kind of darwinian warfare that makes for good television viewing (and has so often been used to justify social inequity). Organisms build their own environments and choose their own niches when they can. If it looks like cooperation, or *feels* like cooperation, then for all practical purposes it *is* cooperation on the level of organization at which it actually occurs. Cooperation *is* fundamental to organized life, from genes on up.

Focusing on how or why cooperation, or cultural inheritance, or social behavior, or organismal selection are "really" just wolves in sheep's clothing may help explain some aspects of life; however, this focus can lead to tunnel vision, drawing attention away from important or even pervasive aspects of how life works. If evolution is the meandering contingent process that it seems to be, the ultimate explanation that cooperation really represents a past history of competition may be true but too generic to explain very much. It is true that our house is made of nails, paint, and boards, but that does not explain our house.

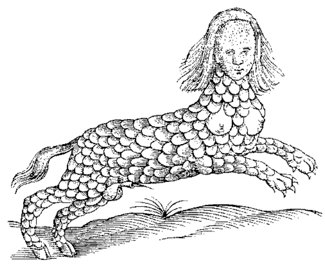
In many ways, the competition-cooperation distinction is one more instance of a false distinction of perspective. Sequestration by itself almost implies cooperation if complex organization is to evolve because the interaction of isolated units like cells is the essence of such complexity. Without "cooperation" between various biological molecules, nothing in life happens. The catalyzing of biological reactions was for much of the 20th century taken to be a definition of the difference between life and nonlife. DNA does not even replicate itself without help, despite that often being stated as a biological fundamental.

Chance is always present as is the potential for natural selection, but the same can be said of organismal selection, a potentially less combative source of adaptation. Organisms sort themselves into local environments depending on what their capabilities are, and over time genetic changes that need not have to do with classical adaptation (e.g., chromosome rearrangements) can produce a species barrier. Indeed, that they are mobile and seeking is one of the traditional definitions of what it means to be an "animal." Facultative searching of environments is also done by single-celled organisms and even by plants.

Like sexual selection, self-sorting by organismal selection can be faster and more precisely related to function than classical natural selection, because organisms know better what they can do best than the crude screen of selection may be able to detect. There is nothing unnatural about this form of sorting and proliferating of genetic and phenotypic variation, perhaps supplemented by genetic assimilation, though it has been treated as a kind of backwater of biological theory. Instead, the search *for* such mechanisms as a way to account for adaptations should be more active.

Not Laws, But Principles, and Why There is no Argus

We have identified a variety of principles that we believe apply generally to evolution as it has occurred, and we think these make a useful addition to the usual darwinian principles. They are not new to us, nor to biological thinking, but they are not usually treated as part of the theory of evolution itself. If our view is justified, we might ask if these principles have any predictive power, one of the criteria for generalizations in science. In fact, we think this is indeed the case.

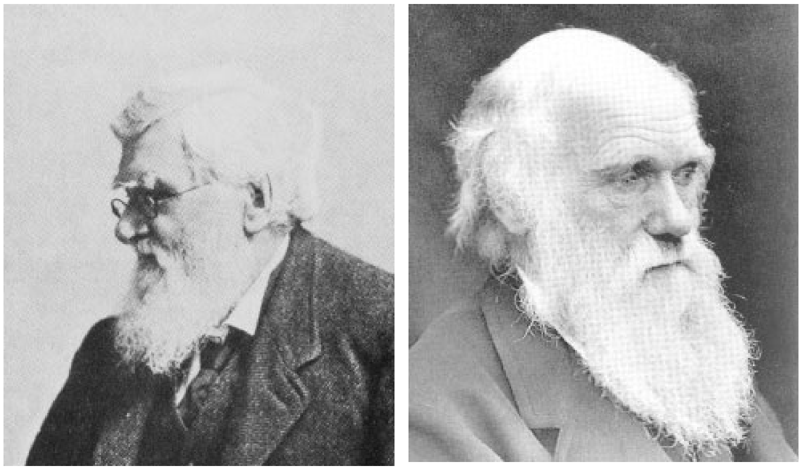


The pattern of the traits that we have described in this book is largely a very orderly one. Traits evolve differences, and new traits arise, out of earlier stages and by reusing processes that already exist. Traits of more ancient origin are more persistent, and derived traits are constrained by that fact. These general statements are true at the gene, morphology, and biochemical levels. As a result we can explain in a fairly formal and rigorous way, why certain patterns occur and others do not. There is much flexibility in these constraints as shown, for example, by the intercalative nature of the apparent re-evolution of traits like eyes. But because of the constraints we have described, we would agree with another observation of Thomas Browne (whom we quoted in Chapter 14 in regard to eyes and Ægles (Browne 1646)), that in "sanguineous" (vertebrate) animals, there are but two eyes, and they are in the head. There can be no Argus. In 1646 he was surmising, but evolutionary biology can explain the reason why.

There are undoubtedly errors in this book, although we hope they are not too many. Errors arise first from our own misunderstanding of existing knowledge in areas we have tried to represent that are beyond our own prior expertise, and then from the incompleteness and rapidly changing nature of that knowledge even if we have interpreted it accurately. However, the broad picture presented in these pages is likely to be robust even in the face of those kinds of error. It is always possible that fundamental new properties of genetic life will be discovered, and new fossil or living species are sure to be found. We cannot know what they might be, but from what we *do* know, it will be very surprising if they do not have the same general properties we have seen so pervasively so far: genealogy, divergence, duplication, reticulation, modularity, sequestration, interaction, functional use and reuse, and chance.

Early in their lives, Darwin and Wallace described the world as a self-directed phenomenon of change driven by competition, unfolding to the present panoply of complex organisms. Later in life, both men had their doubts. Darwin persisted in thinking of a Creator responsible for the initial start who perhaps then left things to go their own way. Wallace studied spiritualism and could never believe that the human mind could be the product of natural selection. Were these the bedside conversions of men facing death or the wishful thinking soft-headedness of age? Unlikely. It is clear from the work of both that these thoughts were present from the beginning. And their brilliant minds are not alone in this experience.

However it works, after billions of years in the making, evolution has produced such grandeur, so difficult to reconstruct in retrospect that it is difficult even for many evolutionary biologists today to accept completely mechanical explanations.



A B

Figure 17-4. (A) Old Wallace; (B) Old Darwin (painting by John Collier, in Down House).

The most reductionist of molecular reductionists have sometimes struggled not to invoke teleological notions but to substitute what comes within a hair's breadth of doing so, for example, via a very determinative view of selection, or Jacques Monod's use of "teleonomy" and the "project" of life to explain a kind of inherent property, if not drive, in the very molecules of life (Monod 1971). René Descartes, who is often credited with starting reductionist science in the first place, said that even if an organism is a machine, it was driven by the spirit. Whatever that is.

Darwin suggested in the *Descent of Man* (Darwin 1871) that despite its importance he may have placed too much stress on the role of natural selection but that he did so to show that natural processes would suffice to explain the living world without the need for special divine intervention. In this book, we have pointed out ways in which selection may be somewhat less necessary than Darwin stressed because other natural processes contribute to plausible explanation. It is remarkable that so modest a number of basic principles can account for both the production *and* evolution of the diversity of life we see on Earth today.

One of the most consistent findings, and a continual source of doubt, is that there are exceptions to these principles. Indeed, viewing them *as* exceptions shows a danger in scientific inference. The exceptions can only be viewed that way if we take our rules too seriously, thinking of them as classic "laws of nature." That there will be exceptions probably is itself one of the fundamental laws of the nature of life.

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